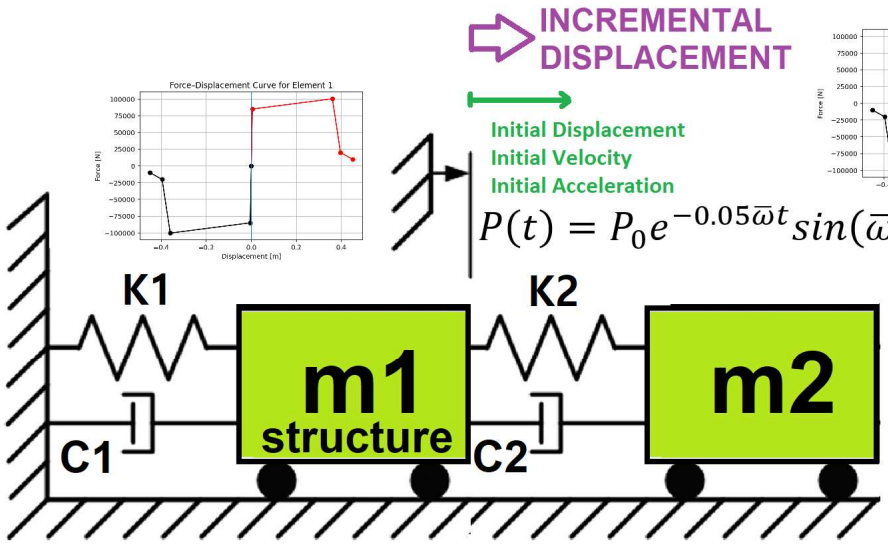
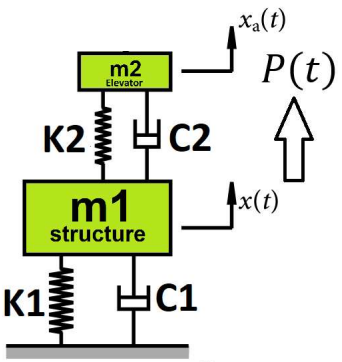
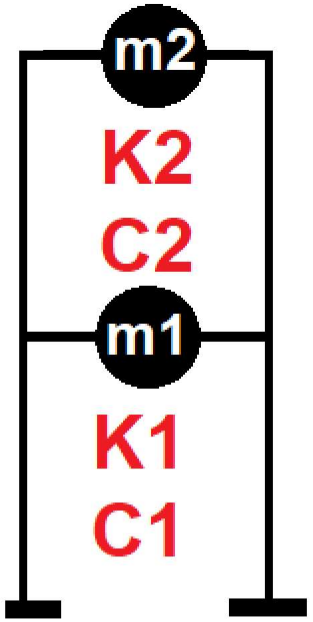
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

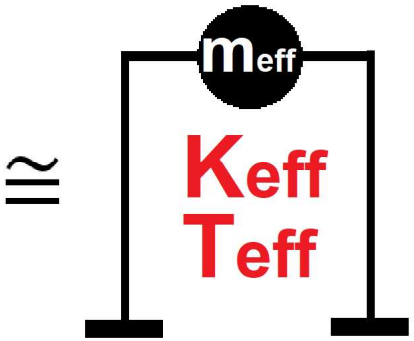
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM PENDULUM & STRUCTURE SYSTEM: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)









$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

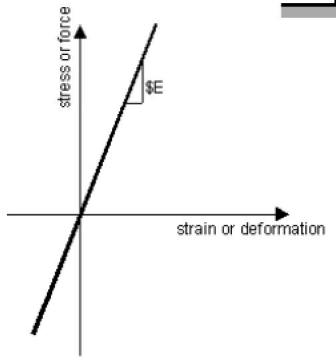
Δ_d = Lateral Displaement from Dynamic Analysis

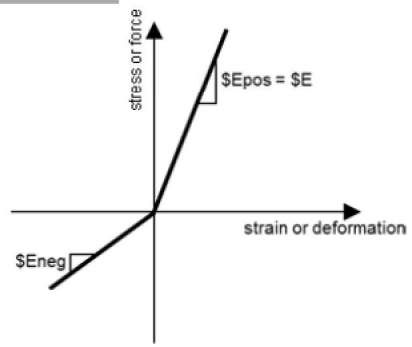
Δ_y = Lateral Yield Displaement from Pushover Analysis

Δ_u = Lateral Ultimate Displaement from Pushover Analysis

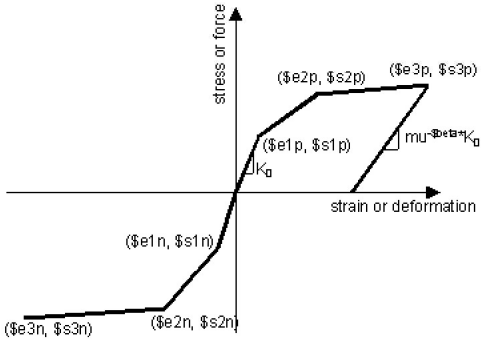
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



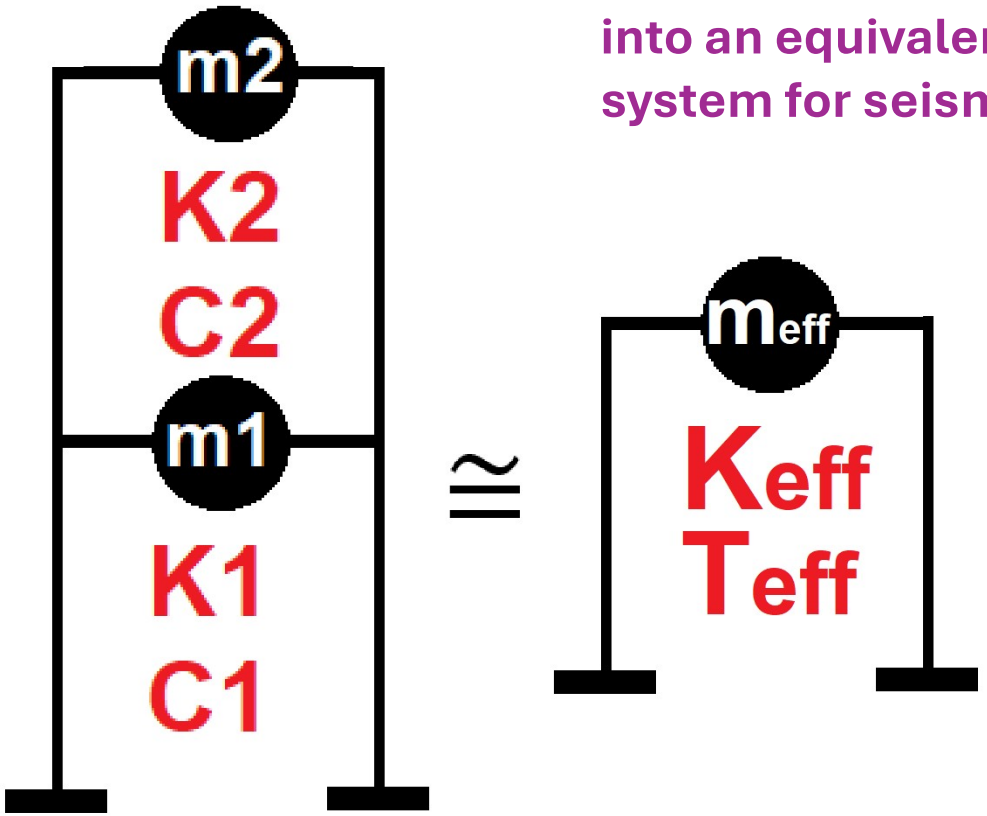


ELASTIC MATERIAL



INELASTIC MATERIAL

Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment



$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i} \qquad m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d} \qquad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CTURE_8_ANA\P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

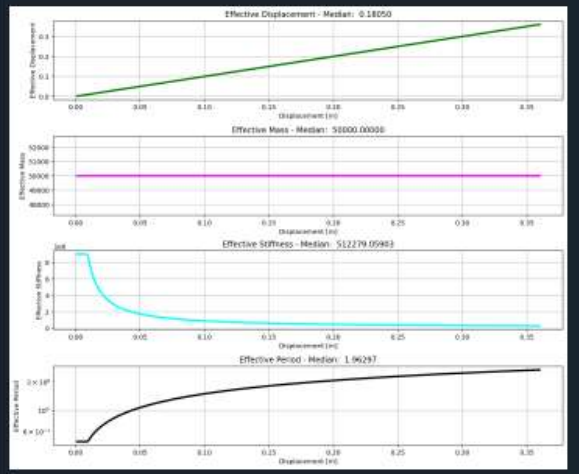
```
#######################################################################################################################
#                    >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#   COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM PENDULUM & STRUCTURE SYSTEM :
#                AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION,
#                          STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS
#-----------------------------------------------------------------------------------------------------------------------
#                EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
#-----------------------------------------------------------------------------------------------------------------------
#          ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacement)
#-----------------------------------------------------------------------------------------------------------------------
#EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANALY
#-----------------------------------------------------------------------------------------------------------------------
#                                                P(t) = P0 sin(wt)
#                                            P(t) = P0 exp(-0.05wt) sin(wt)
#-----------------------------------------------------------------------------------------------------------------------
#         ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
#                                      OF ENERGY DISSIPATION CAPACITY INDEX
#-----------------------------------------------------------------------------------------------------------------------
#                              MARKOV CHAIN FOR EVALUTIONS OF PROBABILITY OF FAILURE
#-----------------------------------------------------------------------------------------------------------------------
#                  THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#                                  EMAIL: salar.d.ghashghaei@gmail.com
#######################################################################################################################
"""
Nonlinear Seismic Performance Assessment of a MDOF Pendulum and Structure System:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake L

This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF Elevator and
 for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity
 (elastic-perfectly plastic or hysteretic steel with strain hardening)
 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
 and large displacements—essential for collapse assessment.

Eight analysis protocols are implemented:
```

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21 %



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Spyder (Python 3.12)

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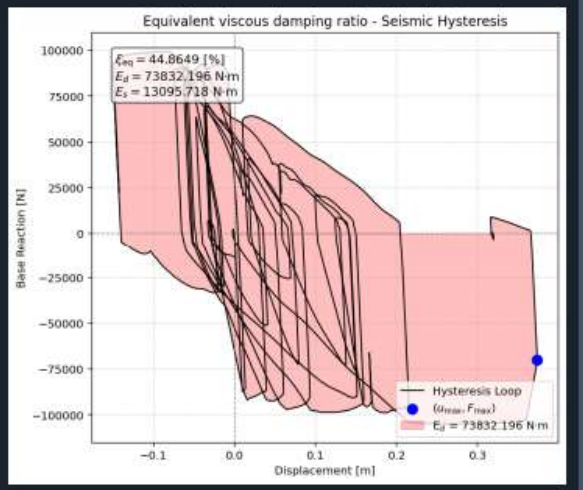
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CTURE_8_ANA\P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

```
#########################################################################################################
#                  >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#   COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM PENDULUM & STRUCTURE SYSTEM :
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IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

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STATIC ANALYSIS

Spyder (Python 3.12)

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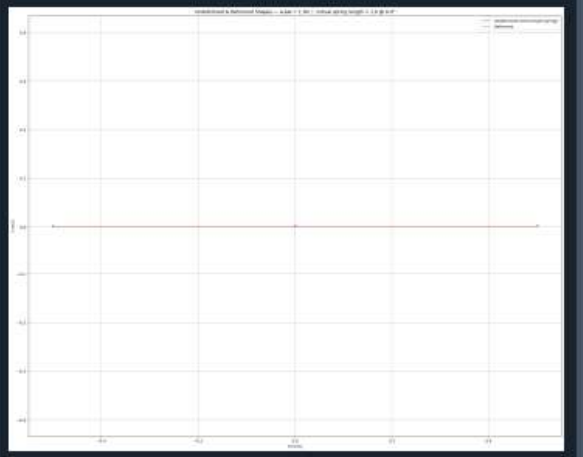
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P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

```
1216  # STATIC ANALYSIS
1217  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1218  ANAL_TYPE = 'STATIC'
1219
1220  DATA = PENDULUM_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1221  (reaction, disp, DI) = DATA
1222
1223  S01.PLOT_1D_SPRING(deformed_scale=1.0, virtual_spring_length=1.0, virtual_spring_angle=0.0)
1224  #%%------------------------------------------------------
1225  # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1226  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1227  ANAL_TYPE = 'PUSHOVER'
1228
1229  DATA = PENDULUM_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1230  (reaction_PUSH, disp_PUSH, DI_PUSH,
1231   ele_force_PUSH, node_displacements_PUSH,
1232   PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
1233
1234
1235  XDATA = disp_PUSH
1236  YDATA = reaction_PUSH
1237  XLABEL = 'Displacement [m]'
1238  YLABEL = 'Base Reaction [N]'
1239  TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1240  COLOR = 'black'
1241  SEMILOGY = False
1242  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1243
1244  DATA = S07.BILNEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
1245  (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
1246
1247  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1248  plt.figure(0, figsize=(12, 8))
1249  plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
```

...ULUM_STRUCTURE_8_ANA+\+61_MDOF_PENDULUM_STRUCTURE_8_ANA

12 %





IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 21, Col 18 | UTF-8 | CRLF | RW | Mem 46%

PUSHOVER ANALYSIS

Spyder (Python 3.12)

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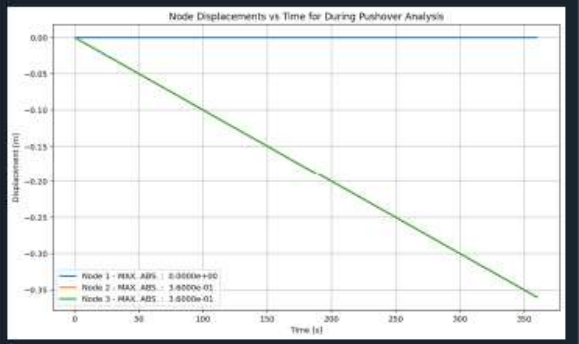
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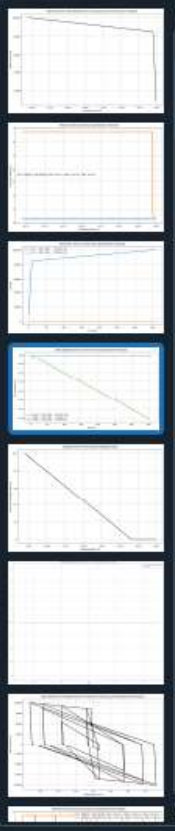
P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

```
1225  # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1226  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1227  ANAL_TYPE = 'PUSHOVER'
1228
1229  DATA = PENDULUM_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1230  (reaction_PUSH, disp_PUSH, DI_PUSH,
1231   ele_force_PUSH, node_displacements_PUSH,
1232   PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
1233
1234
1235  XDATA = disp_PUSH
1236  YDATA = reaction_PUSH
1237  XLABEL = 'Displacement [m]'
1238  YLABEL = 'Base Reaction [N]'
1239  TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1240  COLOR = 'black'
1241  SEMILOGY = False
1242  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1243
1244  DATA = S07.BILNEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
1245  (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
1246
1247  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1248  plt.figure(0, figsize=(12, 8))
1249  plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1250  plt.plot(disp_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1251  plt.title('Period of Structure During Pushover Analysis')
1252  plt.ylabel('Structural Period [s]')
1253  plt.xlabel('Displacement [m]')
1254  #plt.semilogy()
1255  plt.grid()
1256  plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_MIN_
1257              f'PERIOD - MAX VALUES:  Min: {np.min(PERIOD_MAX_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_MAX_
1258              ])
```

...ULUM_STRUCTURE_8_ANA+\+61_MDOF_PENDULUM_STRUCTURE_8_ANA

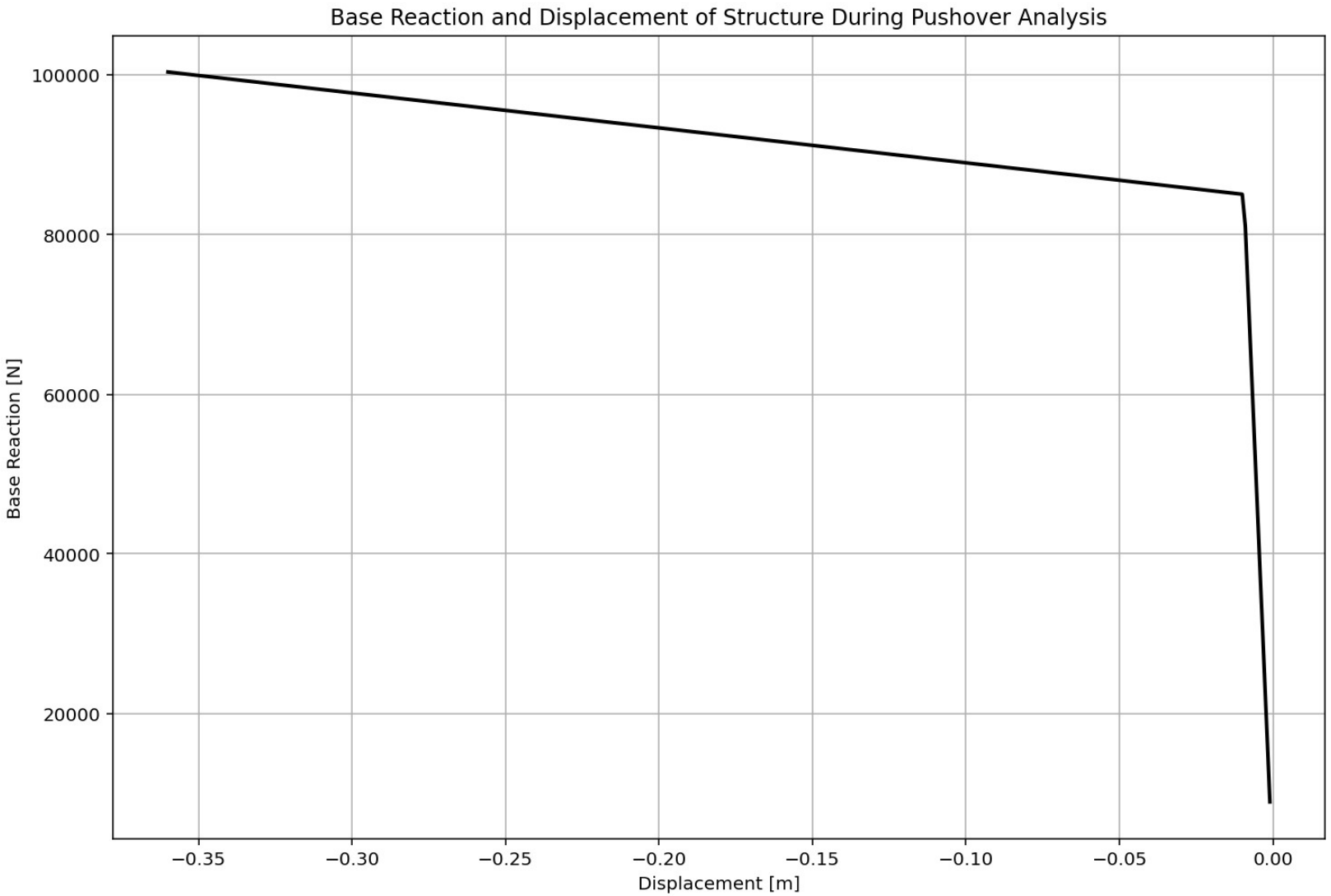
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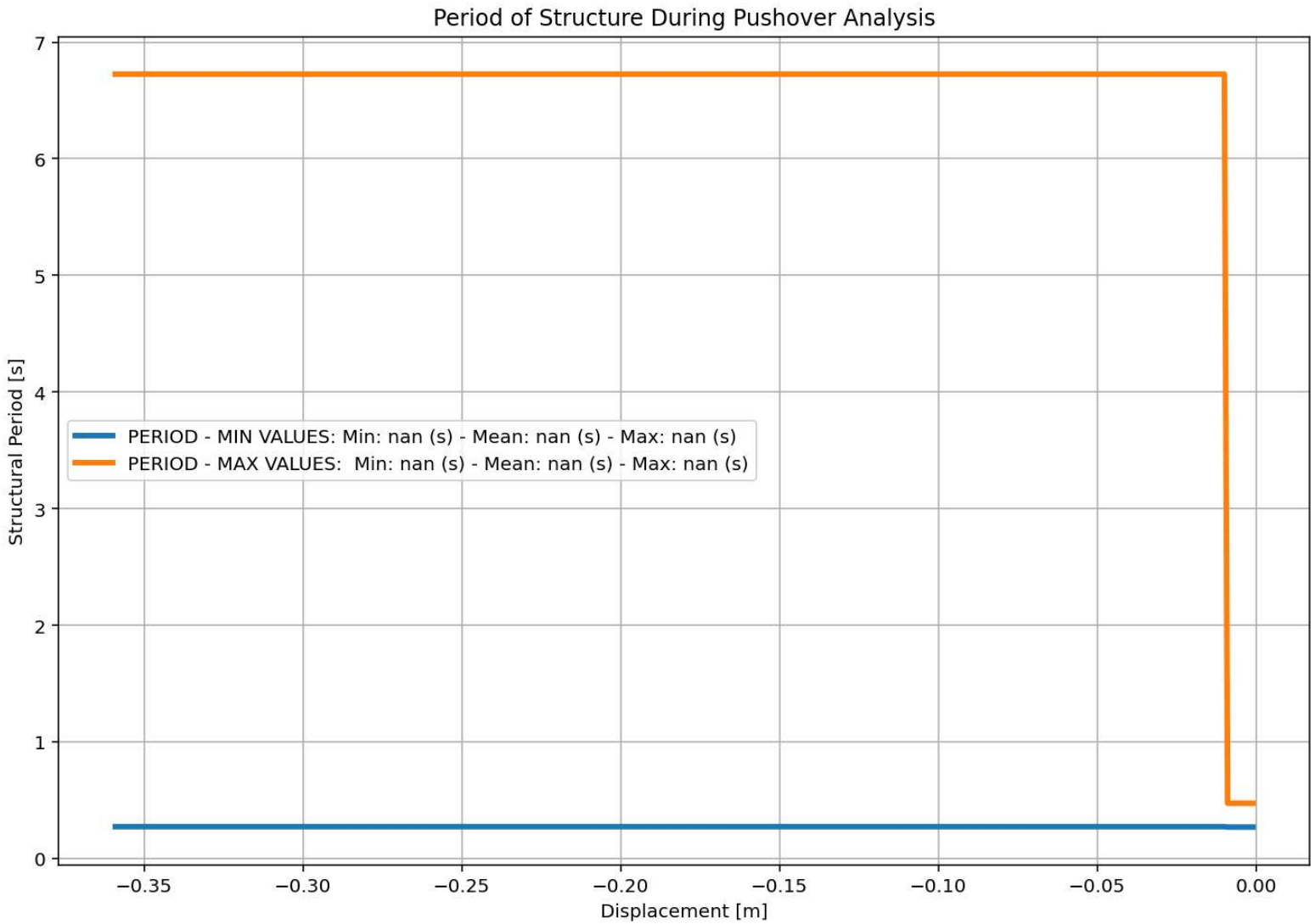


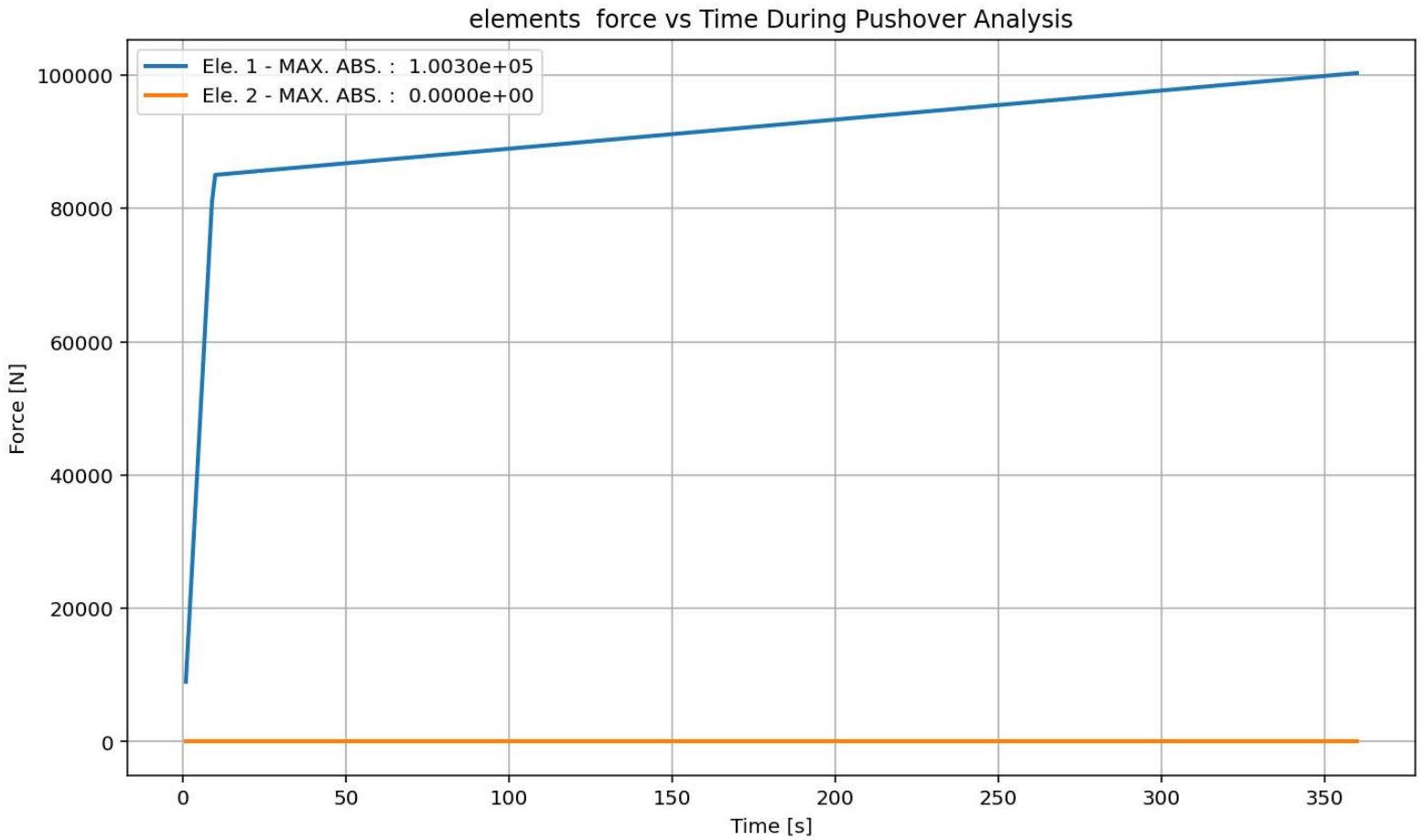


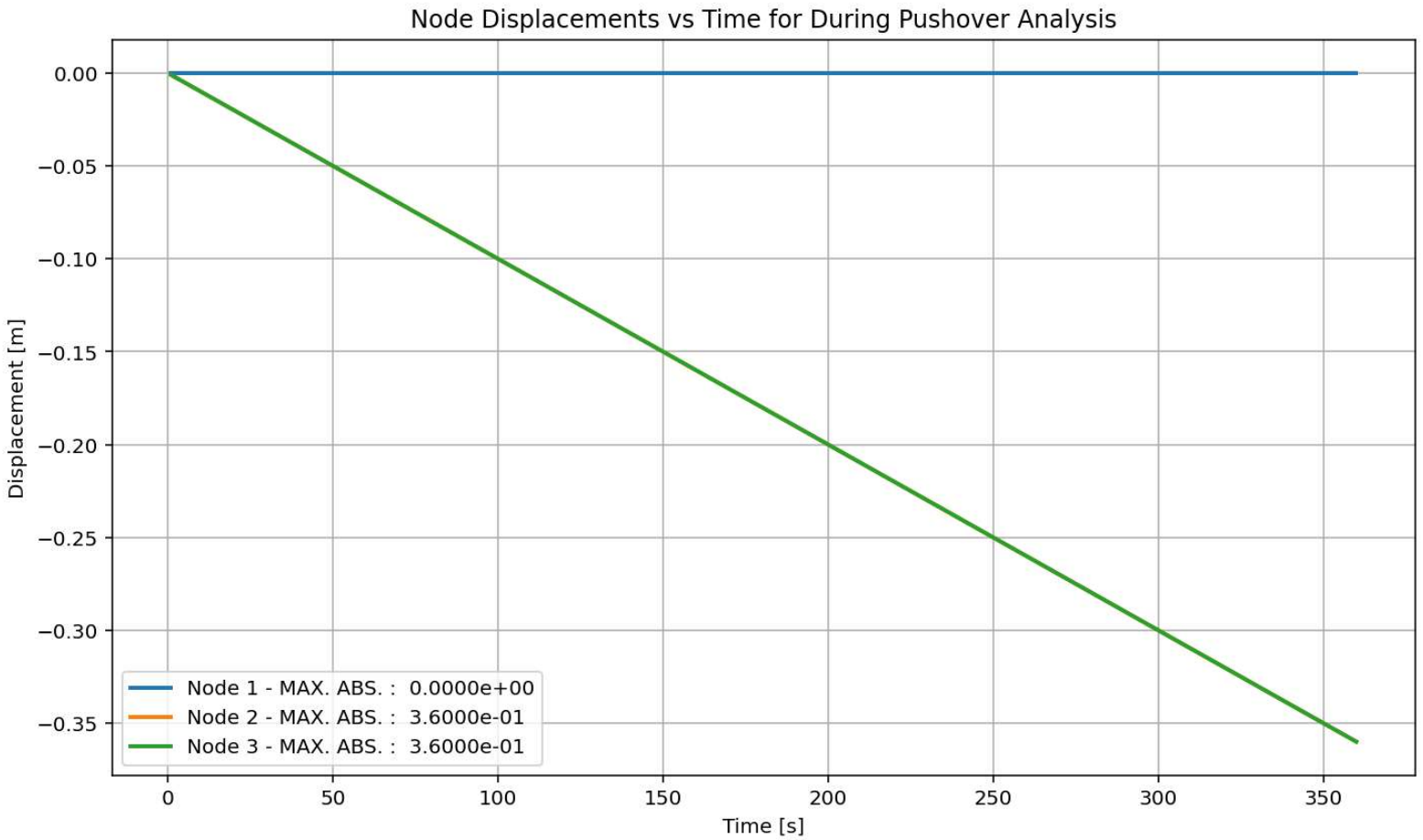
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

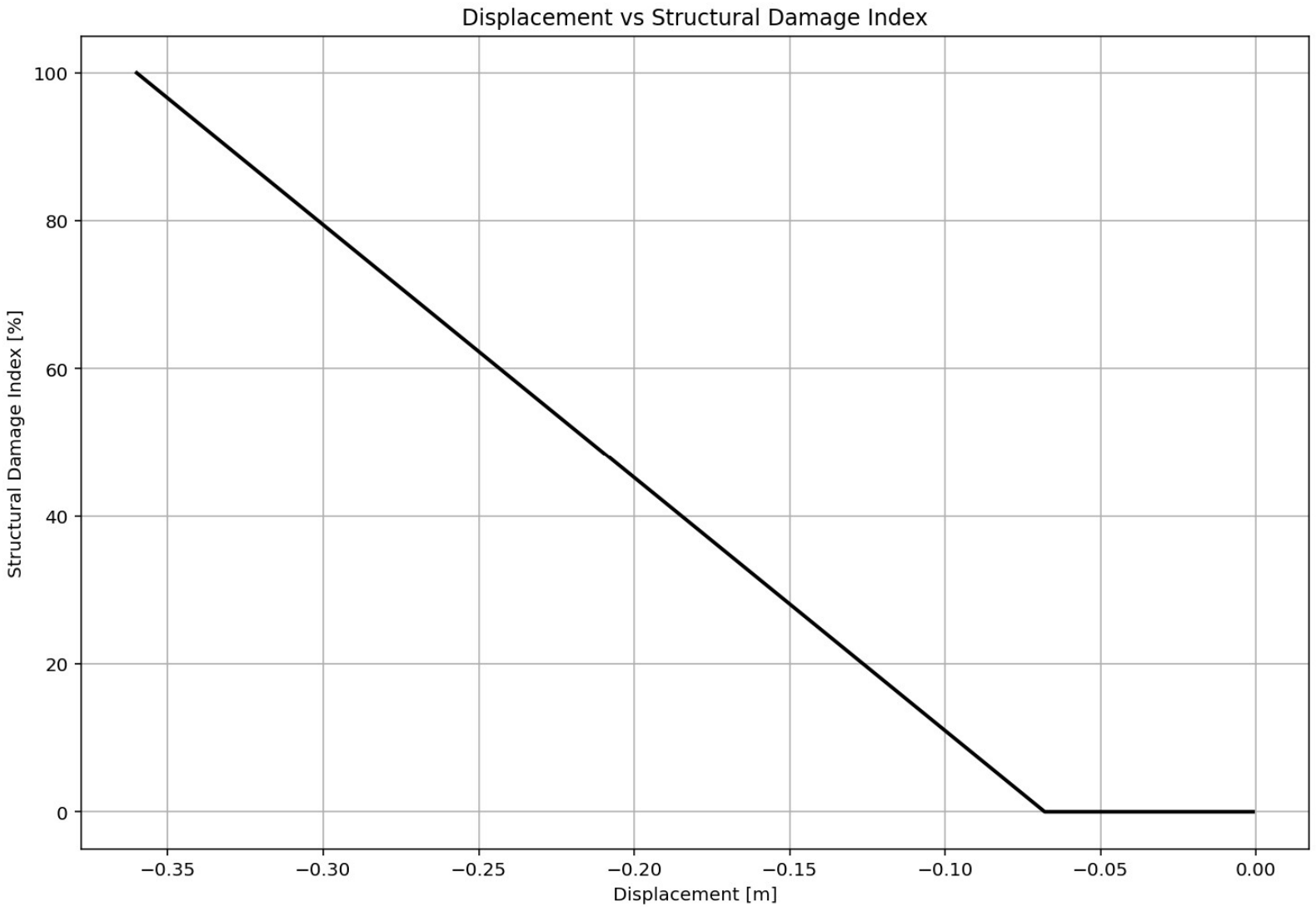
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 21, Col 18 | UTF-8 | CRLF | RW | Mem 47%


Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
100000
80000
60000
40000
20000
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]


Period of Structure During Pushover Analysis
Structural Period [s]
Displacement [m]
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
0
1
2
3
4
5
6
7
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00


elements force vs Time During Pushover Analysis
Ele. 1 - MAX. ABS. : 1.0030e+05
Ele. 2 - MAX. ABS. : 0.0000e+00
Force [N]
100000
80000
60000
40000
20000
0
0
50
100
150
200
250
300
350
Time [s]


Node Displacements vs Time for During Pushover Analysis
Displacement [m]
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
0
50
100
150
200
250
300
350
Time [s]
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.6000e-01
Node 3 - MAX. ABS. : 3.6000e-01


Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]

Spyder (Python 3.12)

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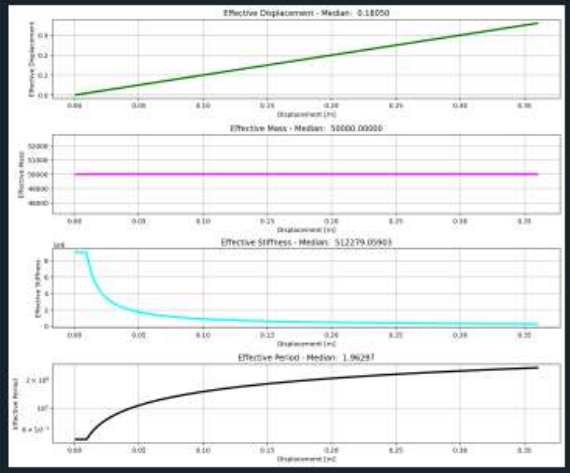
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CTURE_8_ANA\P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

```
1225  # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1226  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1227  ANAL_TYPE = 'PUSHOVER'
1228
1229  DATA = PENDULUM_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1230  (reaction_PUSH, disp_PUSH, DI_PUSH,
1231   ele_force_PUSH, node_displacements_PUSH,
1232   PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
1233
1234
1235  XDATA = disp_PUSH
1236  YDATA = reaction_PUSH
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1240  COLOR = 'black'
1241  SEMILOGY = False
1242  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1243
1244  DATA = S07.BILNEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
1245  (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
1246
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1254  #plt.semilogy()
1255  plt.grid()
1256  plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_MIN_
1257              f'PERIOD - MAX VALUES:  Min: {np.min(PERIOD_MAX_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_MAX_
1258              ])
```

...ULUM_STRUCTURE_8_ANA+\+61_MDOF_PENDULUM_STRUCTURE_8_ANA

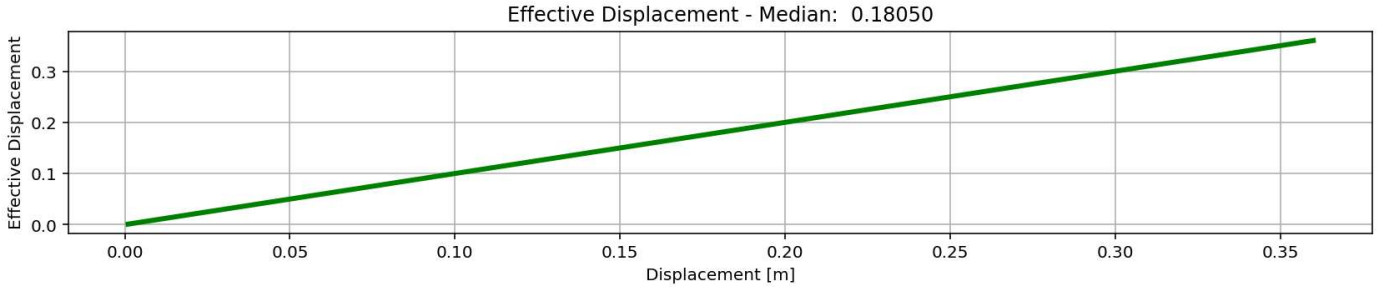
21 %



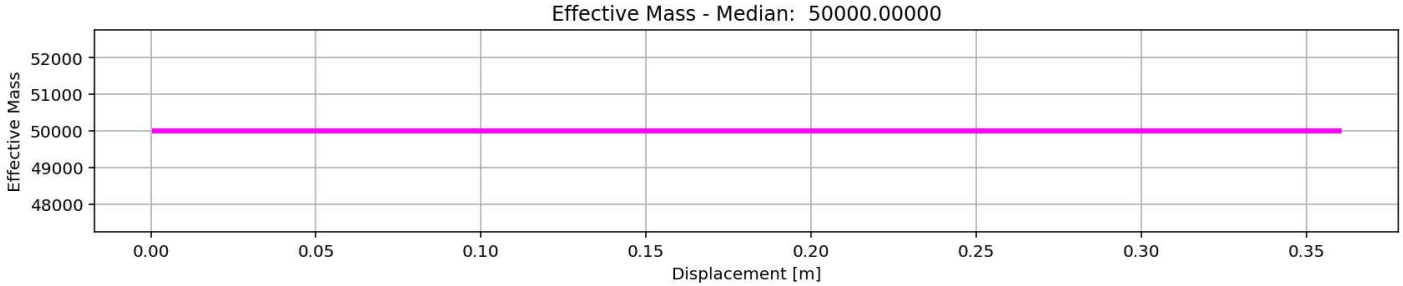


IPython Console Plots Variable Explorer Help Debugger History Files

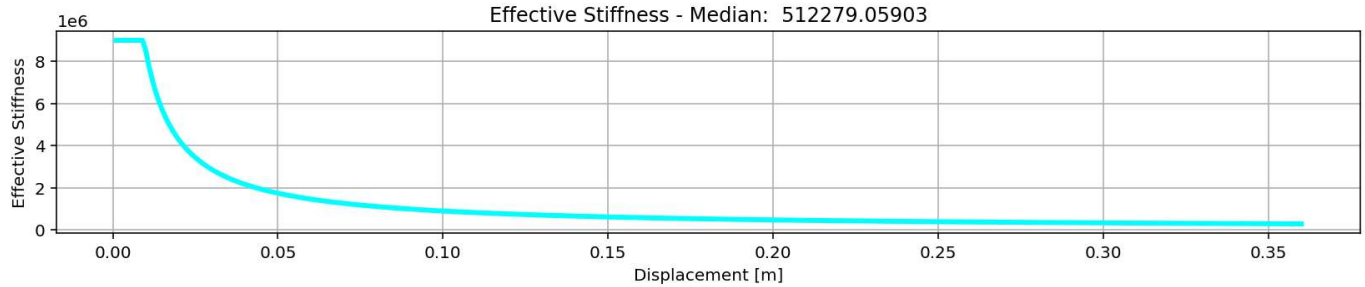
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 21, Col 18 UTF-8 CRLF RW Mem 47%



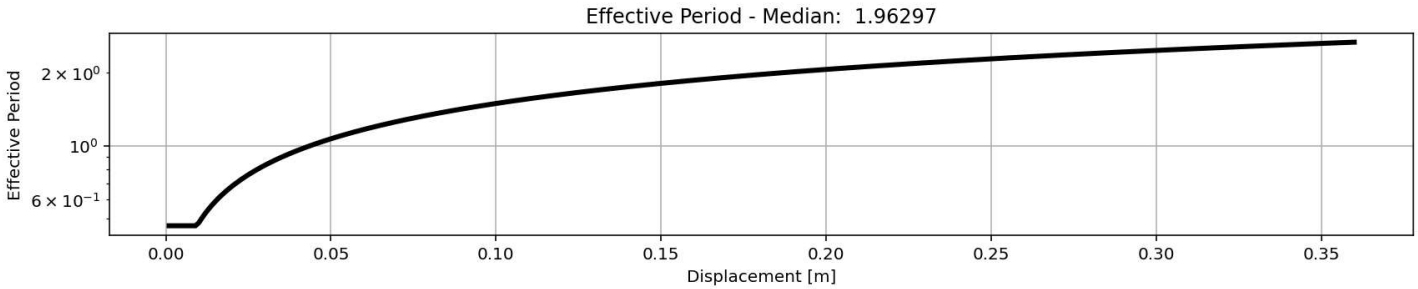
$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i}$$



$$m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$



$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d}$$



$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

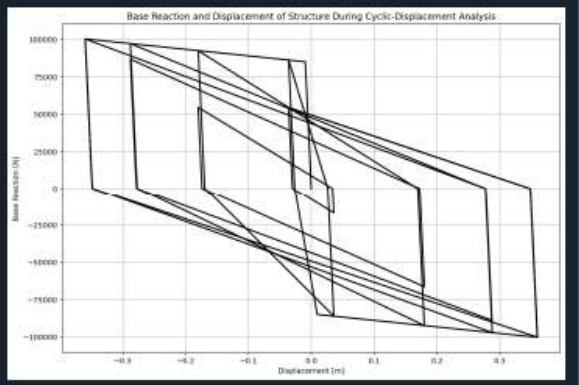
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CTURE_8_ANA\P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

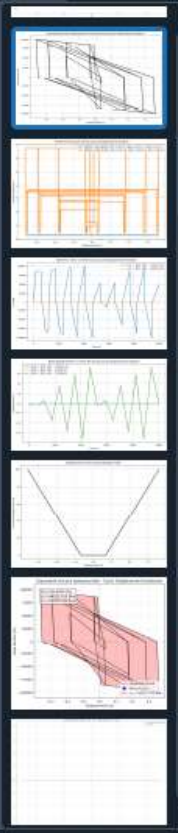
P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

```
1285  # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1286  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1287  ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1288
1289  DATA = PENDULUM_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1290  (reaction_CP, disp_CP, DI_CP,
1291   ele_force_CP, node_displacements_CP,
1292   PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
1293
1294
1295  XDATA = disp_CP
1296  YDATA = reaction_CP
1297  XLABEL = 'Displacement [m]'
1298  YLABEL = 'Base Reaction [N]'
1299  TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
1300  COLOR = 'black'
1301  SEMILOGY = False
1302  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1303
1304  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1305  plt.figure(0, figsize=(12, 8))
1306  plt.plot(disp_CP, PERIOD_MIN_CP, linewidth=3)
1307  plt.plot(disp_CP, PERIOD_MAX_CP, linewidth=3)
1308  plt.title('Period of Structure During Cyclic Displacement Analysis')
1309  plt.ylabel('Structural Period [s]')
1310  plt.xlabel('Displacement [m]')
1311  #plt.semilogy()
1312  plt.grid()
1313  plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - Mean: {np.mean(PERIOD_MIN_CP,
1314              f'PERIOD - MAX VALUES:  Min: {np.min(PERIOD_MAX_CP):.3f} (s) - Mean: {np.mean(PERIOD_MAX_CP
1315              ])
1316  plt.show()
1317
1318
```

...ULUM_STRUCTURE_8_ANA+\+61_MDOF_PENDULUM_STRUCTURE_8_ANA

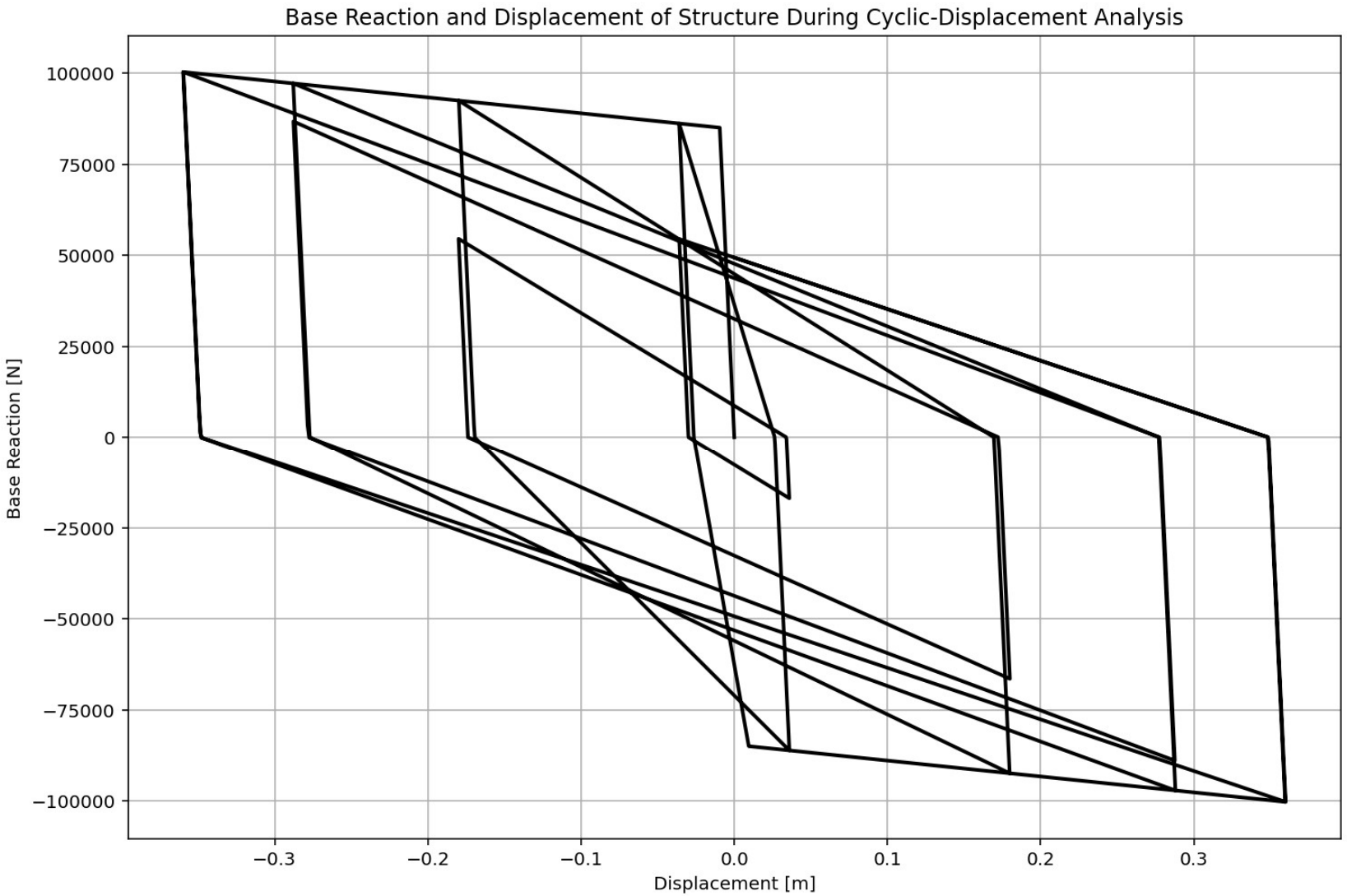
24 %

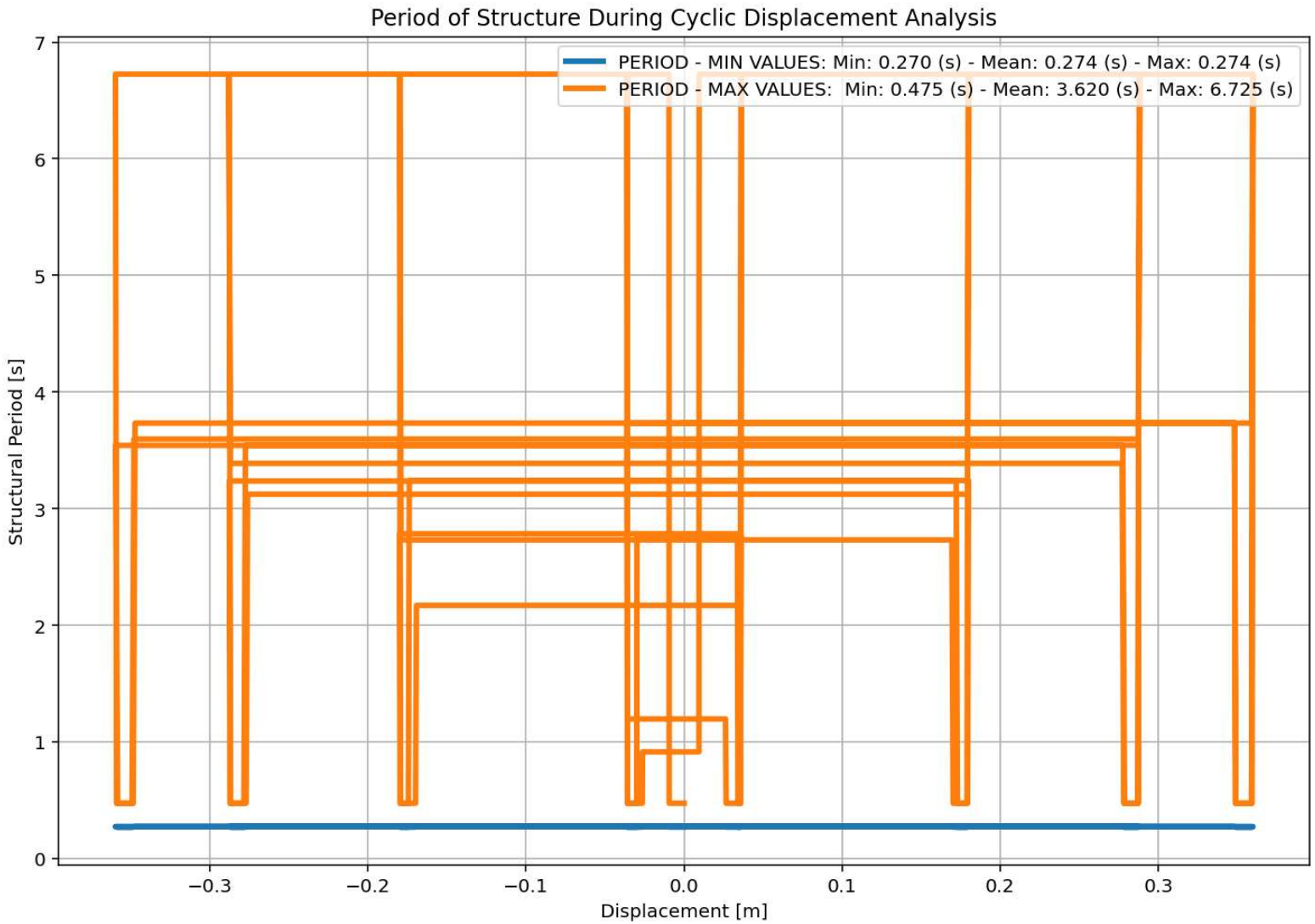


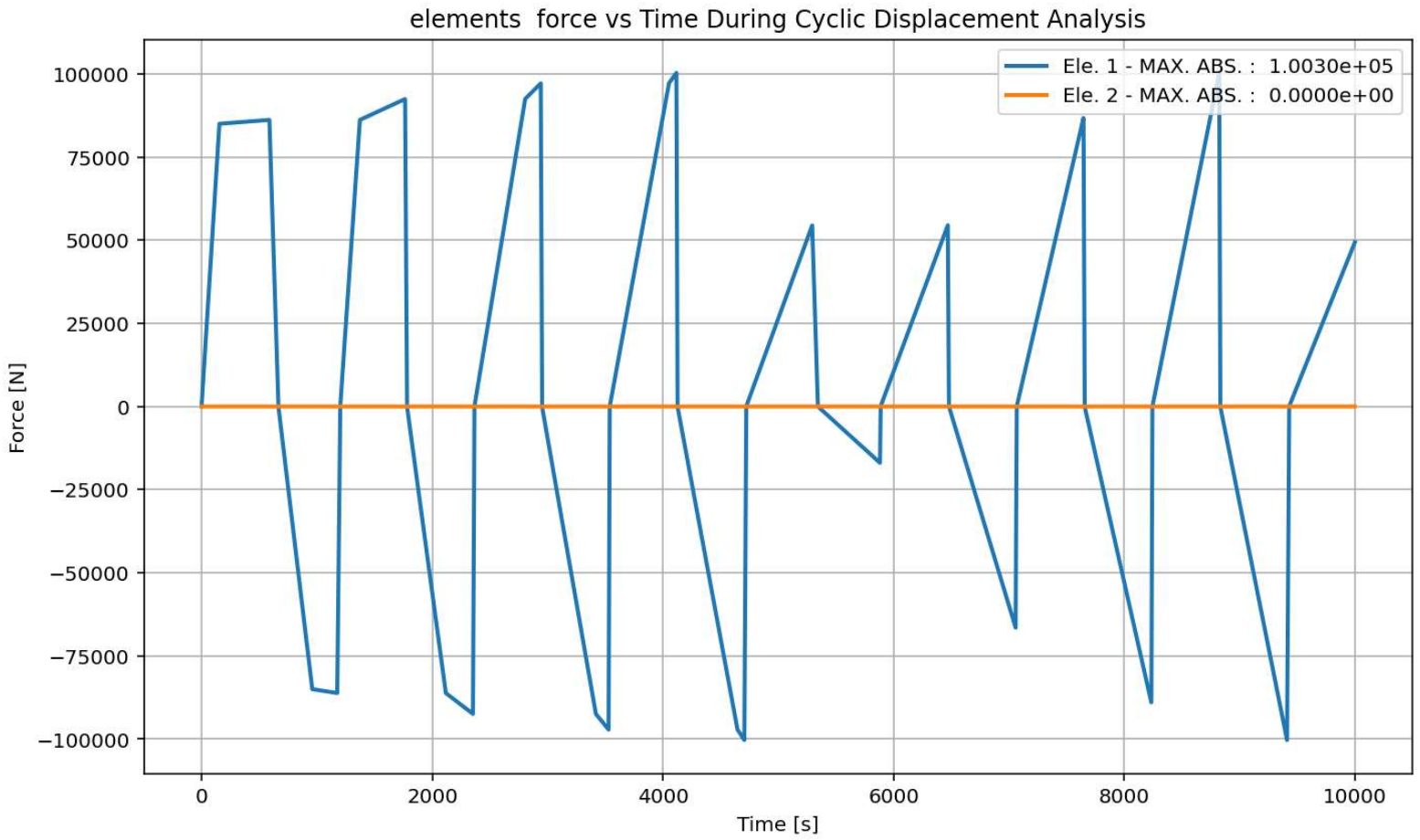


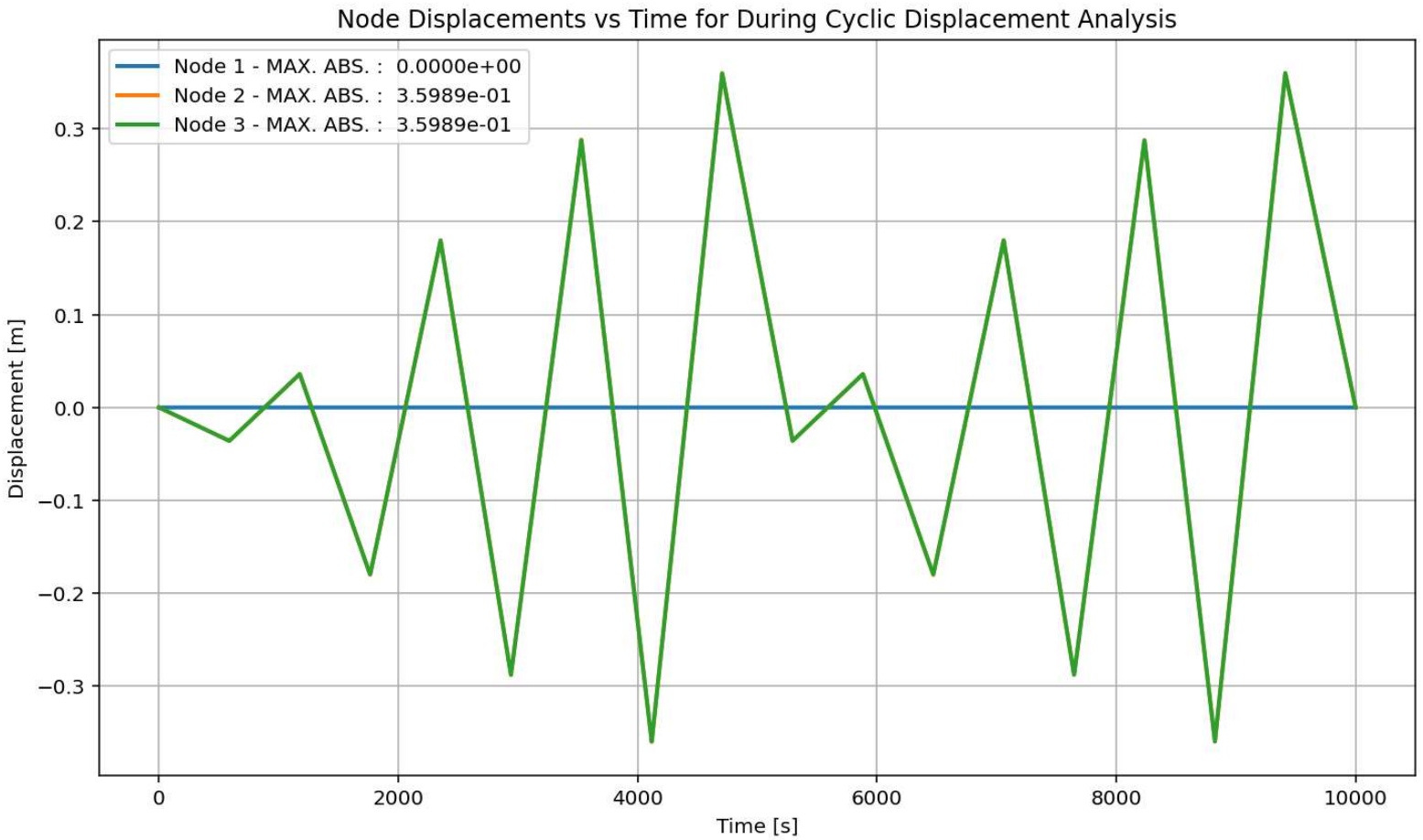
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

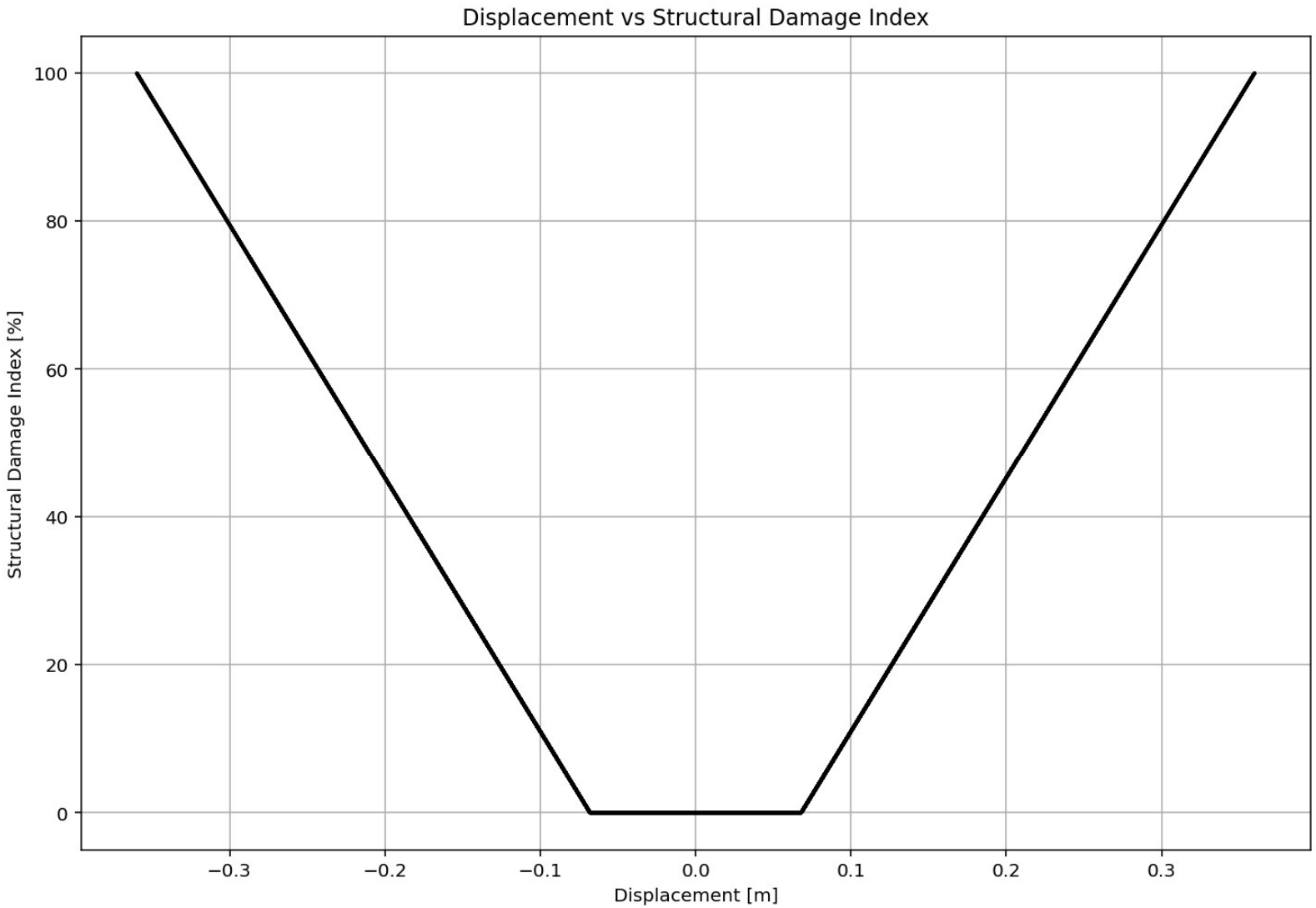
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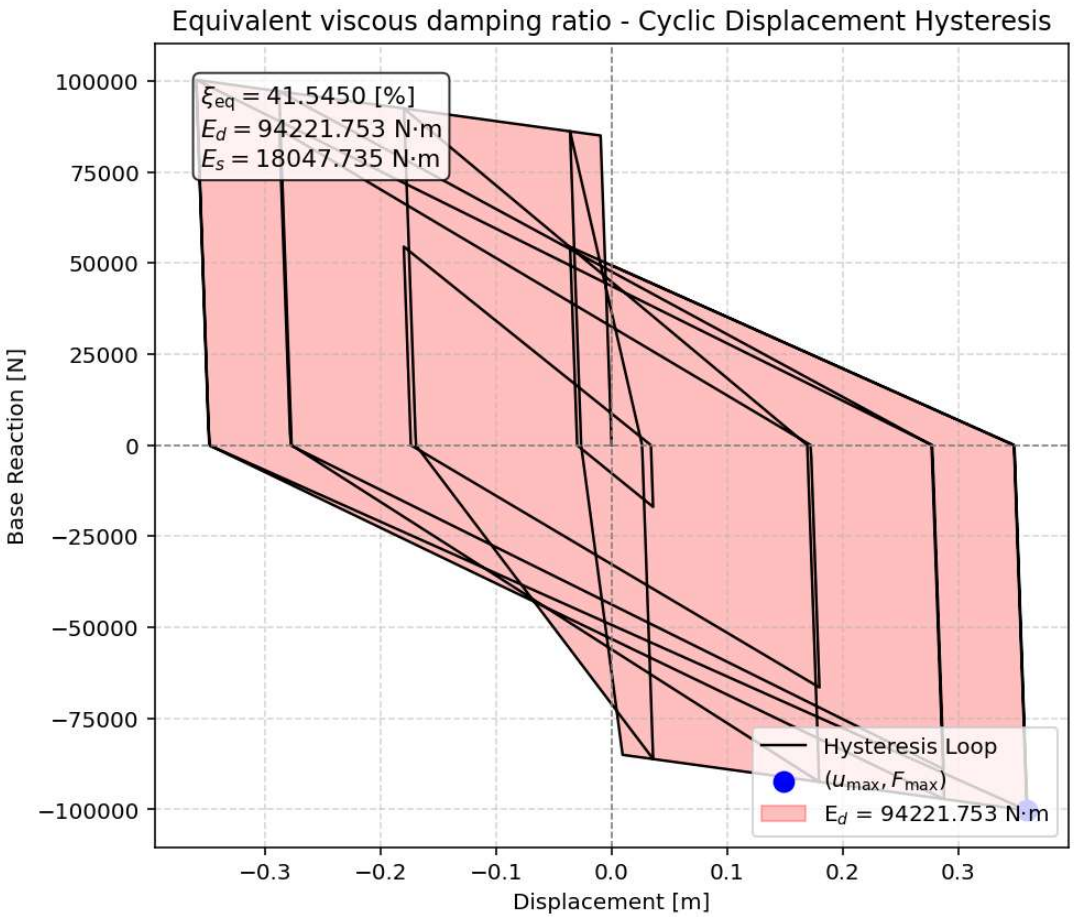

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]


Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: 0.270 (s) - Mean: 0.274 (s) - Max: 0.274 (s)
PERIOD - MAX VALUES: Min: 0.475 (s) - Mean: 3.620 (s) - Max: 6.725 (s)
Structural Period [s]
Displacement [m]
0
1
2
3
4
5
6
7
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3


elements force vs Time During Cyclic Displacement Analysis
Ele. 1 - MAX. ABS. : 1.0030e+05
Ele. 2 - MAX. ABS. : 0.0000e+00
Force [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
0
2000
4000
6000
8000
10000
Time [s]


Node Displacements vs Time for During Cyclic Displacement Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.5989e-01
Node 3 - MAX. ABS. : 3.5989e-01
Displacement [m]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
2000
4000
6000
8000
10000
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 41.5450$ [%]
$E_d = 94221.753$ N·m
$E_s = 18047.735$ N·m
Base Reaction [N]
Displacement [m]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 94221.753 N·m

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

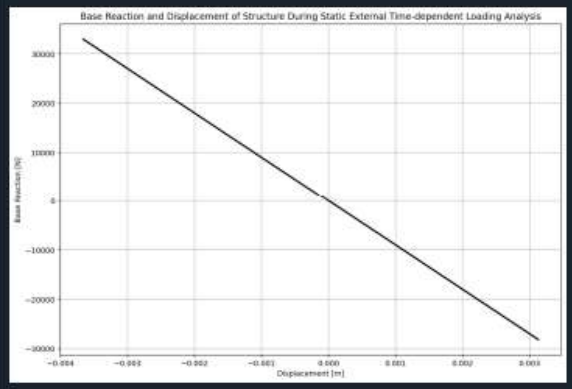
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CTURE_8_ANA\P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

```
1351  # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1352  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1353  ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1354
1355  DATA = PENDULUM_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1356
1357  (reaction_ETDLS, disp_ETDLS, DI_ETDLS,
1358   ele_force_ETDLS, node_displacements_ETDLS,
1359   PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
1360
1361
1362  XDATA = disp_ETDLS
1363  YDATA = reaction_ETDLS
1364  XLABEL = 'Displacement [m]'
1365  YLABEL = 'Base Reaction [N]'
1366  TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent Loading Ana
1367  COLOR = 'black'
1368  SEMILOGY = False
1369  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1370
1371  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1372  plt.figure(0, figsize=(12, 8))
1373  plt.plot(disp_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1374  plt.plot(disp_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1375  plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
1376  plt.ylabel('Structural Period [s]')
1377  plt.xlabel('Displacement [m]')
1378  #plt.semilogy()
1379  plt.grid()
1380  plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD_MIN_
1381              f'PERIOD - MAX VALUES:  Min: {np.min(PERIOD_MAX_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD_MA
1382              ])
1383  plt.show()
1384
```

...ULUM_STRUCTURE_8_ANA+\+61_MDOF_PENDULUM_STRUCTURE_8_ANA

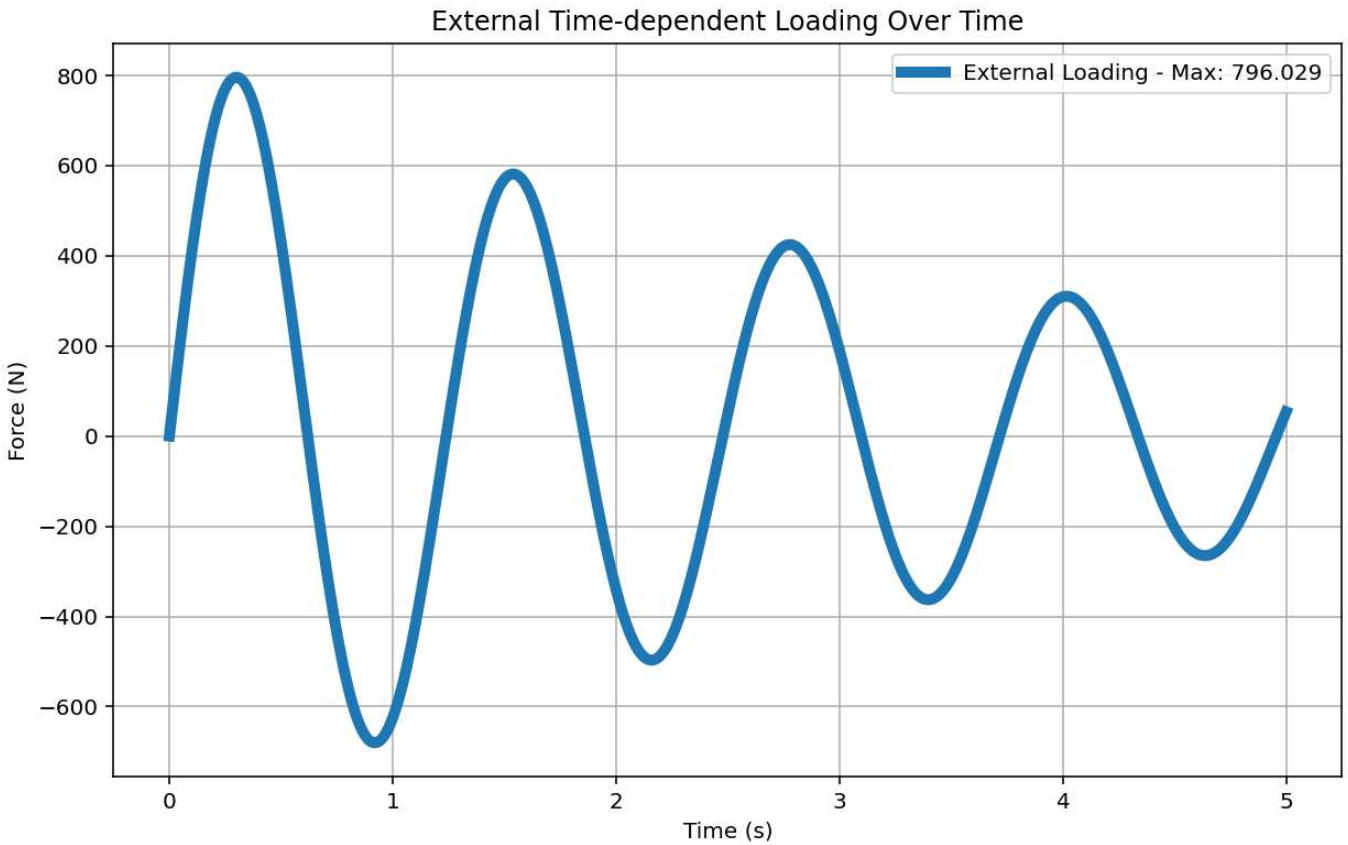
24 %

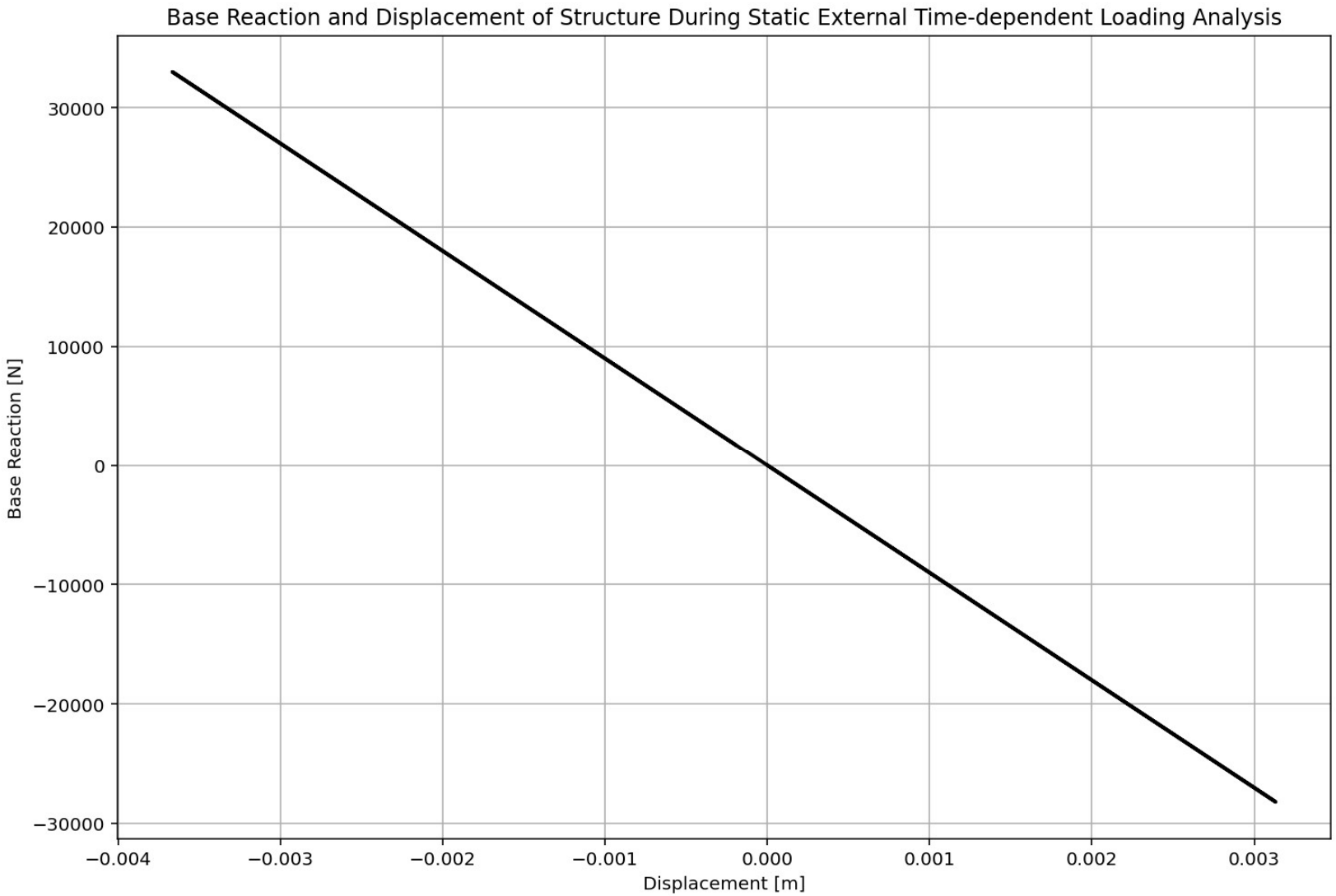


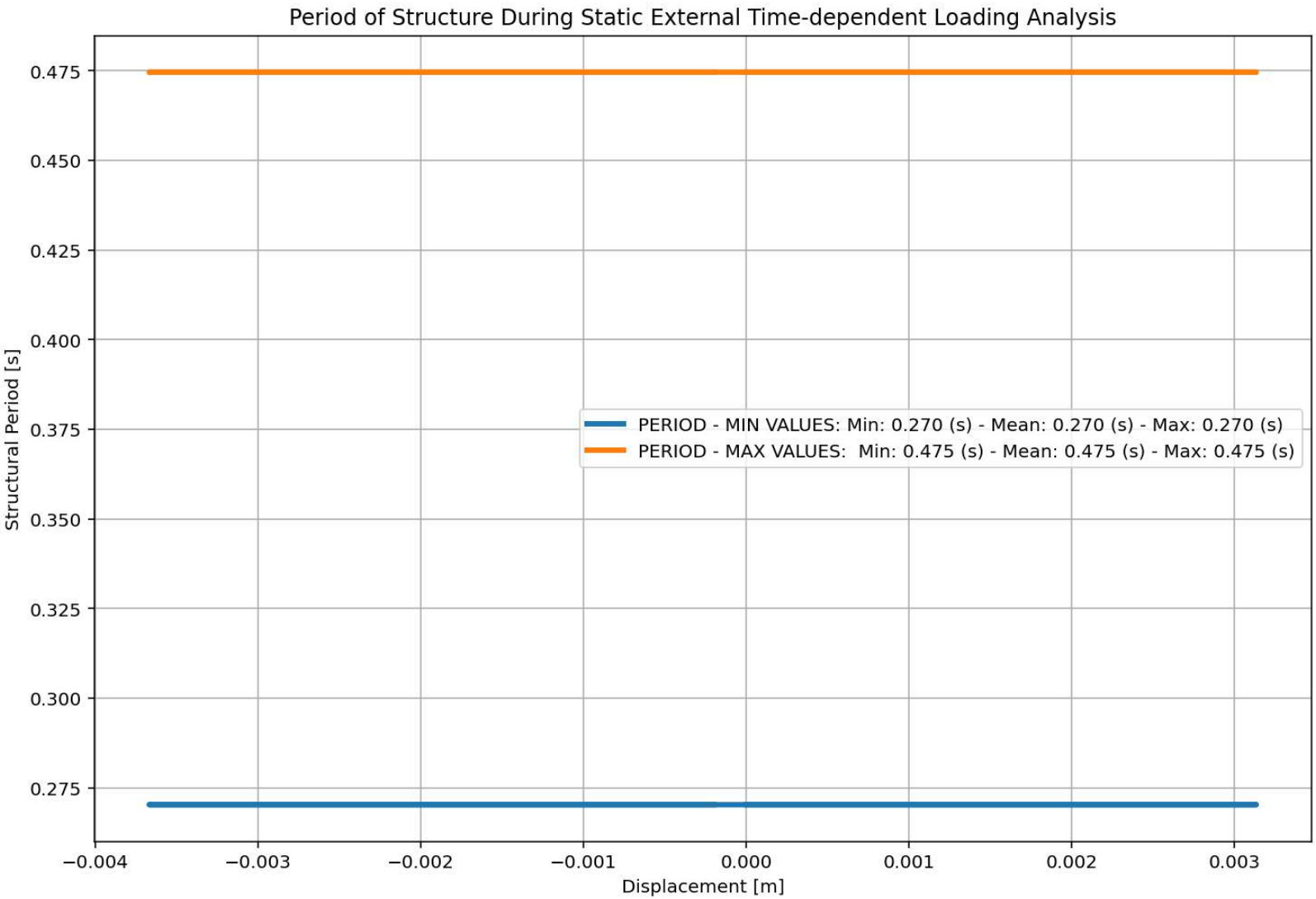


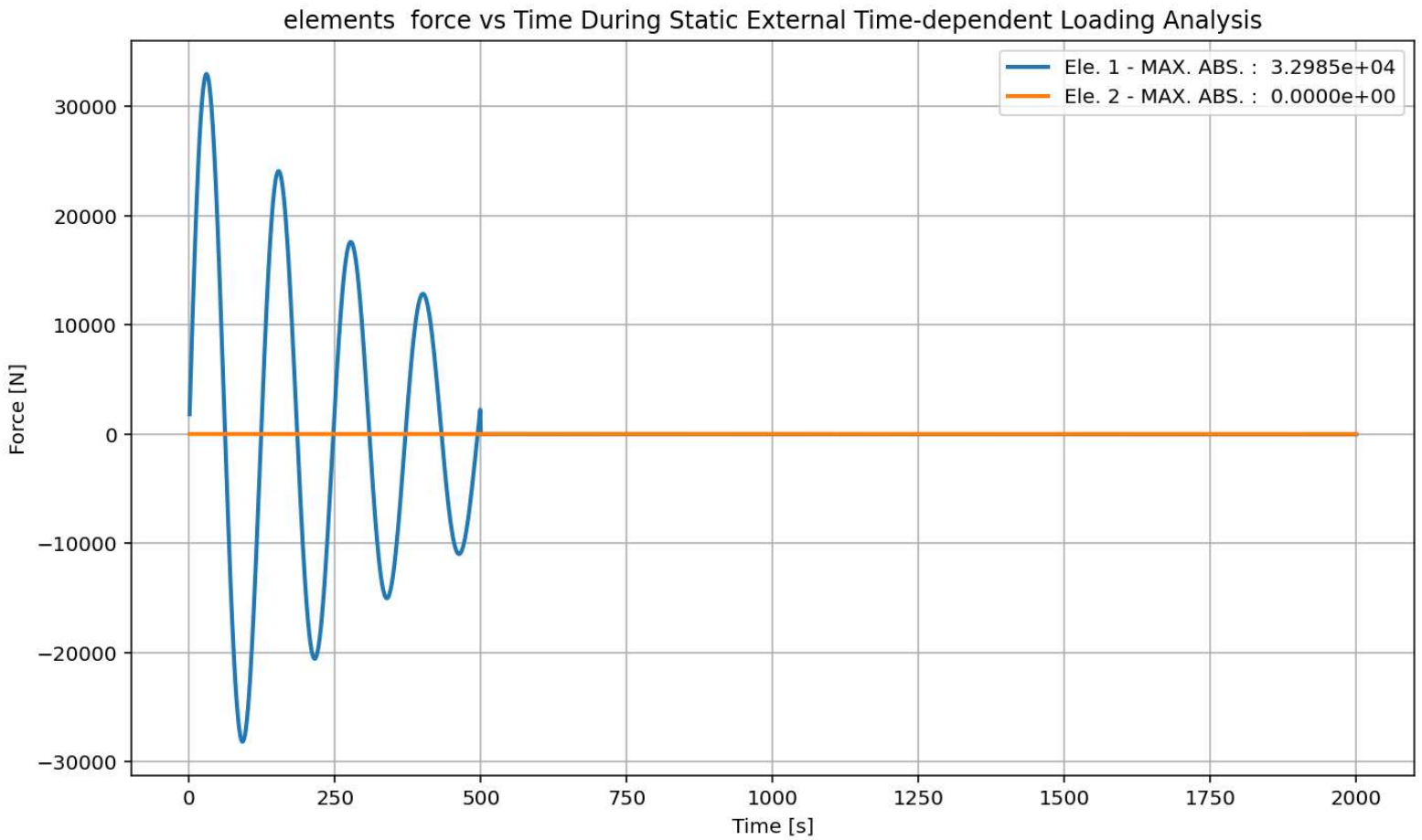
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

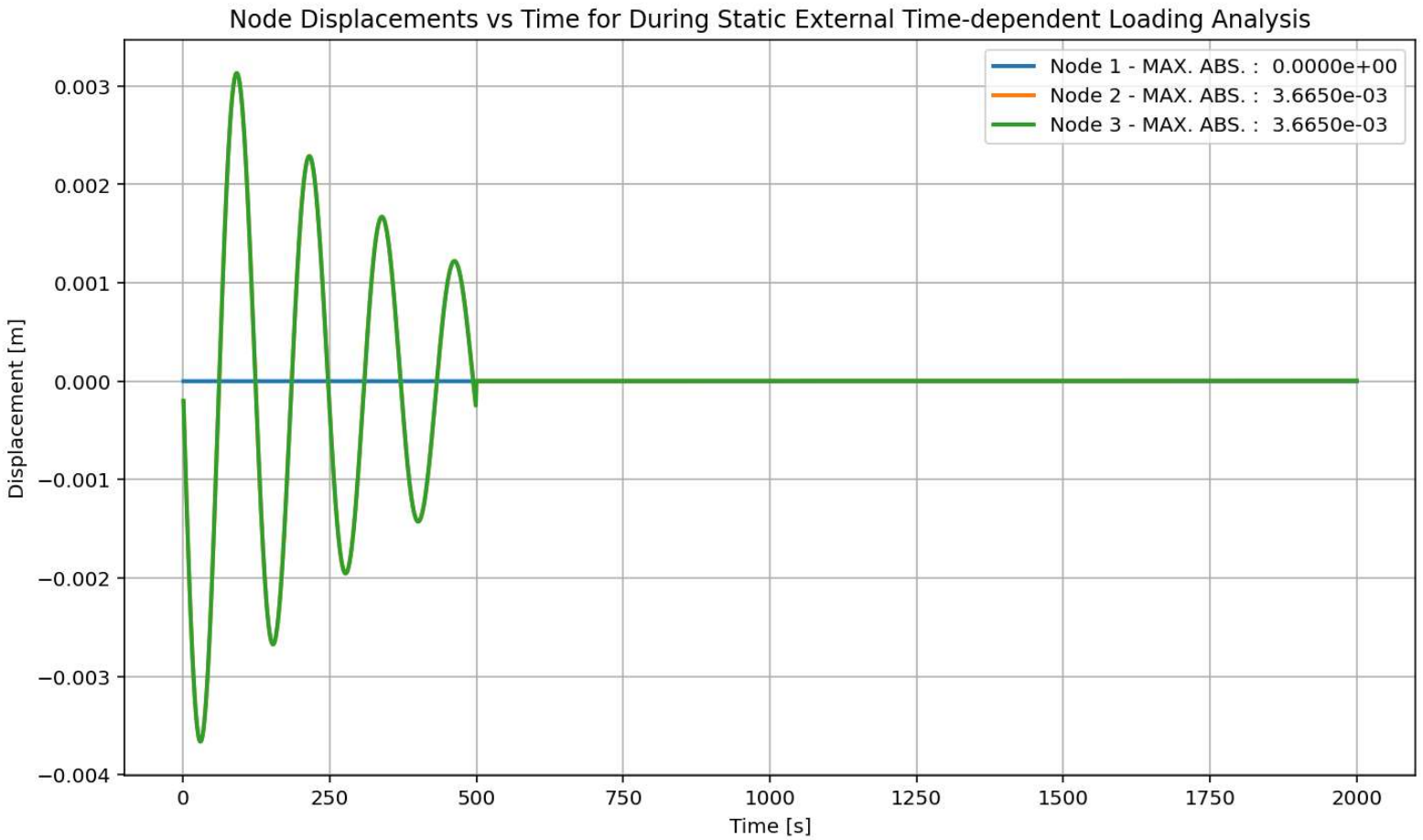
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 21, Col 18 | UTF-8 | CRLF | RW | Mem 47%

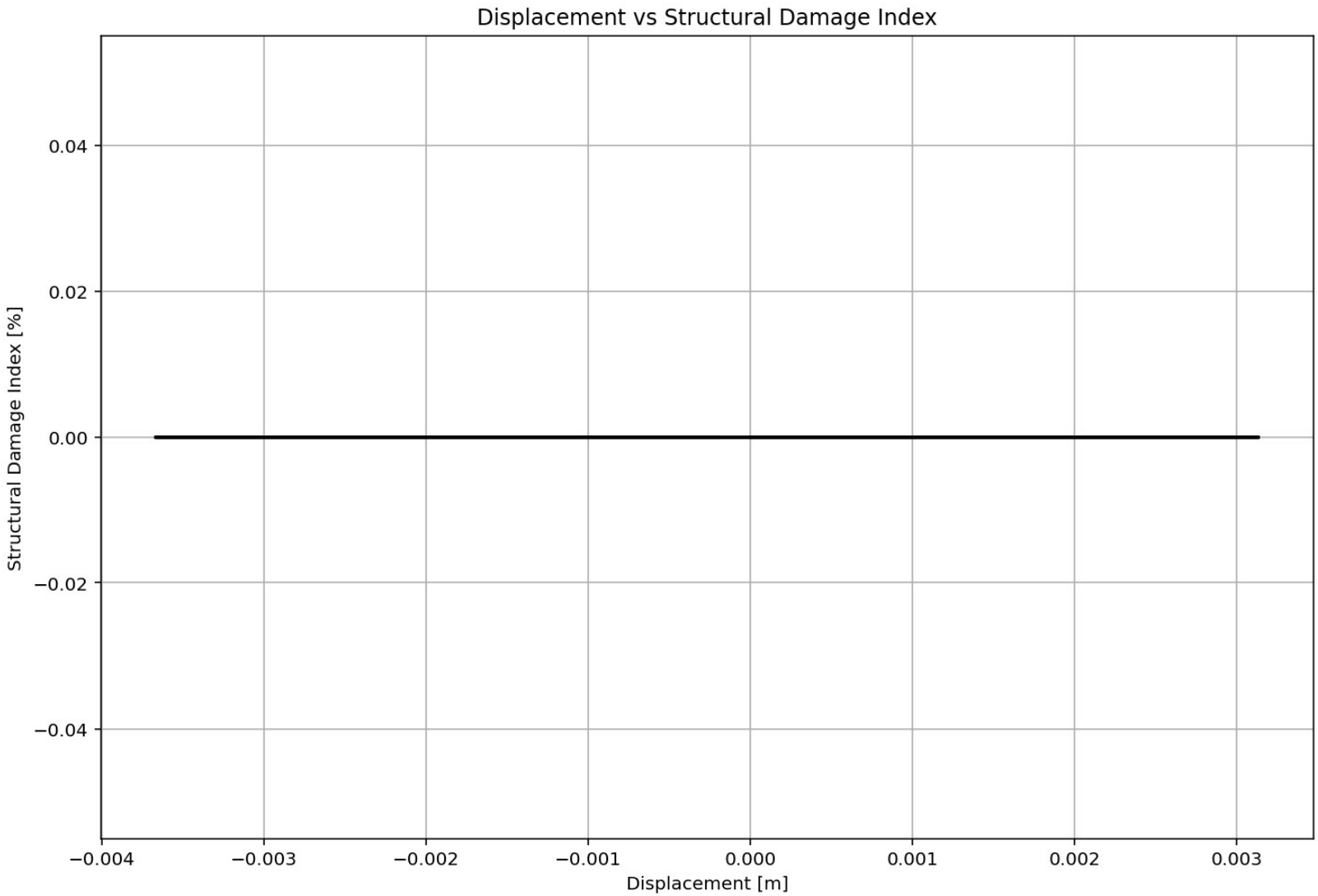

External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5
Time (s)


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
30000
20000
10000
0
−10000
−20000
−30000
−0.004
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [m]


Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.300
0.275
−0.004
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [m]
PERIOD - MIN VALUES: Min: 0.270 (s) - Mean: 0.270 (s) - Max: 0.270 (s)
PERIOD - MAX VALUES: Min: 0.475 (s) - Mean: 0.475 (s) - Max: 0.475 (s)


elements force vs Time During Static External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 3.2985e+04
Ele. 2 - MAX. ABS. : 0.0000e+00
Force [N]
30000
20000
10000
0
−10000
−20000
−30000
0
250
500
750
1000
1250
1500
1750
2000
Time [s]


Node Displacements vs Time for During Static External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.6650e-03
Node 3 - MAX. ABS. : 3.6650e-03
Displacement [m]
0.003
0.002
0.001
0.000
−0.001
−0.002
−0.003
−0.004
0
250
500
750
1000
1250
1500
1750
2000
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.004
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [m]

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

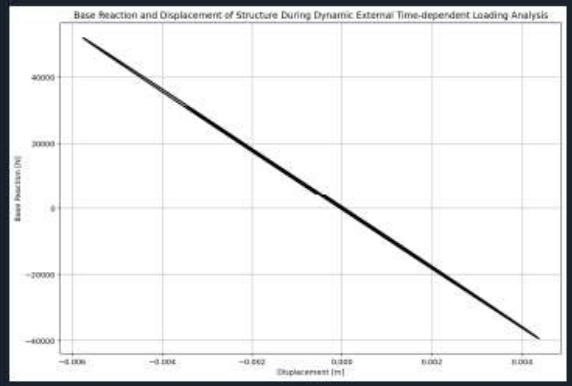
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CTURE_8_ANA\P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

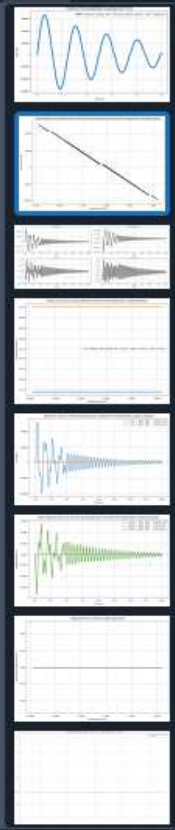
P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

```
1411  # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1412  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1413  ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1414
1415  DATA = PENDULUM_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1416
1417  (time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD,  DI_ETDLD,
1418   ele_force_ETDLD, node_displacements_ETDLD,
1419  stiffness, PERIOD, damping_ratio,
1420  PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
1421
1422
1423
1424  XDATA = disp_ETDLD
1425  YDATA = reaction_ETDLD
1426  XLABEL = 'Displacement [m]'
1427  YLABEL = 'Base Reaction [N]'
1428  TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Anc
1429  COLOR = 'black'
1430  SEMILOGY = False
1431  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1432
1433  PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD)
1434
1435  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1436  plt.figure(0, figsize=(12, 8))
1437  plt.plot(disp_ETDLD, PERIOD_MIN_ETDLD, linewidth=3)
1438  plt.plot(disp_ETDLD, PERIOD_MAX_ETDLD, linewidth=3)
1439  plt.title('Period of Structure During Dynamic External Time-dependent Loading Analysis')
1440  plt.ylabel('Structural Period [s]')
1441  plt.xlabel('Displacement [m]')
1442  #plt.semilogy()
1443  plt.grid()
1444  plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLD):.3f} (s) - Mean: {np.mean(PERIOD_MIN_
```

...ULUM_STRUCTURE_8_ANA+\+61_MDOF_PENDULUM_STRUCTURE_8_ANA

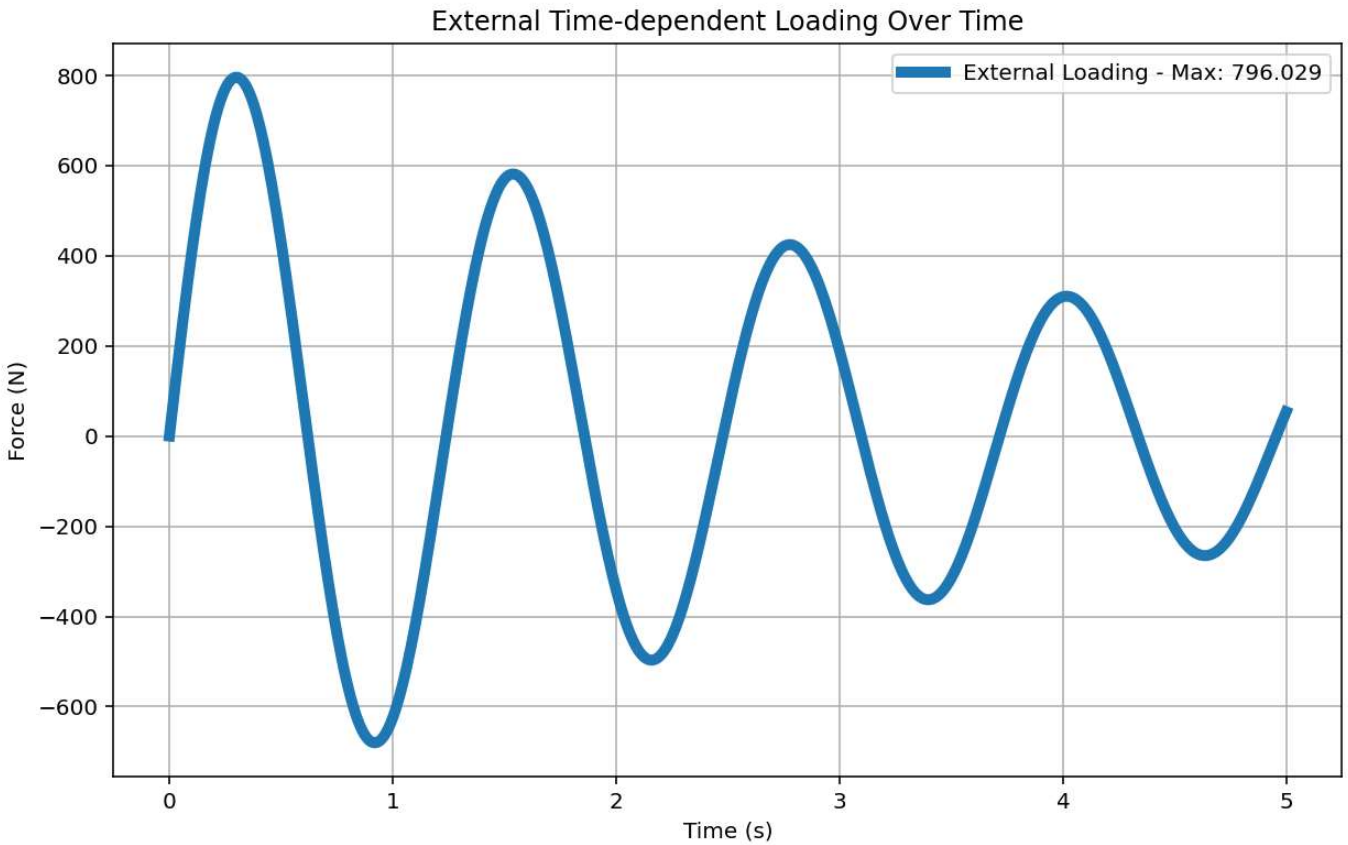
24 %

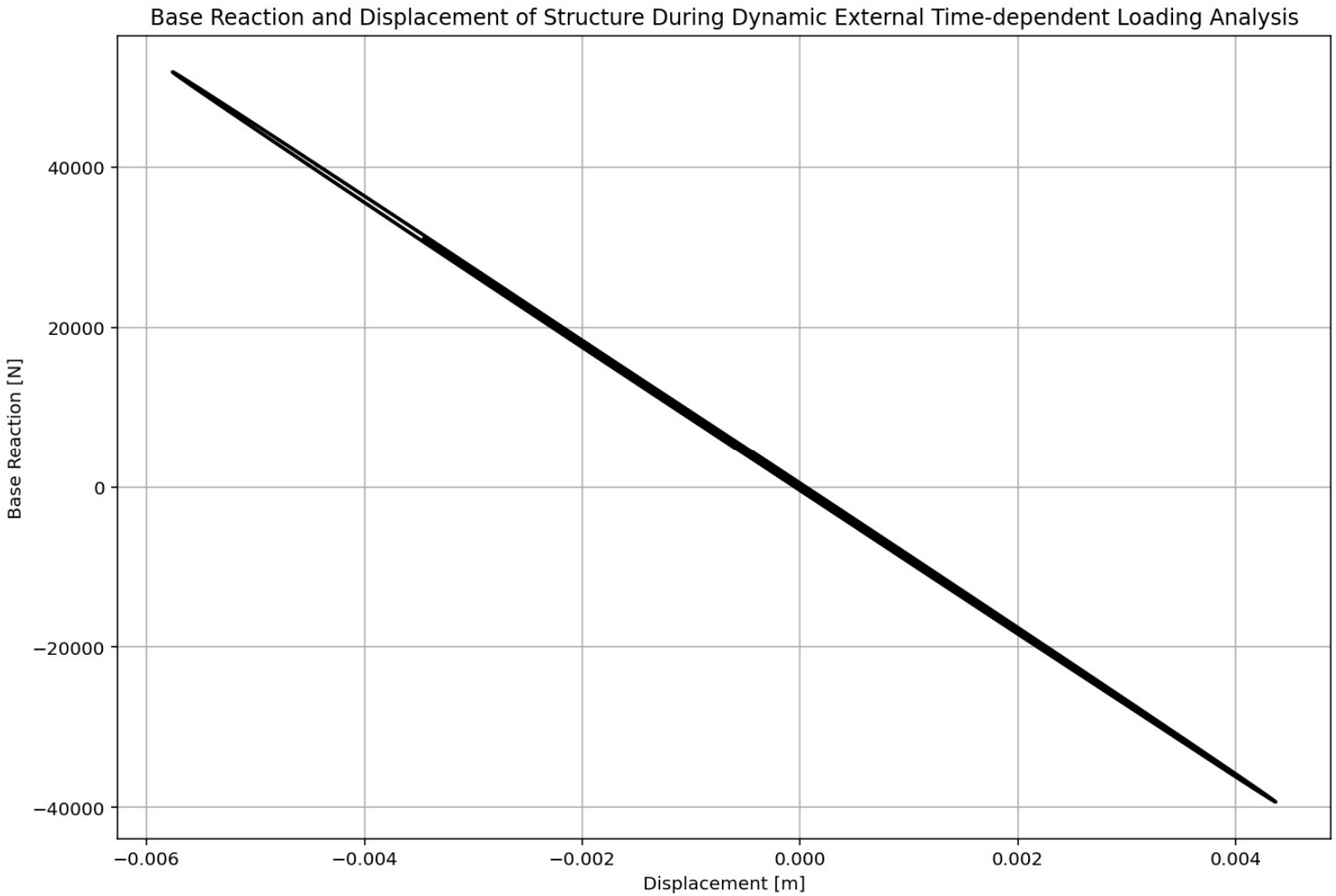


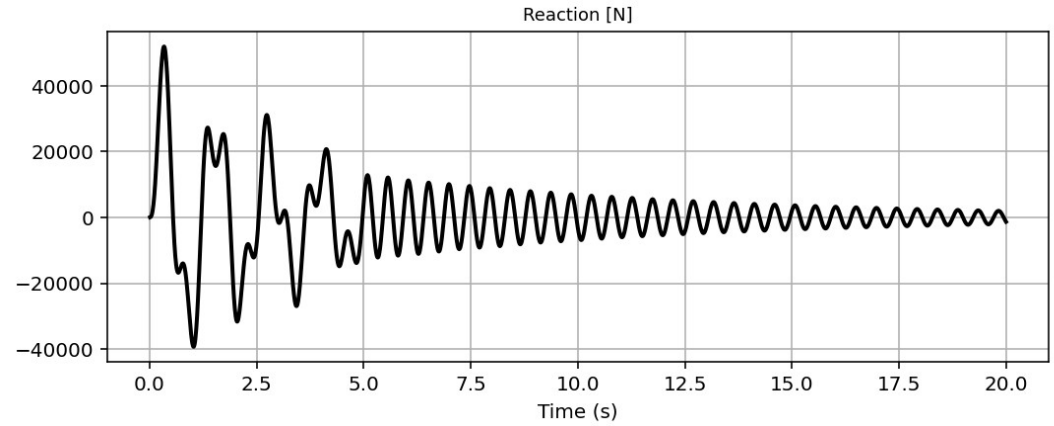


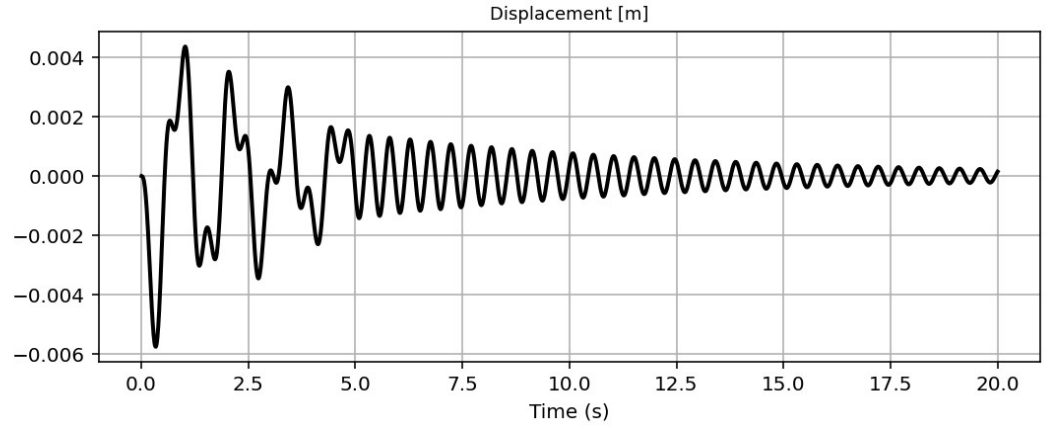
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

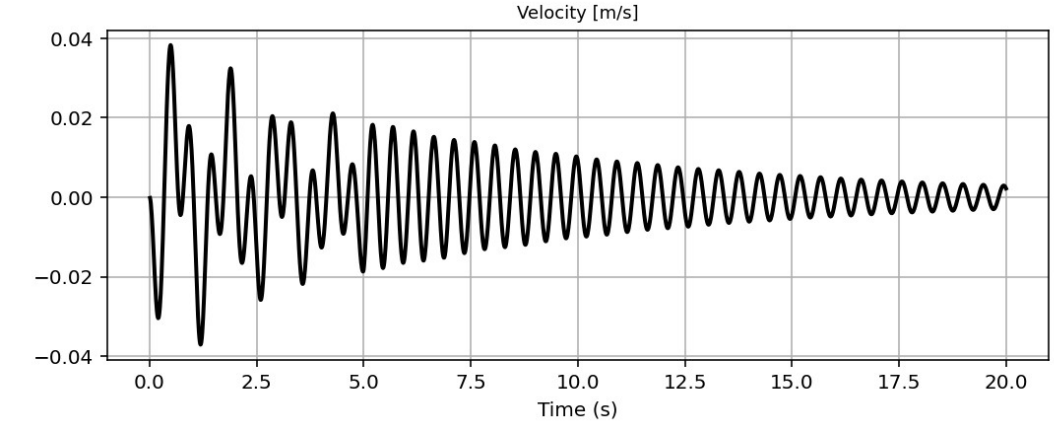
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 21, Col 18 | UTF-8 | CRLF | RW | Mem 48%

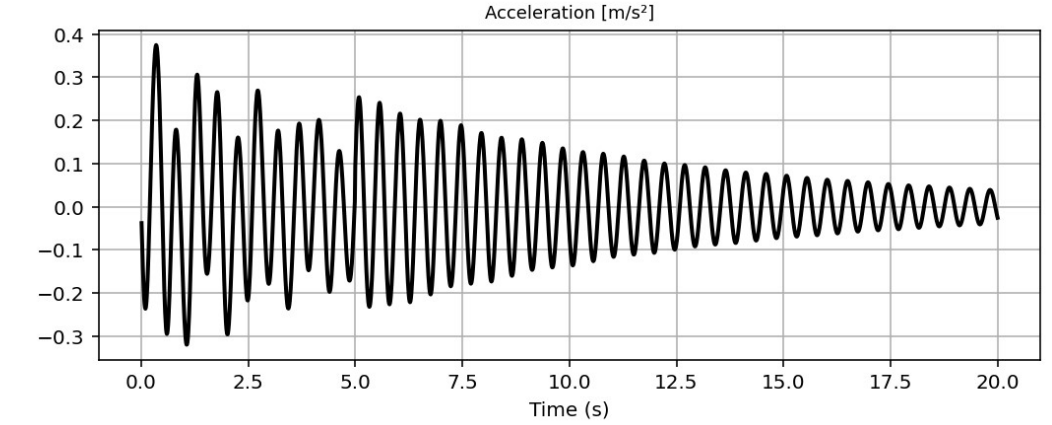

External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5
Time (s)

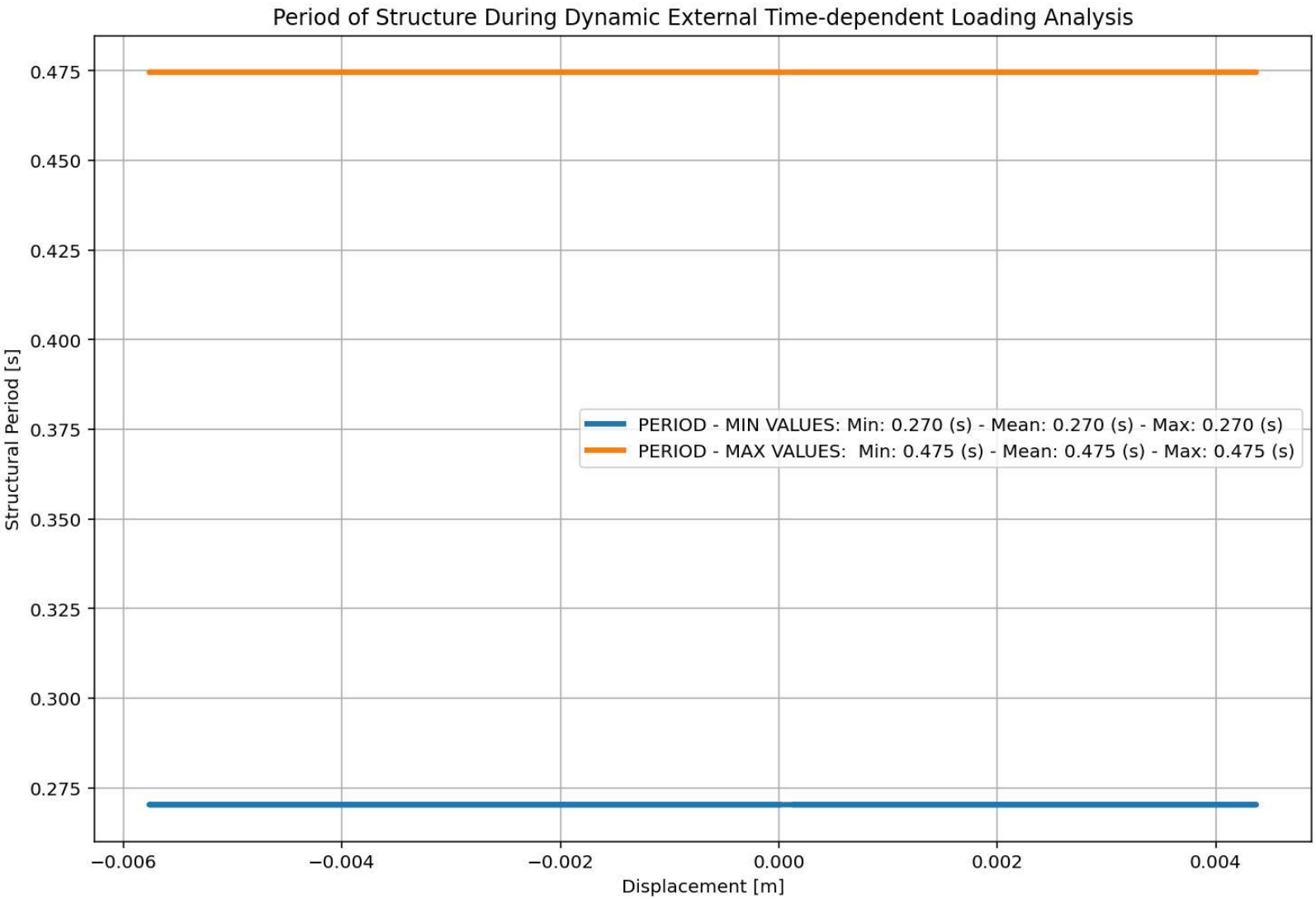

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
40000
20000
0
−20000
−40000
−0.006
−0.004
−0.002
0.000
0.002
0.004
Displacement [m]

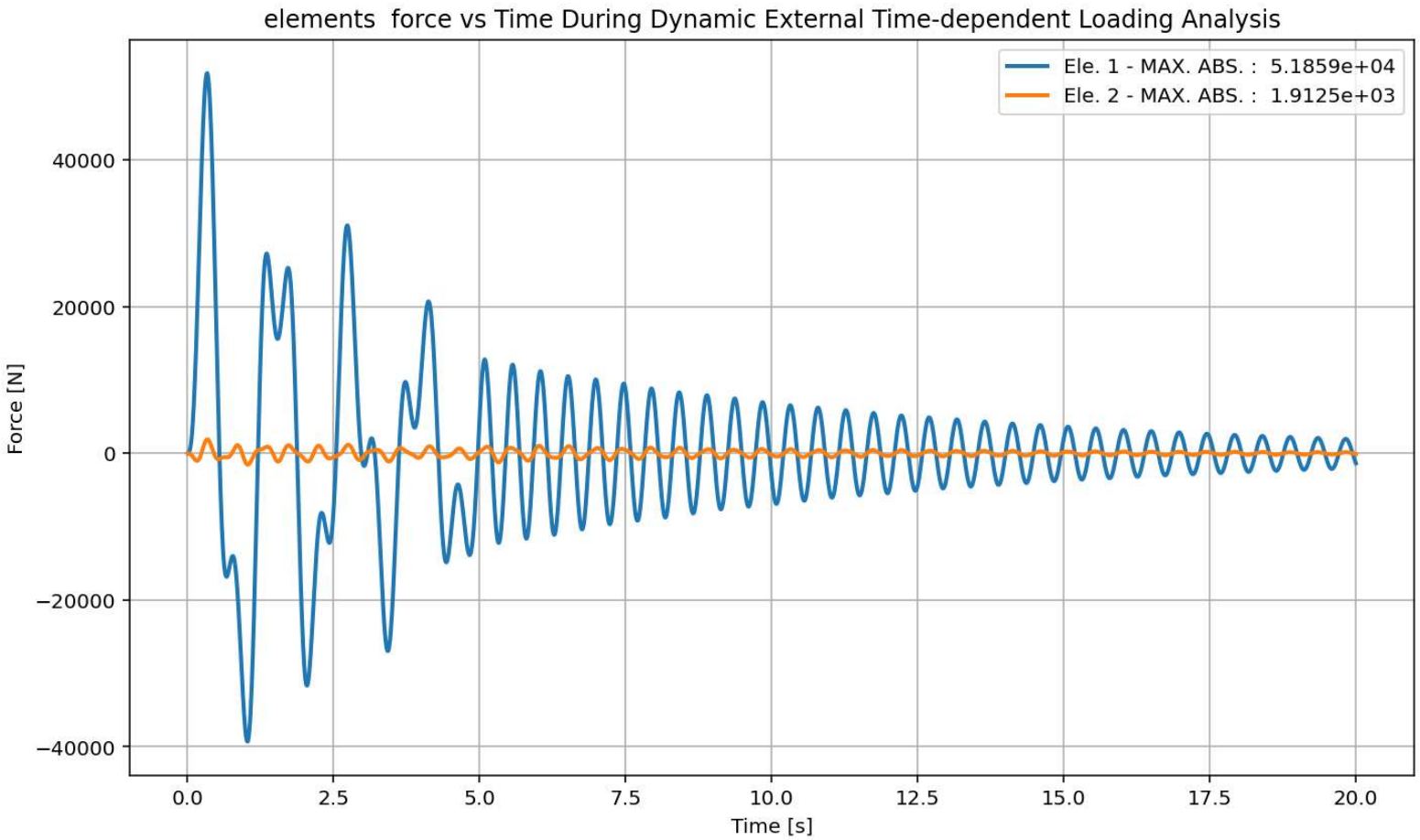

Reaction [N]
40000
20000
0
−20000
−40000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

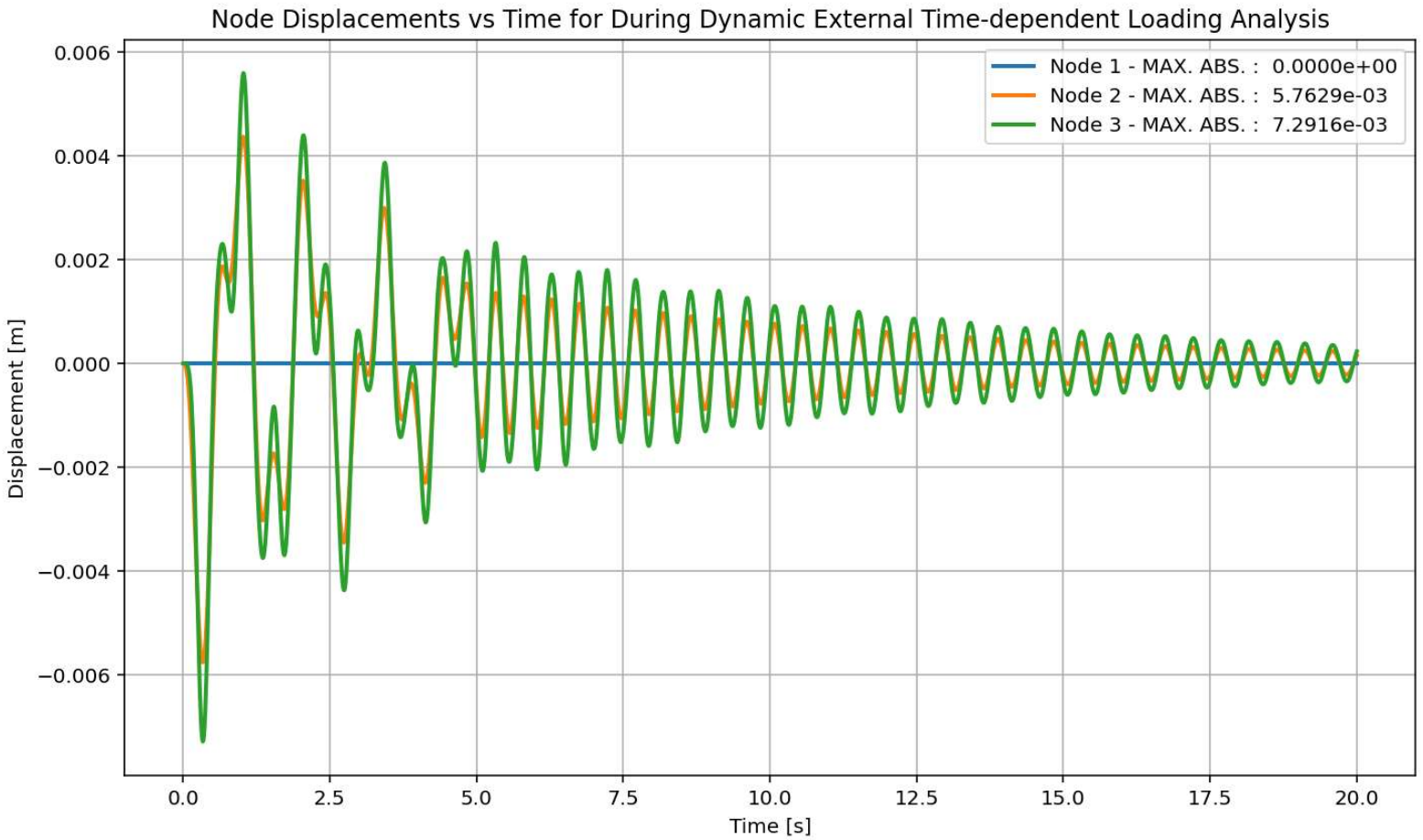

Displacement [m]
0.004
0.002
0.000
−0.002
−0.004
−0.006
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

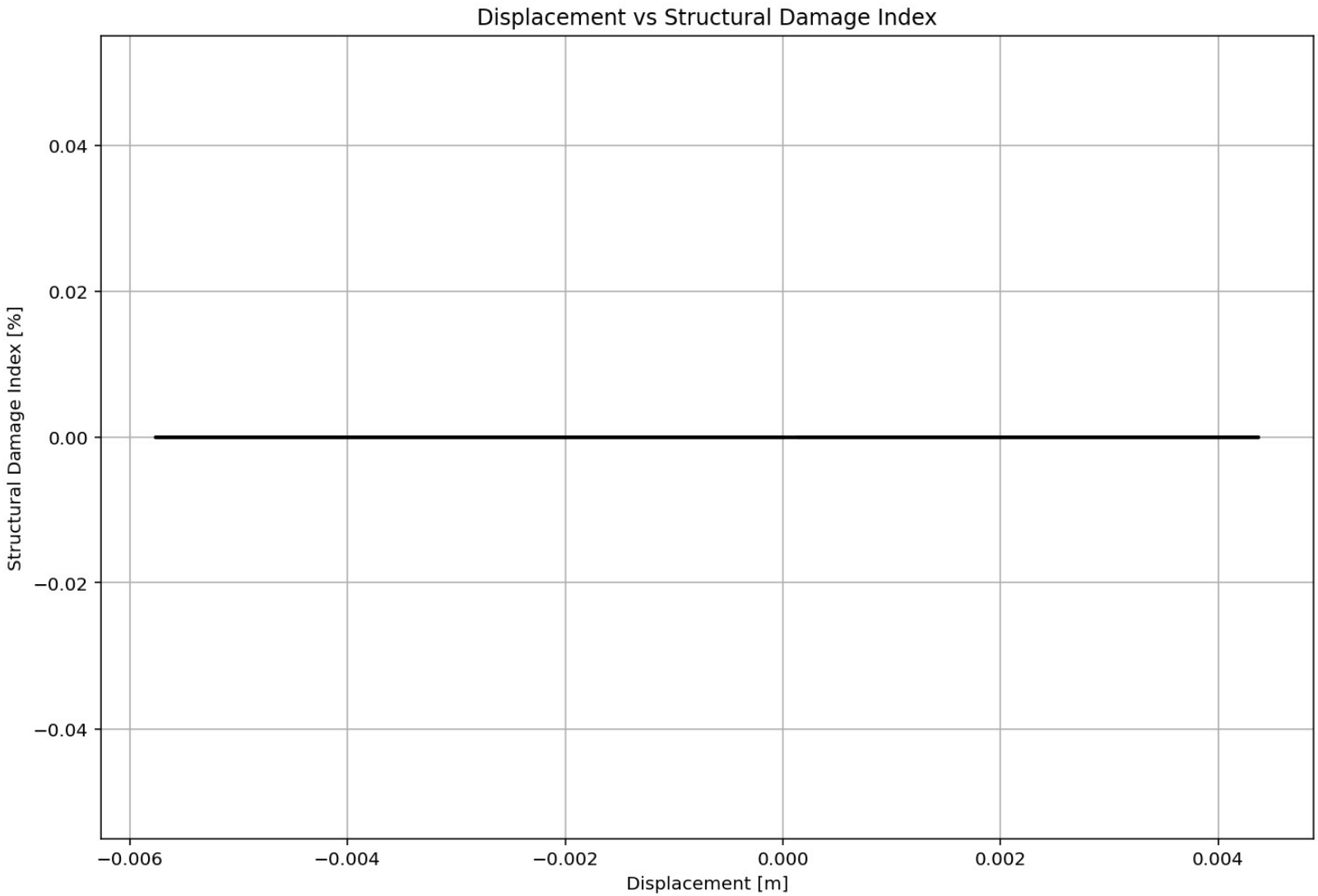

Velocity [m/s]
0.04
0.02
0.00
−0.02
−0.04
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Acceleration [m/s²]
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.300
0.275
−0.006
−0.004
−0.002
0.000
0.002
0.004
Displacement [m]
PERIOD - MIN VALUES: Min: 0.270 (s) - Mean: 0.270 (s) - Max: 0.270 (s)
PERIOD - MAX VALUES: Min: 0.475 (s) - Mean: 0.475 (s) - Max: 0.475 (s)


elements force vs Time During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 5.1859e+04
Ele. 2 - MAX. ABS. : 1.9125e+03
Force [N]
40000
20000
0
−20000
−40000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Node Displacements vs Time for During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 5.7629e-03
Node 3 - MAX. ABS. : 7.2916e-03
Displacement [m]
0.006
0.004
0.002
0.000
−0.002
−0.004
−0.006
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.006
−0.004
−0.002
0.000
0.002
0.004
Displacement [m]

FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

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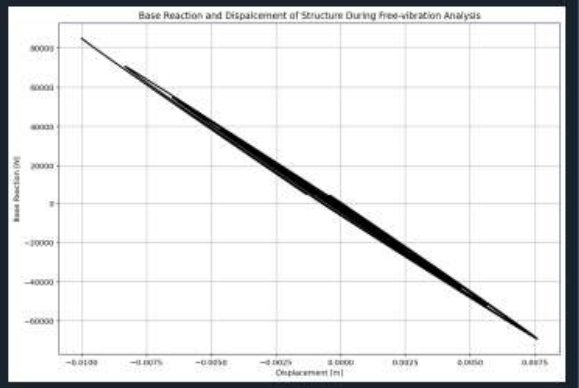
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CTURE_8_ANA\P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

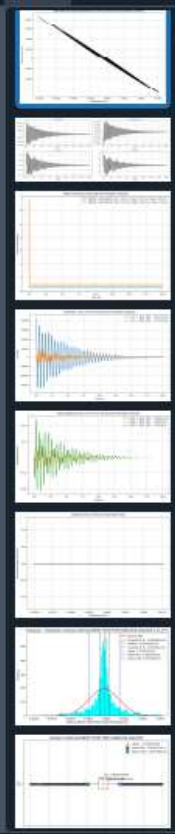
P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

```
1467  # FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1468  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1469  ANAL_TYPE = 'FREE-VIBRATION'
1470  DATA = PENDULUM_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1471
1472  (time_FV, reaction_FV, disp_FV, velo_FV, acc_FV, DI_FV,
1473   ele_force_FV, node_displacements_FV,
1474  stiffness_FV, PERIOD_FV, damping_ratio_FV,
1475  PERIOD_MIN_FV, PERIOD_MAX_FV) = DATA
1476
1477  XDATA = disp_FV
1478  YDATA = reaction_FV
1479  XLABEL = 'Displacement [m]'
1480  YLABEL = 'Base Reaction [N]'
1481  TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration Analysis'
1482  COLOR = 'black'
1483  SEMILOGY = False
1484  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1485
1486  PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_FV, velo_FV, acc_FV)
1487
1488  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1489  plt.figure(0, figsize=(12, 8))
1490  plt.plot(time_FV, PERIOD_MIN_FV, linewidth=3)
1491  plt.plot(time_FV, PERIOD_MAX_FV, linewidth=3)
1492  plt.title('Period of Structure During Free-vibration Analysis')
1493  plt.ylabel('Structural Period [s]')
1494  plt.xlabel('Time [s]')
1495  #plt.semilogy()
1496  plt.grid()
1497  plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_FV):.3f} (s) - Mean: {np.mean(PERIOD_MIN_FV)
1498              f'PERIOD - MAX VALUES:  Min: {np.min(PERIOD_MAX_FV):.3f} (s) - Mean: {np.mean(PERIOD_MAX_FV)
1499              ])
1500  plt.show()
```

...ULUM_STRUCTURE_8_ANA+\+61_MDOF_PENDULUM_STRUCTURE_8_ANA

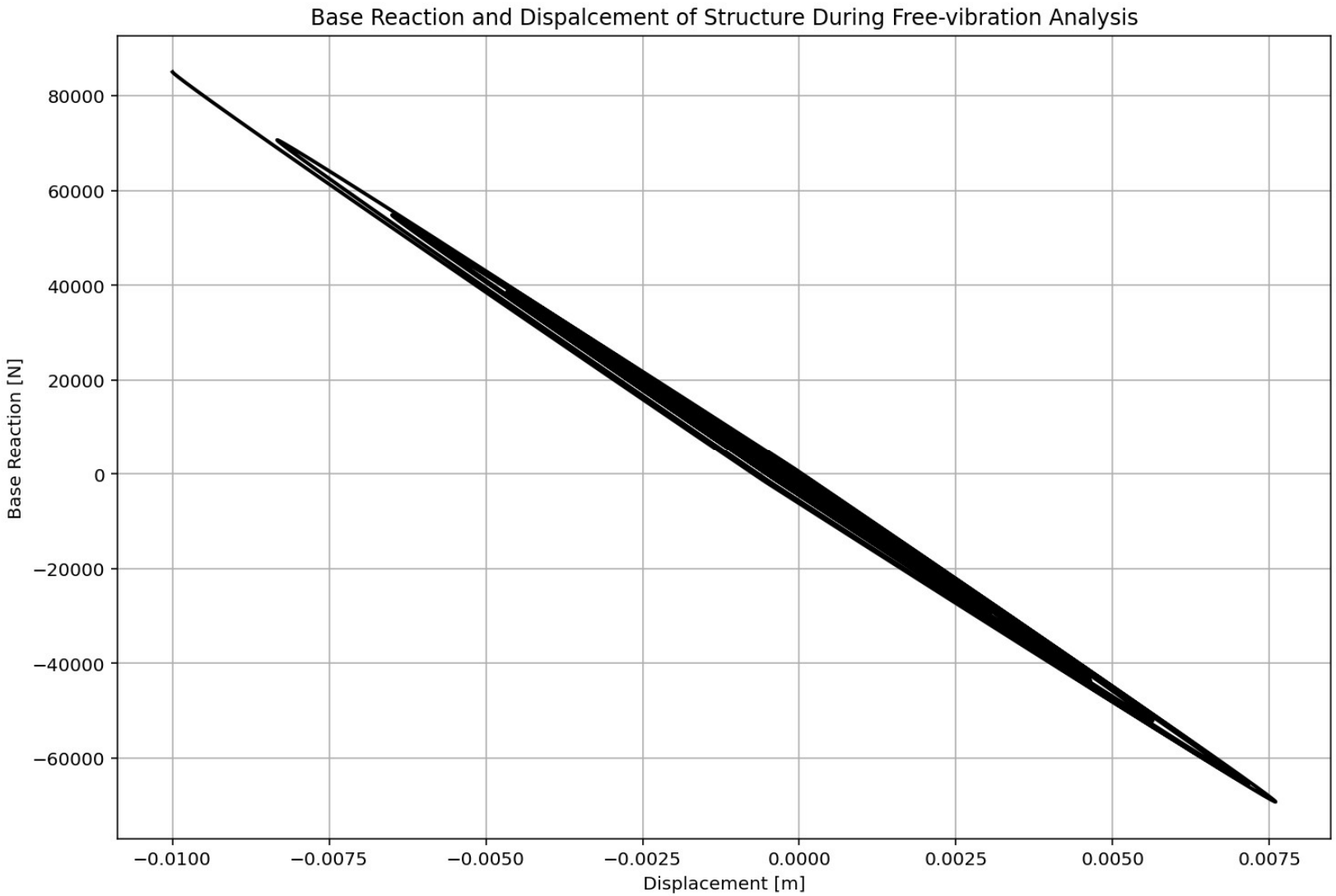
24 %

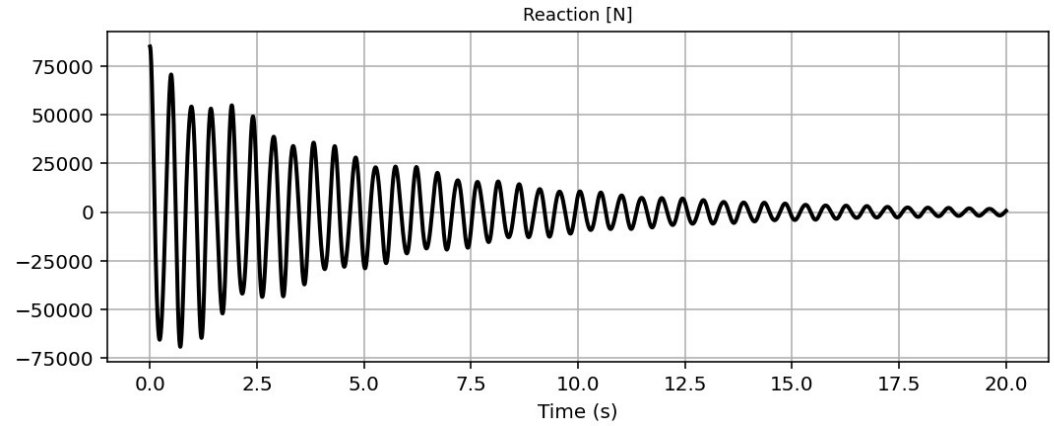


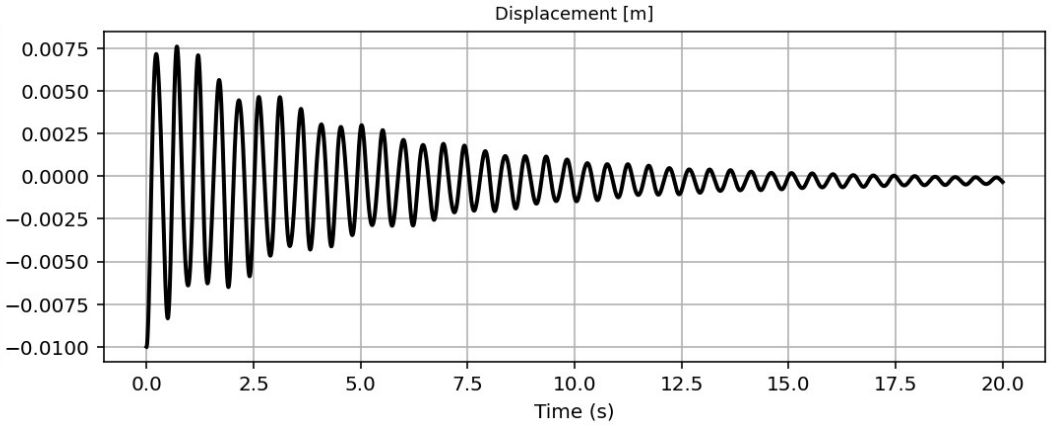


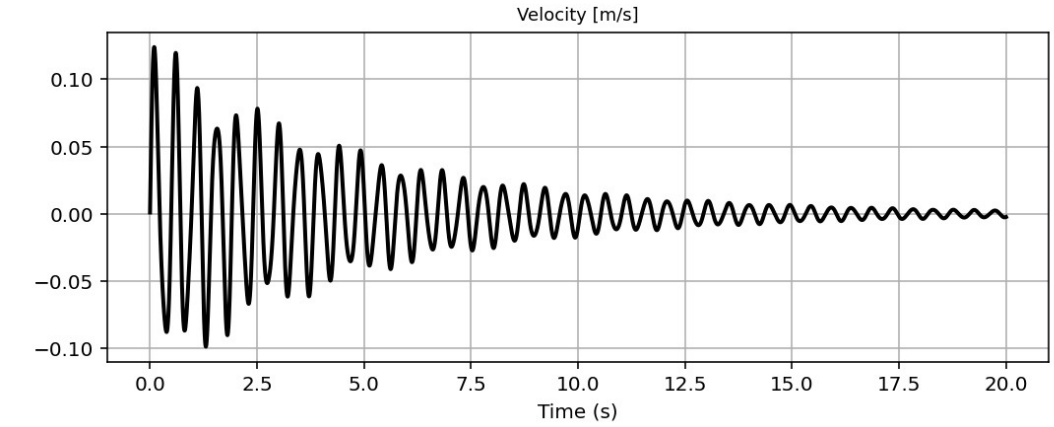
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

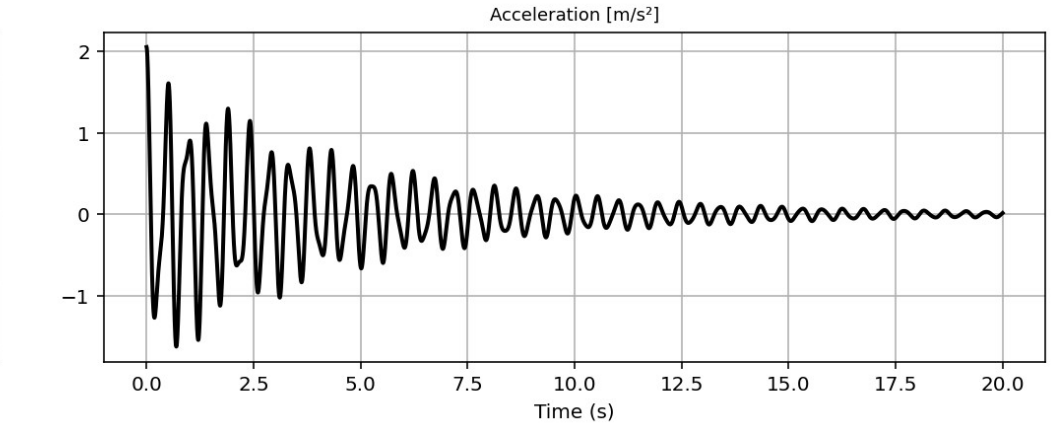
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 21, Col 18 | UTF-8 | CRLF | RW | Mem 48%

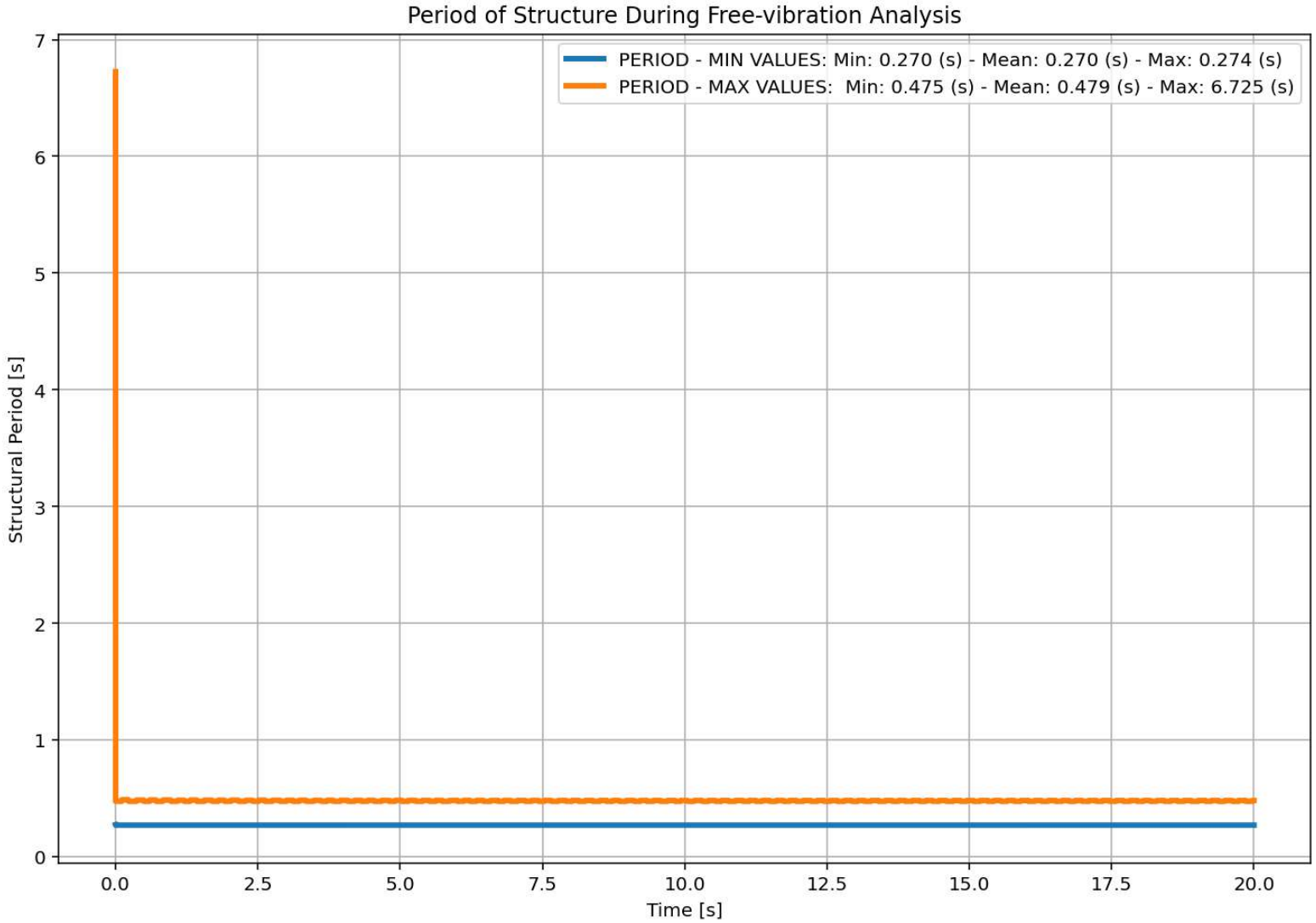

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
80000
60000
40000
20000
0
−20000
−40000
−60000
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
Displacement [m]

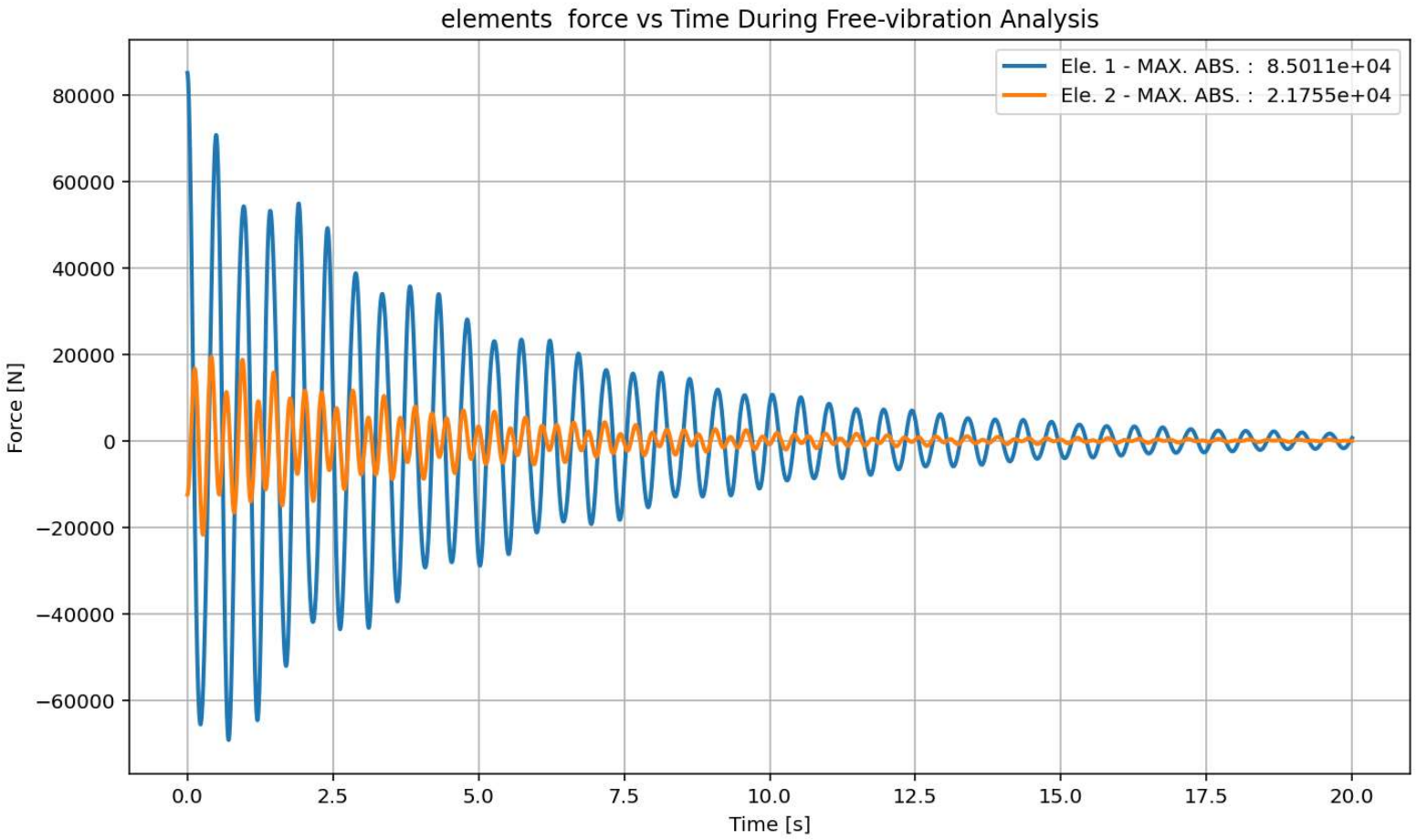

Reaction [N]
75000
50000
25000
0
−25000
−50000
−75000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

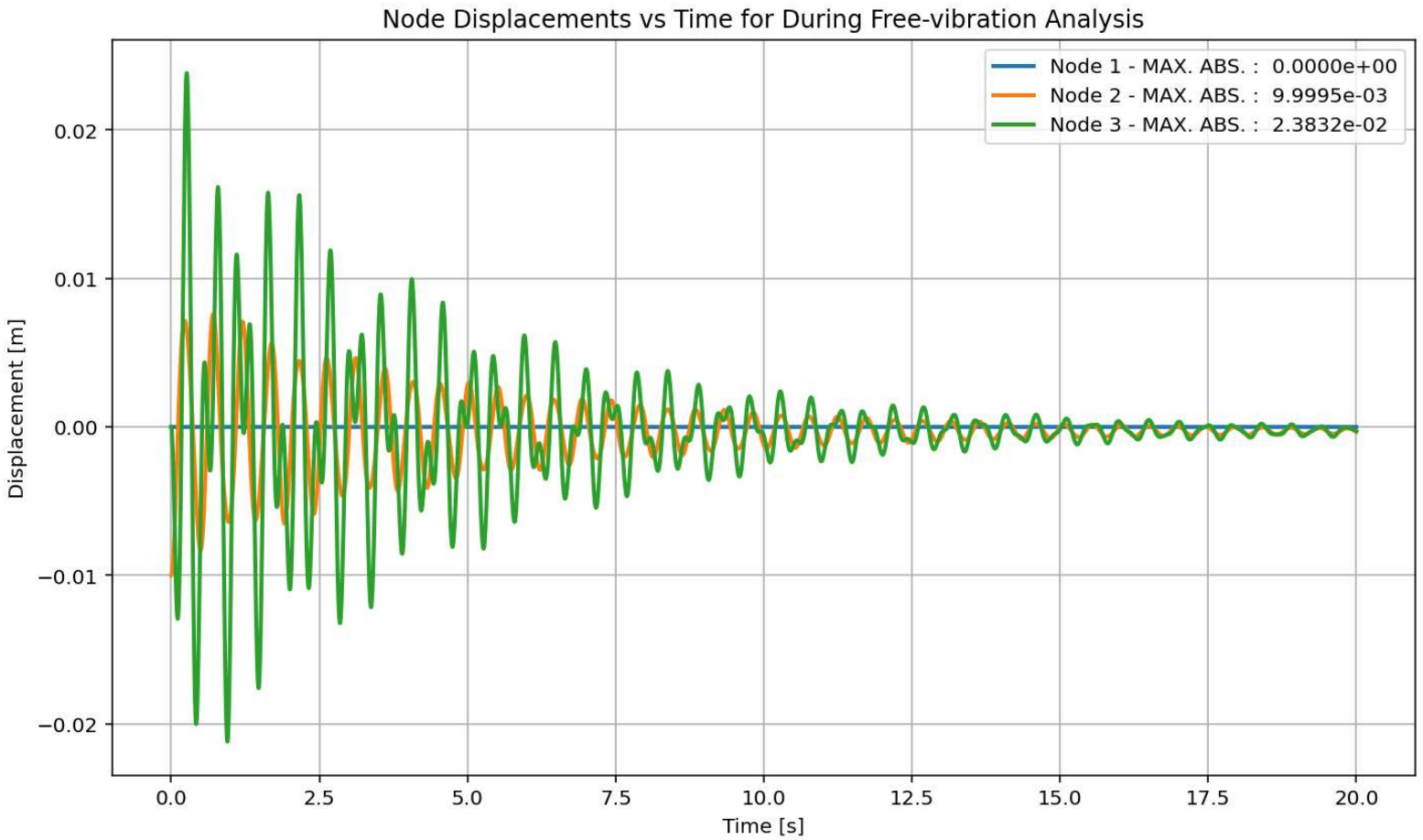

Displacement [m]
0.0075
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
−0.0100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

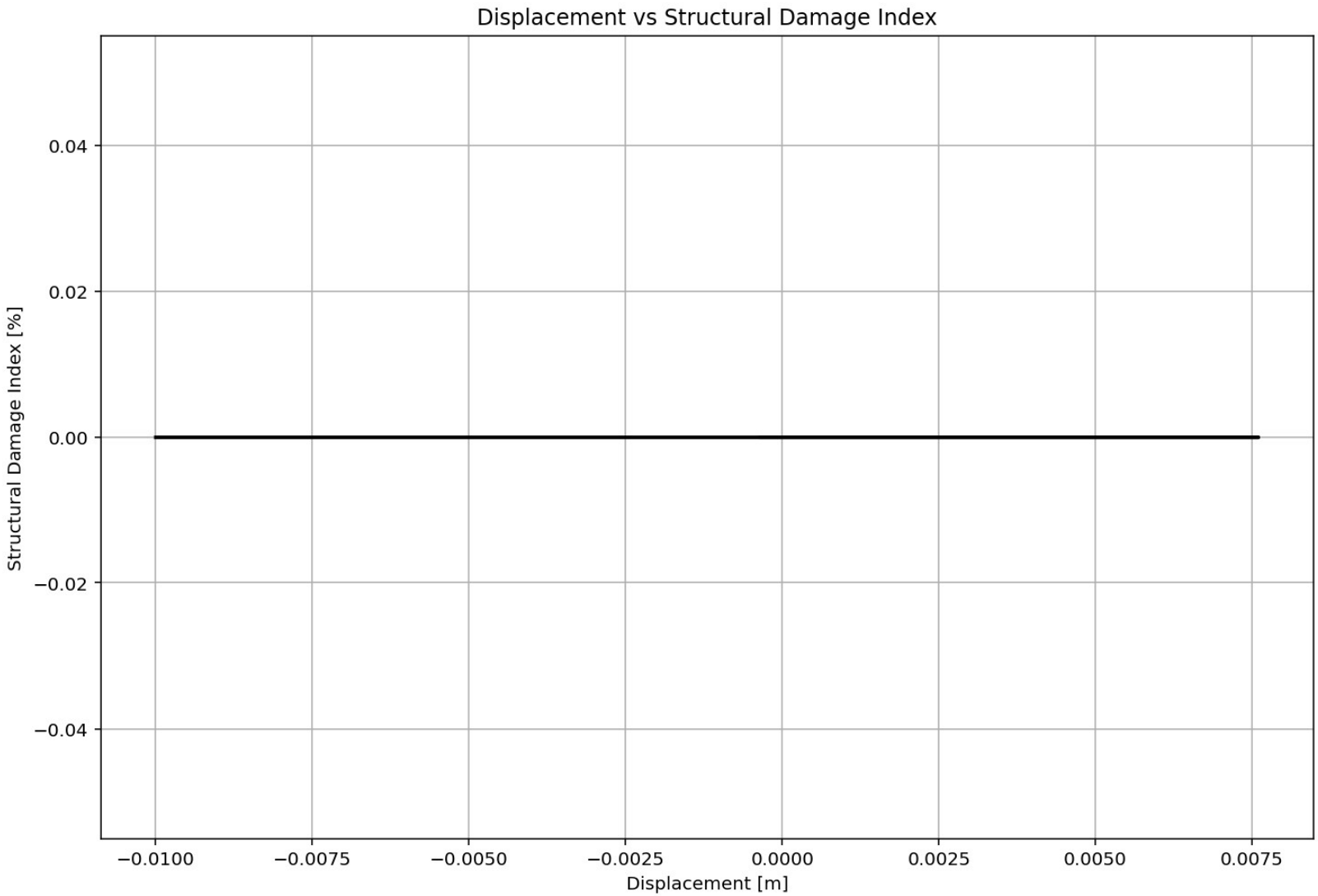

Velocity [m/s]
0.10
0.05
0.00
−0.05
−0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

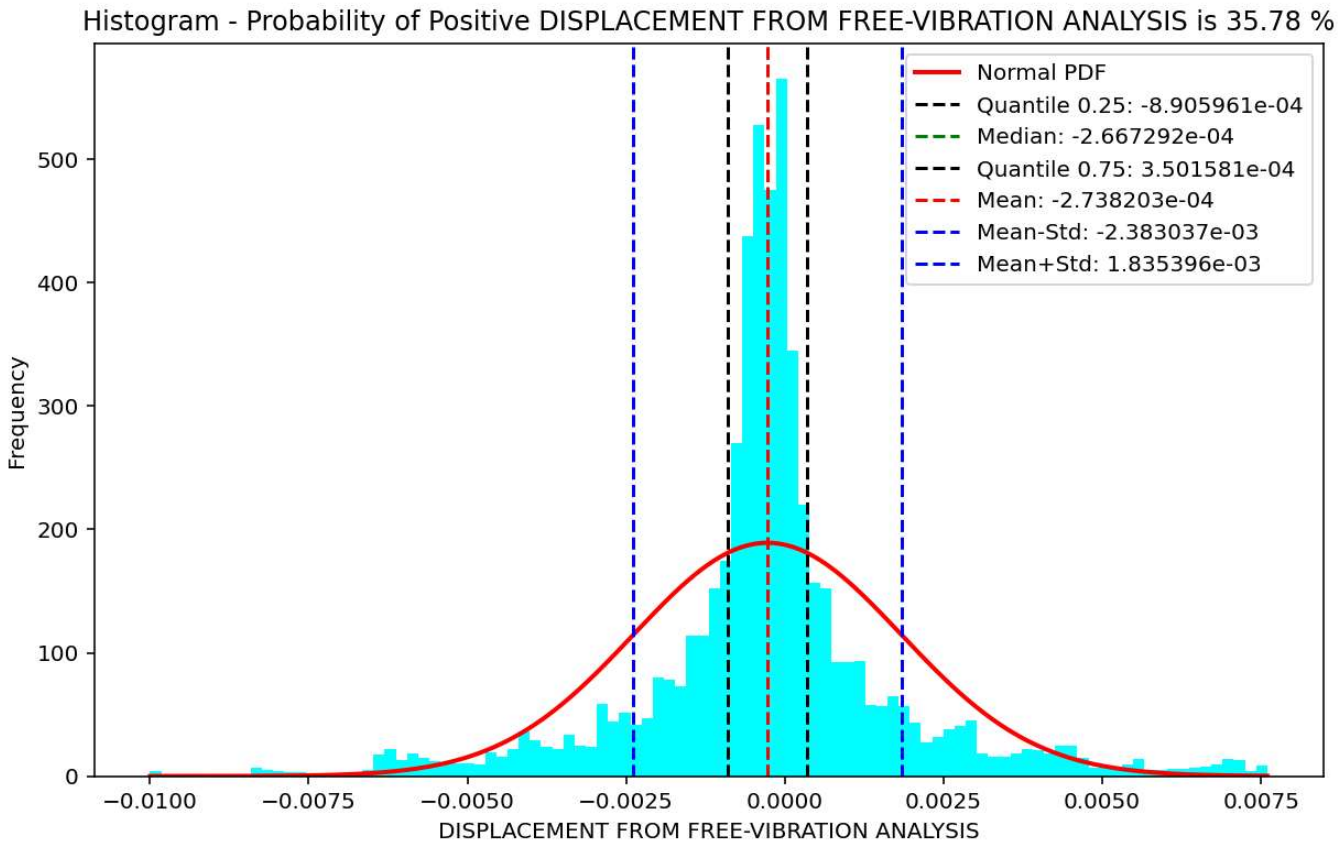

Acceleration [m/s²]
2
1
0
−1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

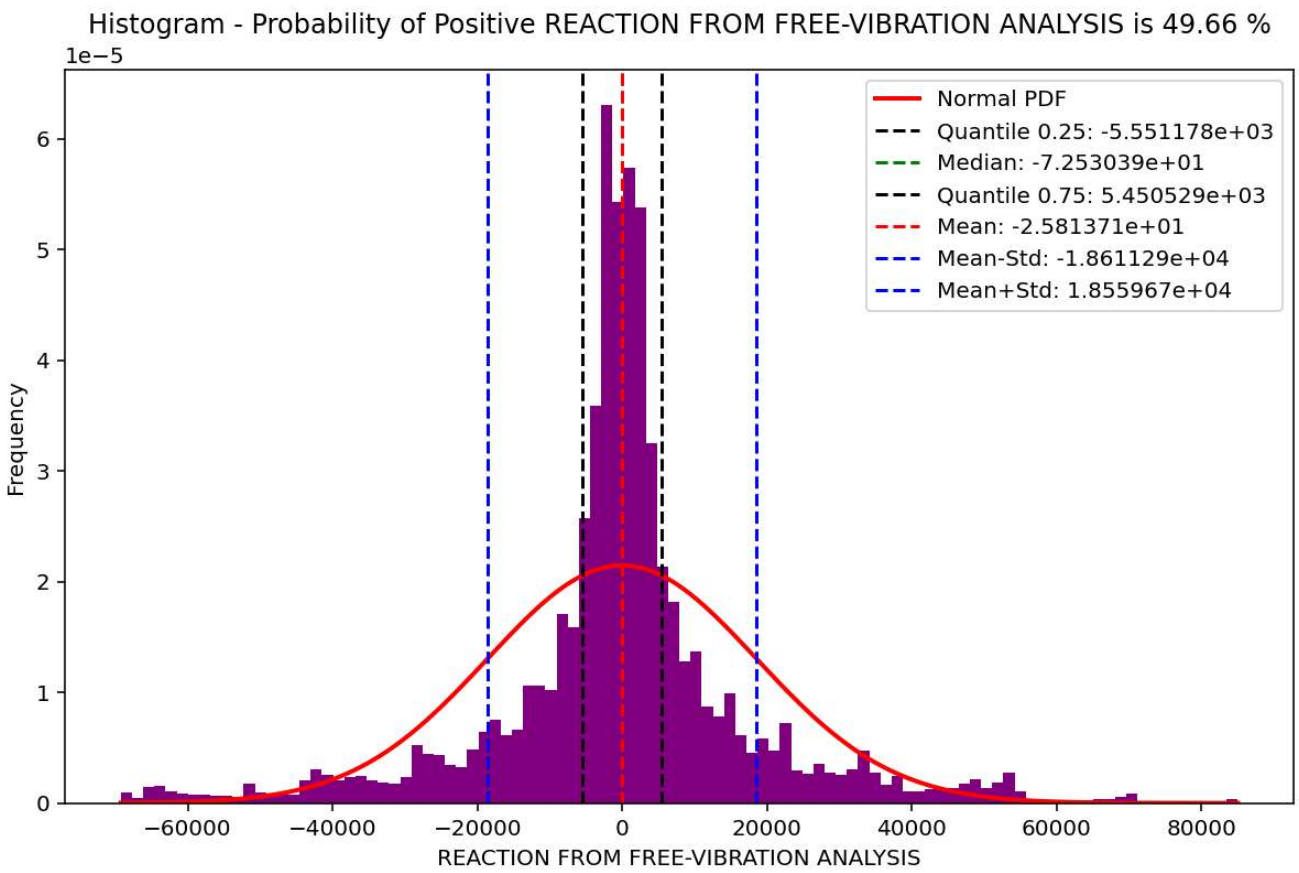

Period of Structure During Free-vibration Analysis
PERIOD - MIN VALUES: Min: 0.270 (s) - Mean: 0.270 (s) - Max: 0.274 (s)
PERIOD - MAX VALUES: Min: 0.475 (s) - Mean: 0.479 (s) - Max: 6.725 (s)
Structural Period [s]
Time [s]
0
1
2
3
4
5
6
7
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

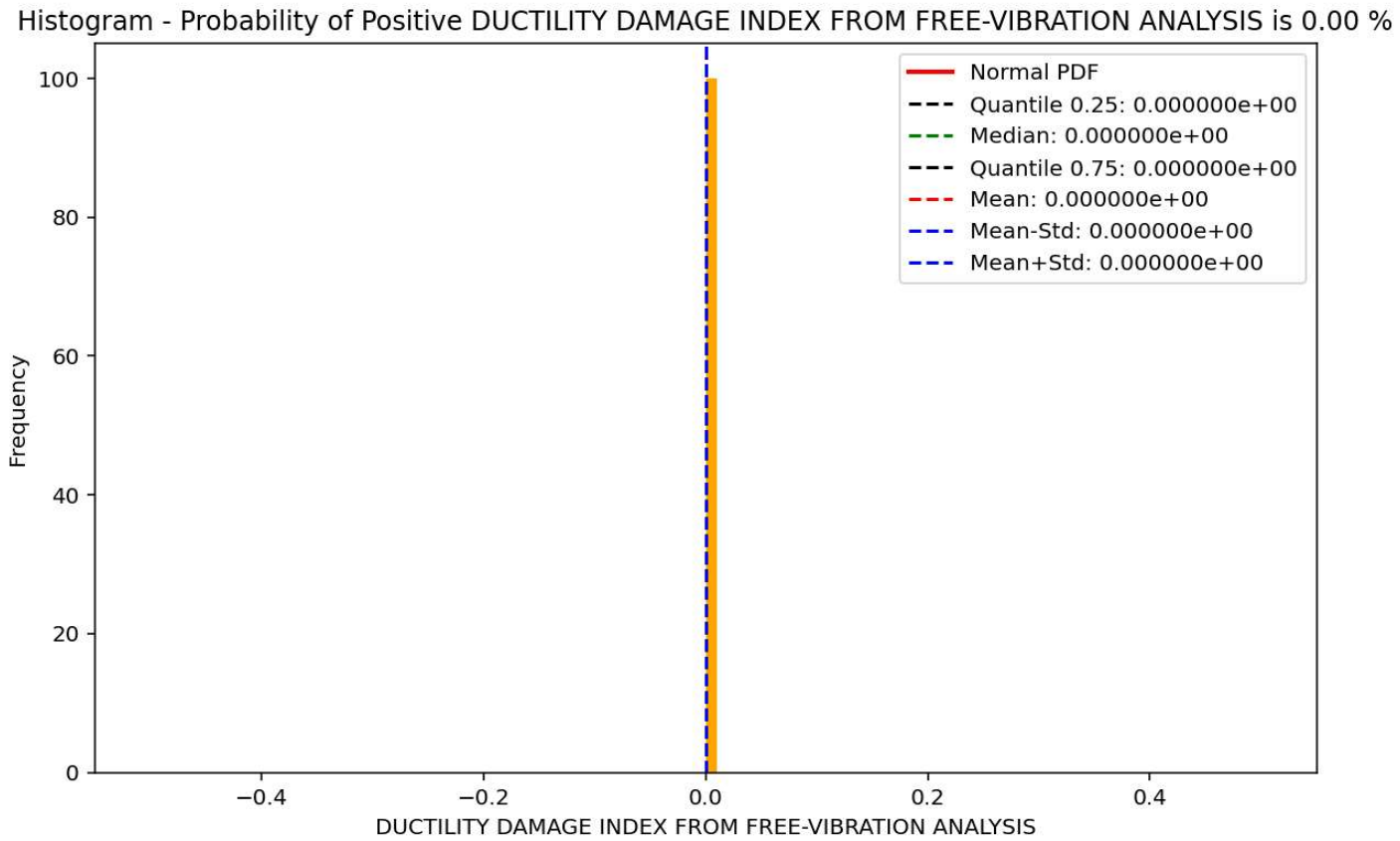

elements force vs Time During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 8.5011e+04
Ele. 2 - MAX. ABS. : 2.1755e+04
Force [N]
80000
60000
40000
20000
0
−20000
−40000
−60000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

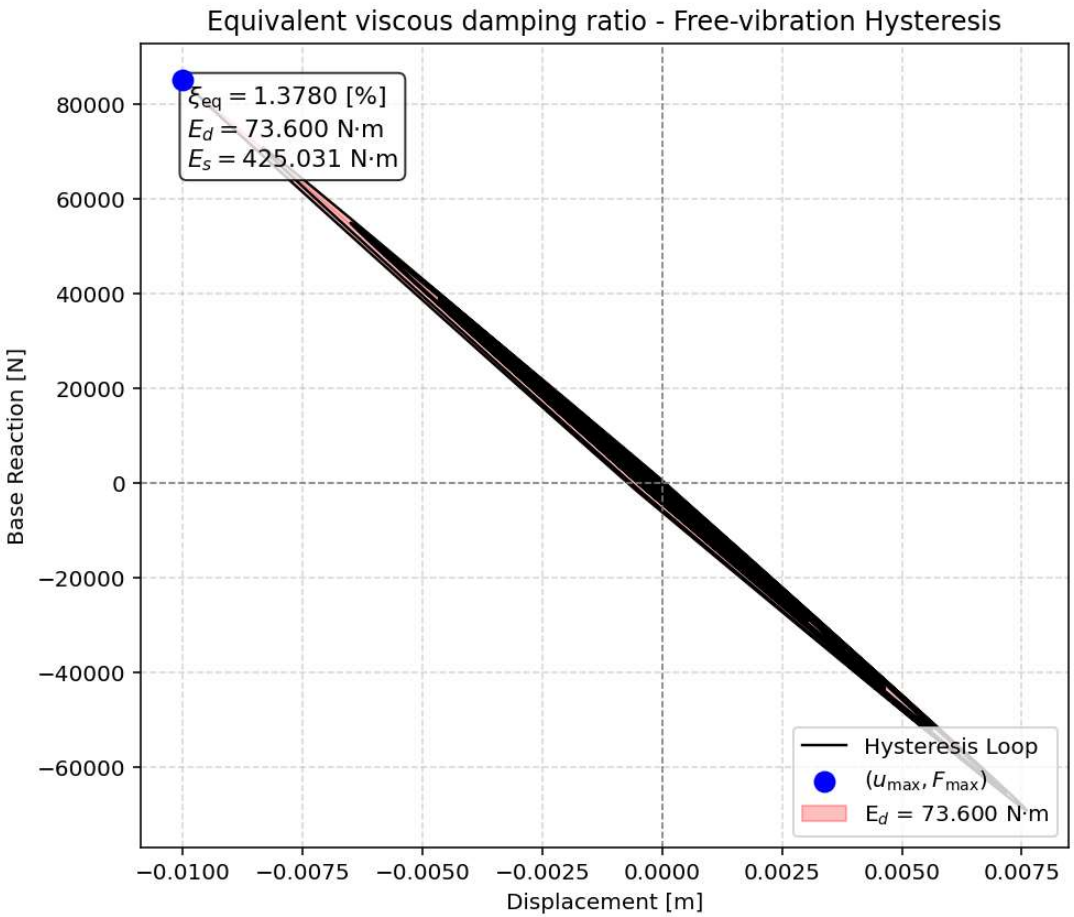

Node Displacements vs Time for During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 9.9995e-03
Node 3 - MAX. ABS. : 2.3832e-02
Displacement [m]
0.02
0.01
0.00
−0.01
−0.02
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
Displacement [m]


Histogram - Probability of Positive DISPLACEMENT FROM FREE-VIBRATION ANALYSIS is 35.78 %
Normal PDF
Quantile 0.25: -8.905961e-04
Median: -2.667292e-04
Quantile 0.75: 3.501581e-04
Mean: -2.738203e-04
Mean-Std: -2.383037e-03
Mean+Std: 1.835396e-03
Frequency
0
100
200
300
400
500
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
DISPLACEMENT FROM FREE-VIBRATION ANALYSIS


Histogram - Probability of Positive REACTION FROM FREE-VIBRATION ANALYSIS is 49.66 %
1e−5
Frequency
REACTION FROM FREE-VIBRATION ANALYSIS
Normal PDF
Quantile 0.25: -5.551178e+03
Median: -7.253039e+01
Quantile 0.75: 5.450529e+03
Mean: -2.581371e+01
Mean-Std: -1.861129e+04
Mean+Std: 1.855967e+04


Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %
Normal PDF
Quantile 0.25: 0.000000e+00
Median: 0.000000e+00
Quantile 0.75: 0.000000e+00
Mean: 0.000000e+00
Mean-Std: 0.000000e+00
Mean+Std: 0.000000e+00
Frequency
100
80
60
40
20
0
−0.4
−0.2
0.0
0.2
0.4
DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 1.3780$ [%]
$E_d = 73.600$ N·m
$E_s = 425.031$ N·m
Base Reaction [N]
Displacement [m]
80000
60000
40000
20000
0
−20000
−40000
−60000
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
Hysteresis Loop
(u_{max}, F_{max})
E_d = 73.600 N·m

SEISMIC ANALYSIS

Spyder (Python 3.12)

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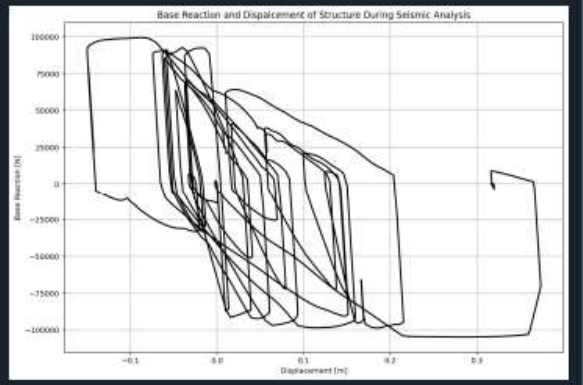
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CTURE_8_ANA\P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

P(t)_MDOF_PENDULUM_STRUCTURE_8_ANA.py

```
1534  # SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1535  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1536  ANAL_TYPE = 'SEISMIC'
1537
1538  DATA = PENDULUM_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1539
1540  (time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI, DI_SEI,
1541   ele_force_SEI, node_displacements_SEI,
1542   stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
1543   PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
1544
1545
1546  XDATA = disp_SEI
1547  YDATA = reaction_SEI
1548  XLABEL = 'Displacement [m]'
1549  YLABEL = 'Base Reaction [N]'
1550  TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
1551  COLOR = 'black'
1552  SEMILOGY = False
1553  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1554
1555  PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI)
1556
1557  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1558  plt.figure(0, figsize=(12, 8))
1559  plt.plot(time_SEI, PERIOD_MIN_SEI, linewidth=3)
1560  plt.plot(time_SEI, PERIOD_MAX_SEI, linewidth=3)
1561  plt.title('Period of Structure During Seismic Analysis')
1562  plt.ylabel('Structural Period [s]')
1563  plt.xlabel('Time [s]')
1564  #plt.semilogy()
1565  plt.grid()
1566  plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_SEI):.3f} (s) - Mean: {np.mean(PERIOD_MIN_SE
1567              f'PERIOD - MAX VALUES:  Min: {np.min(PERIOD_MAX_SEI):.3f} (s) - Mean: {np.mean(PERIOD_MAX_S
```

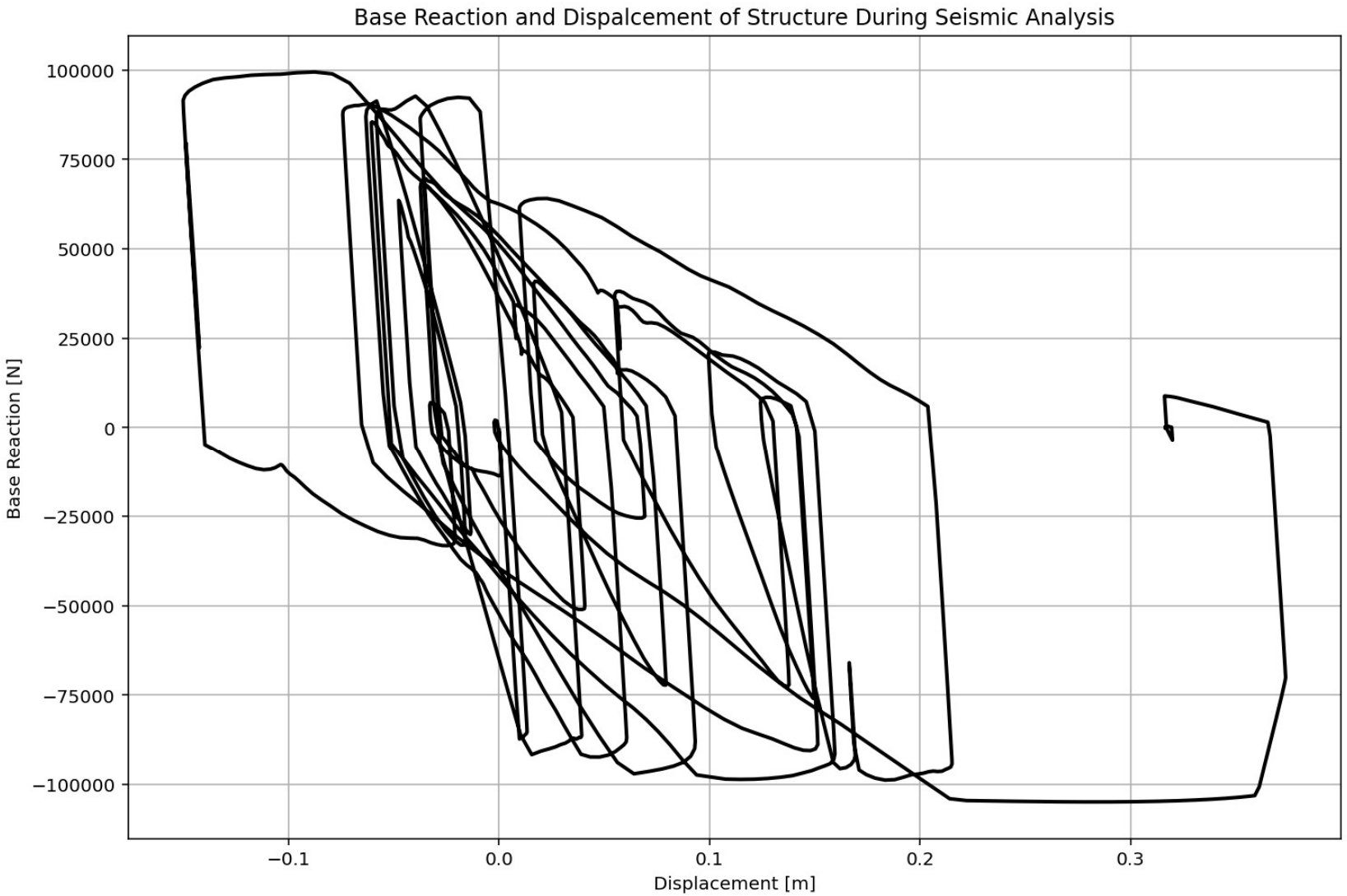
...ULUM_STRUCTURE_8_ANA+\+61_MDOF_PENDULUM_STRUCTURE_8_ANA

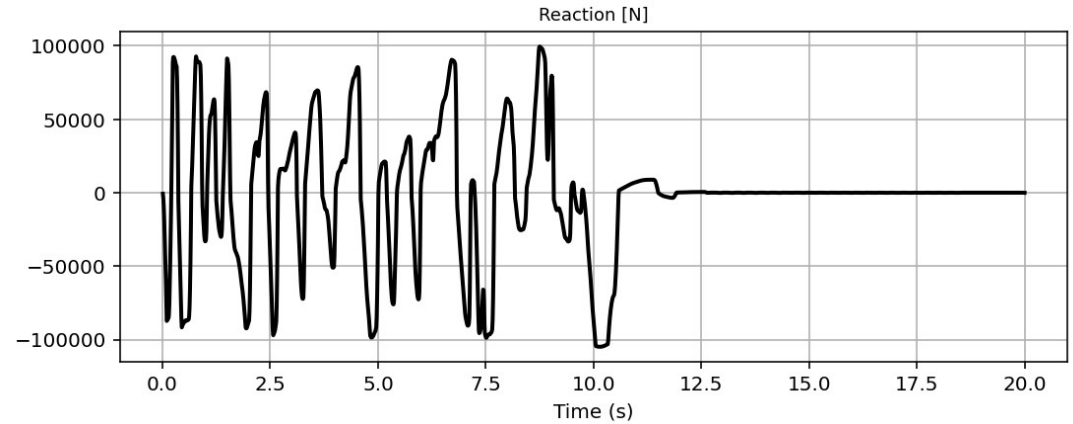
24 %

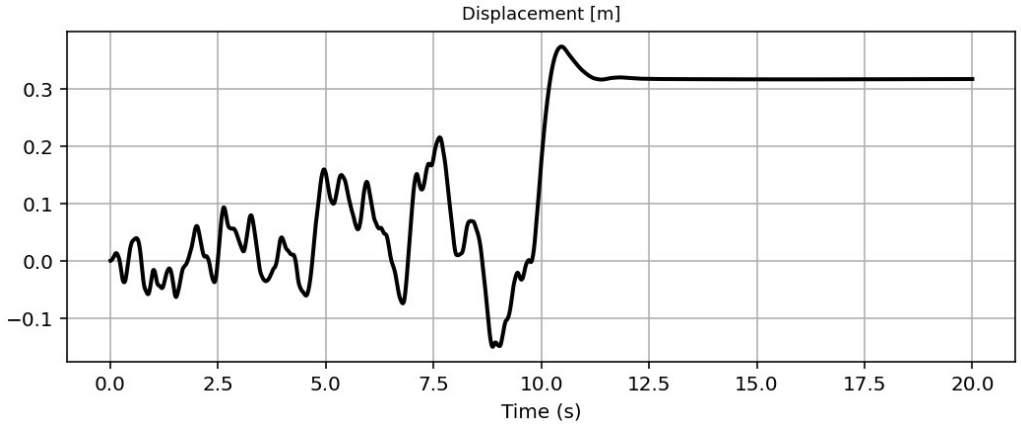


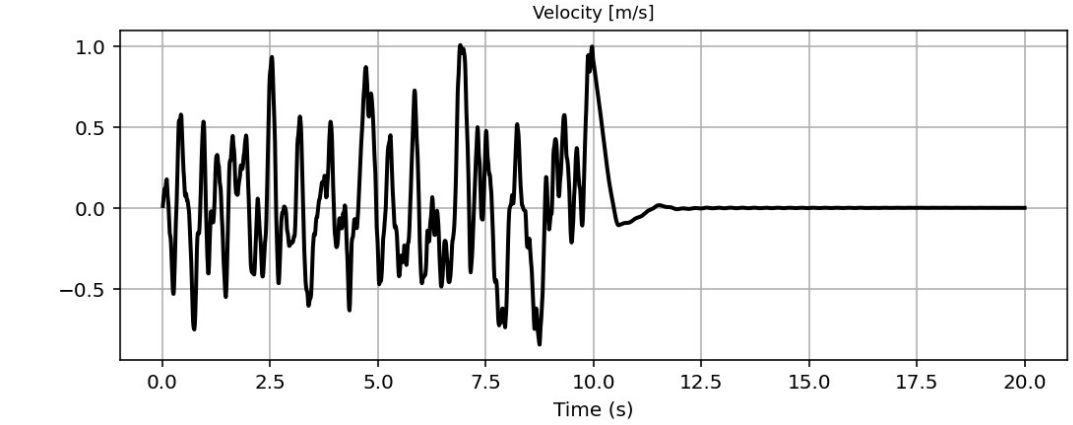
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

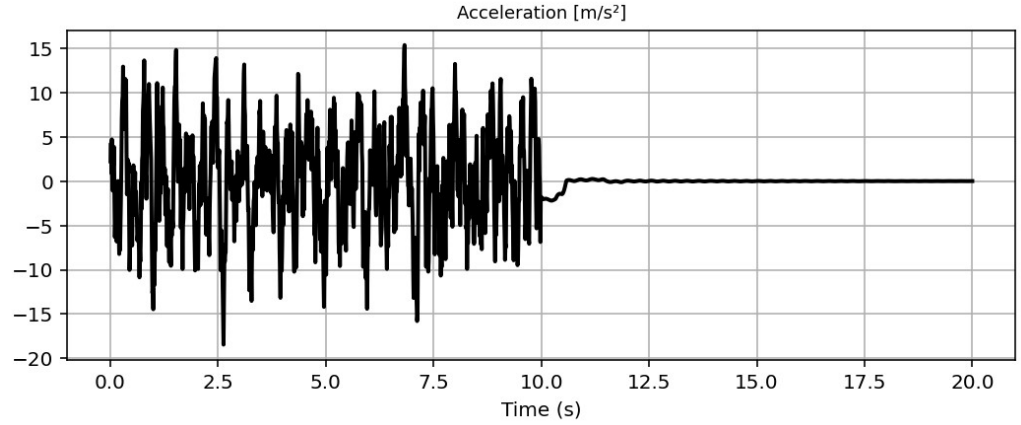
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 21, Col 18 | UTF-8 | CRLF | RW | Mem 48%

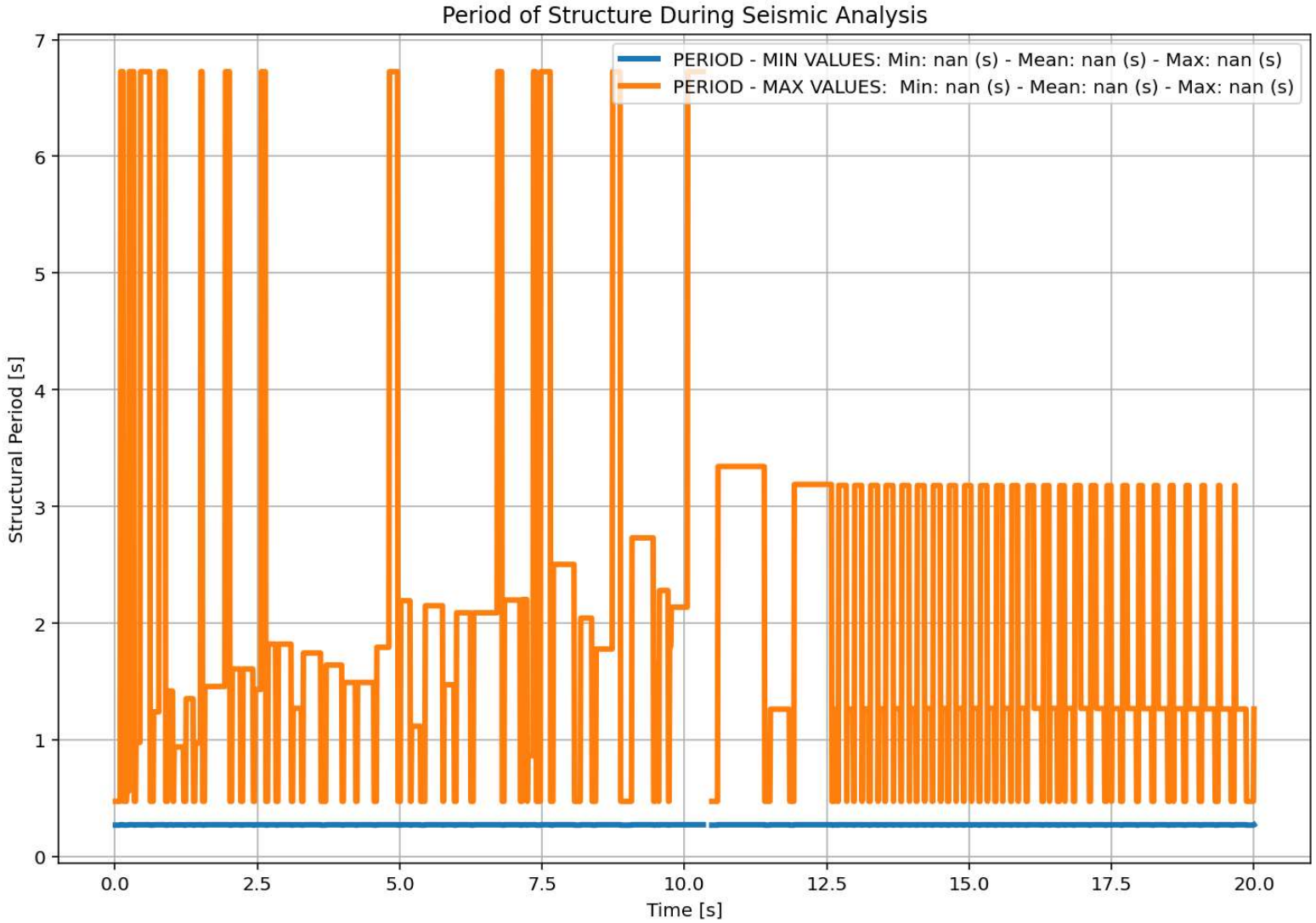

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.1
0.0
0.1
0.2
0.3
Displacement [m]

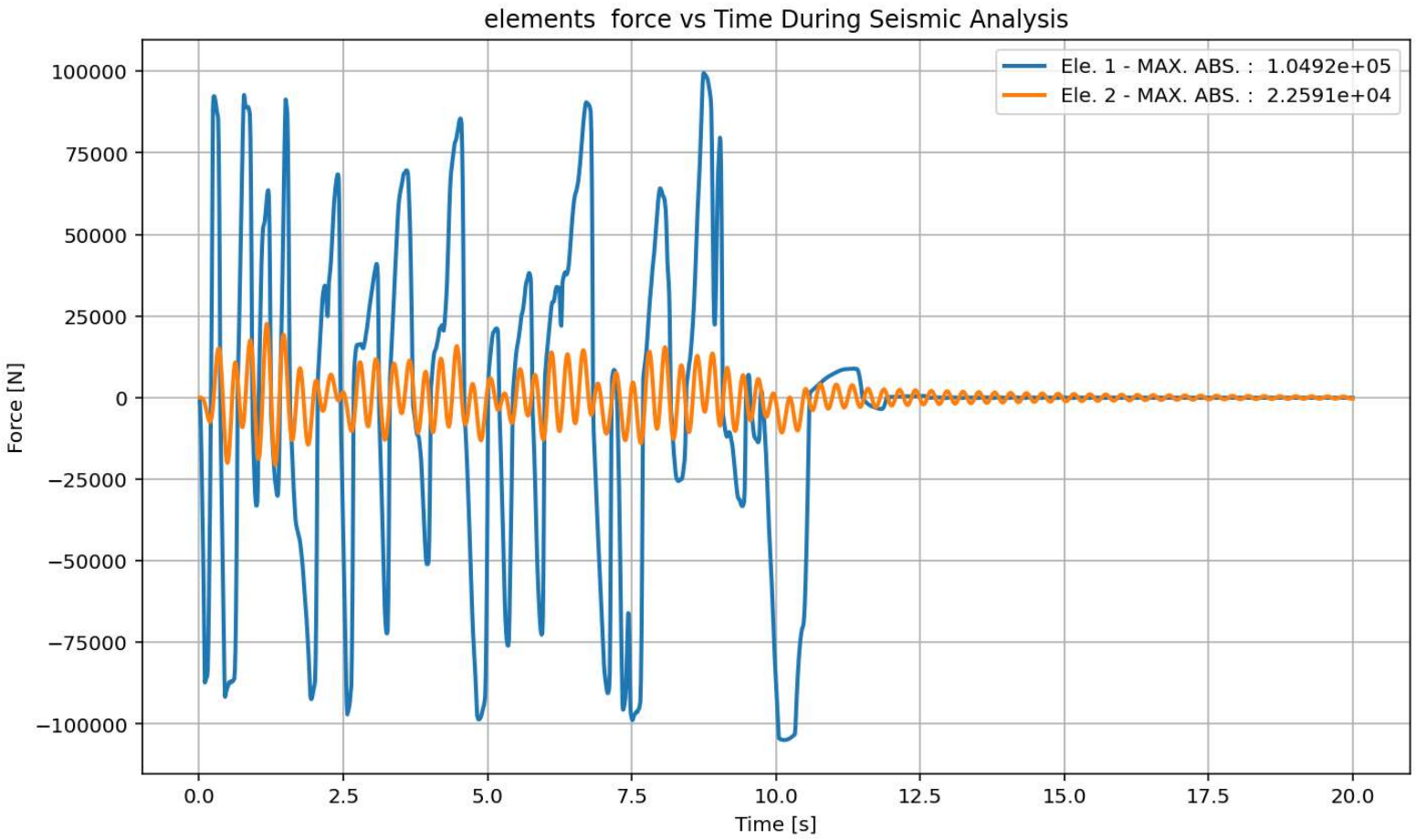

Reaction [N]
100000
50000
0
−50000
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

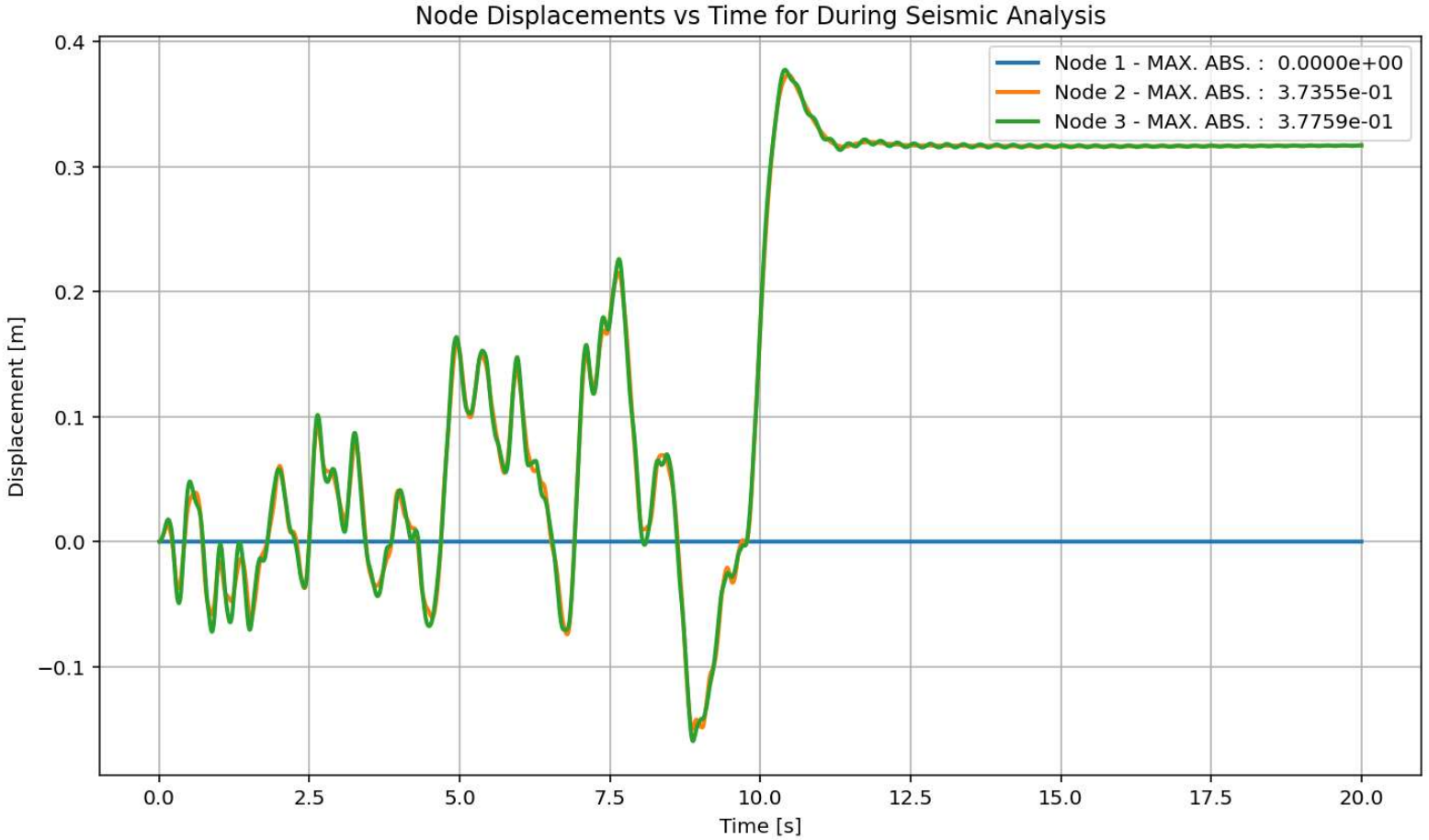

Displacement [m]
0.3
0.2
0.1
0.0
−0.1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

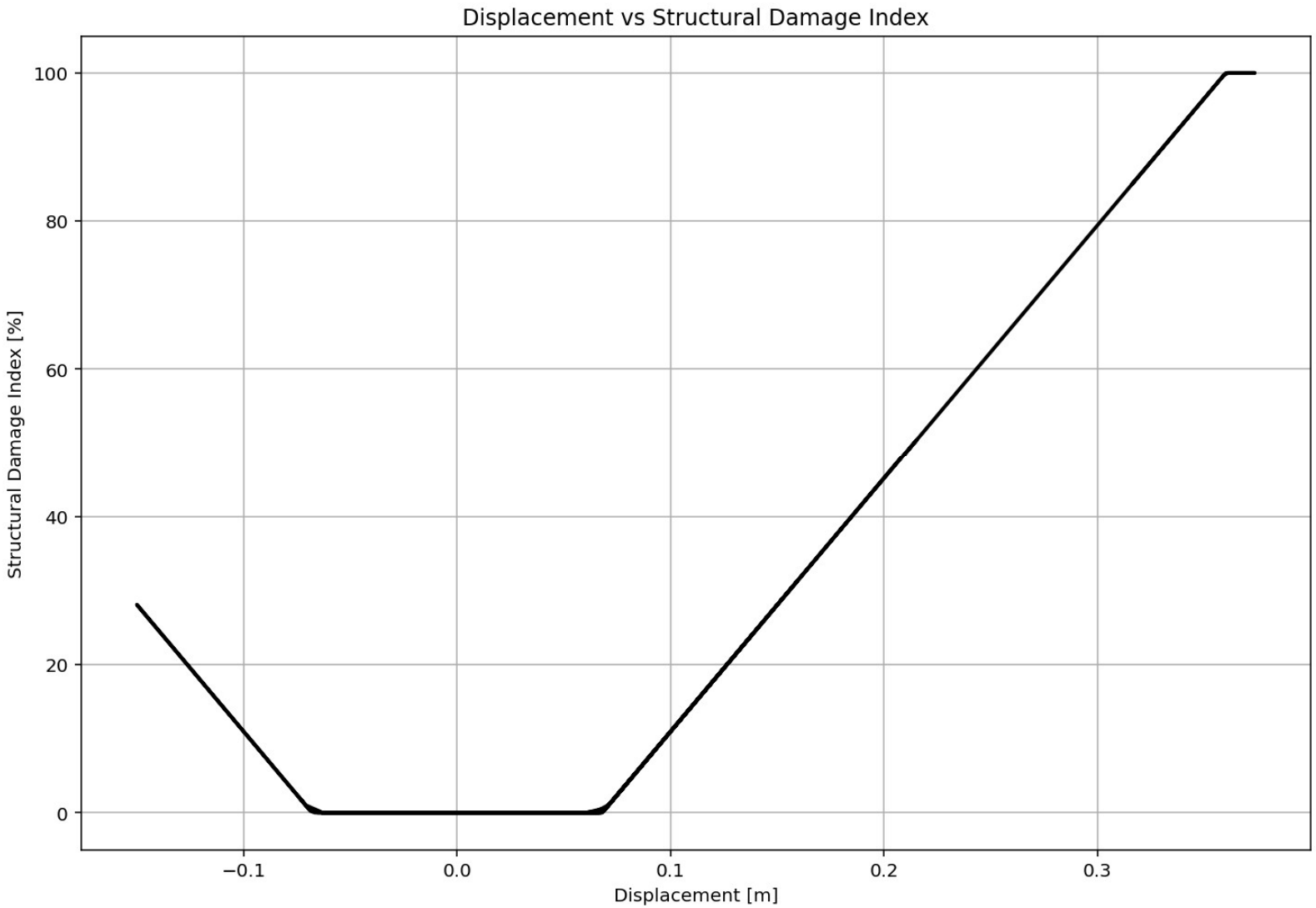

Velocity [m/s]
1.0
0.5
0.0
−0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

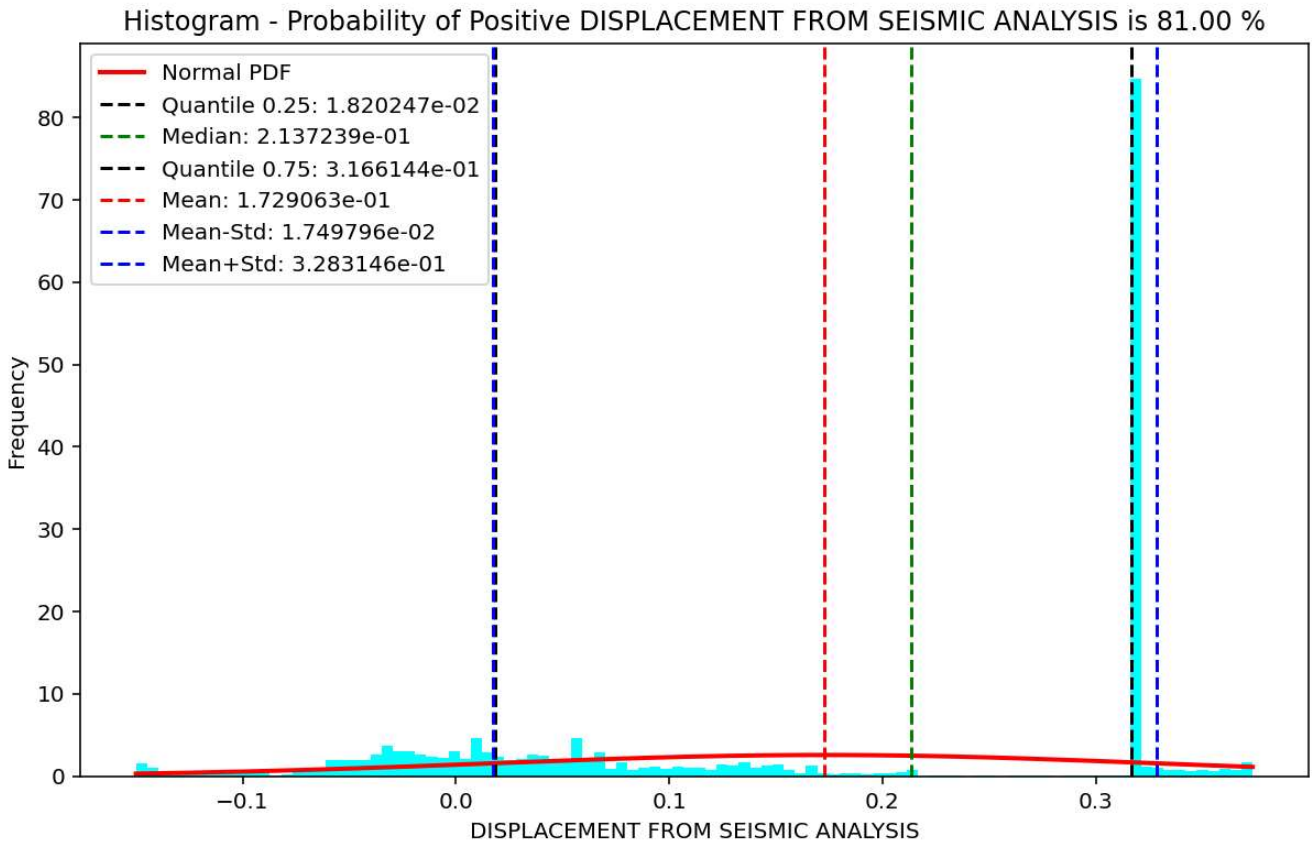

Acceleration [m/s²]
15
10
5
0
−5
−10
−15
−20
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

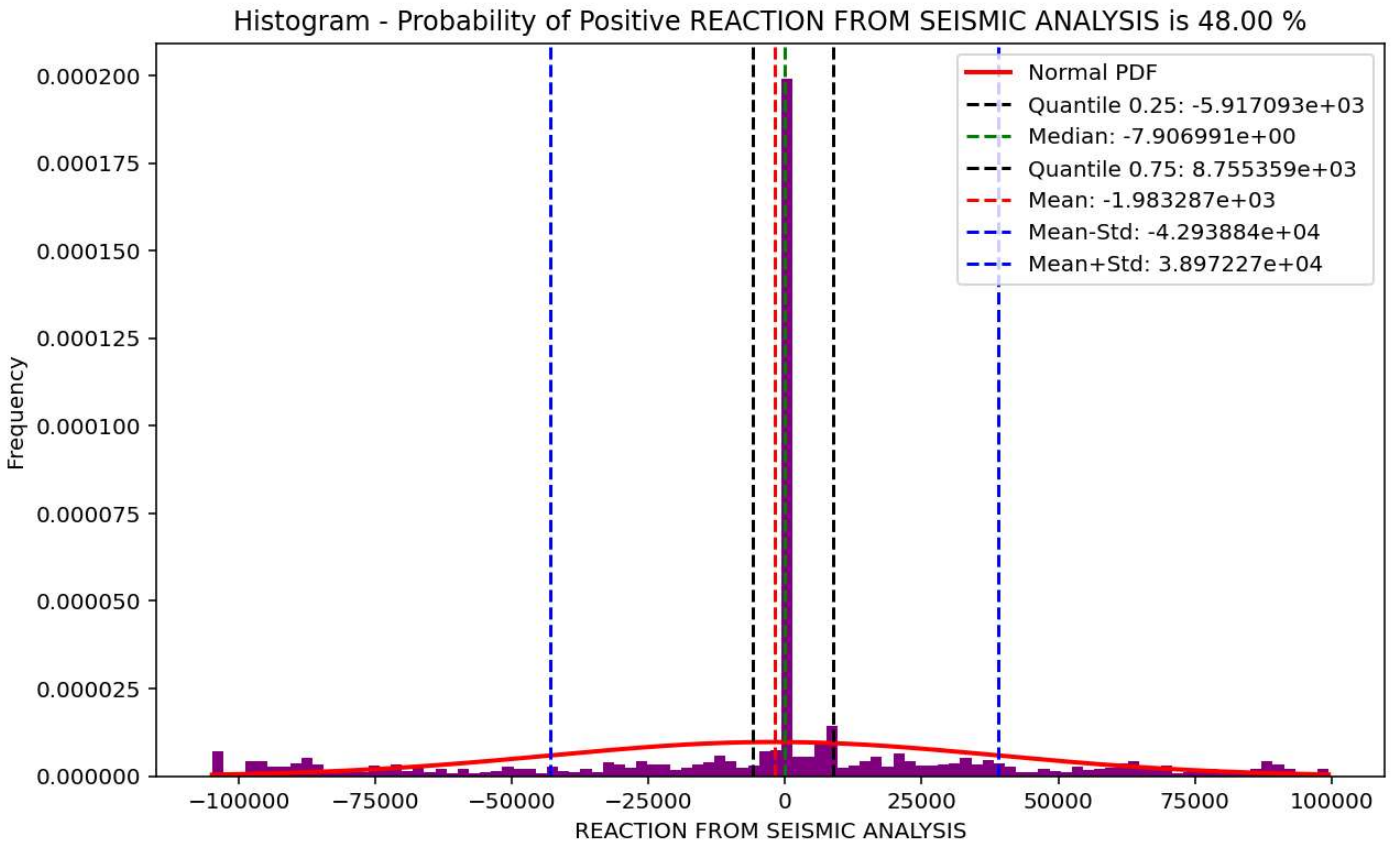

Period of Structure During Seismic Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Time [s]
0
1
2
3
4
5
6
7
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

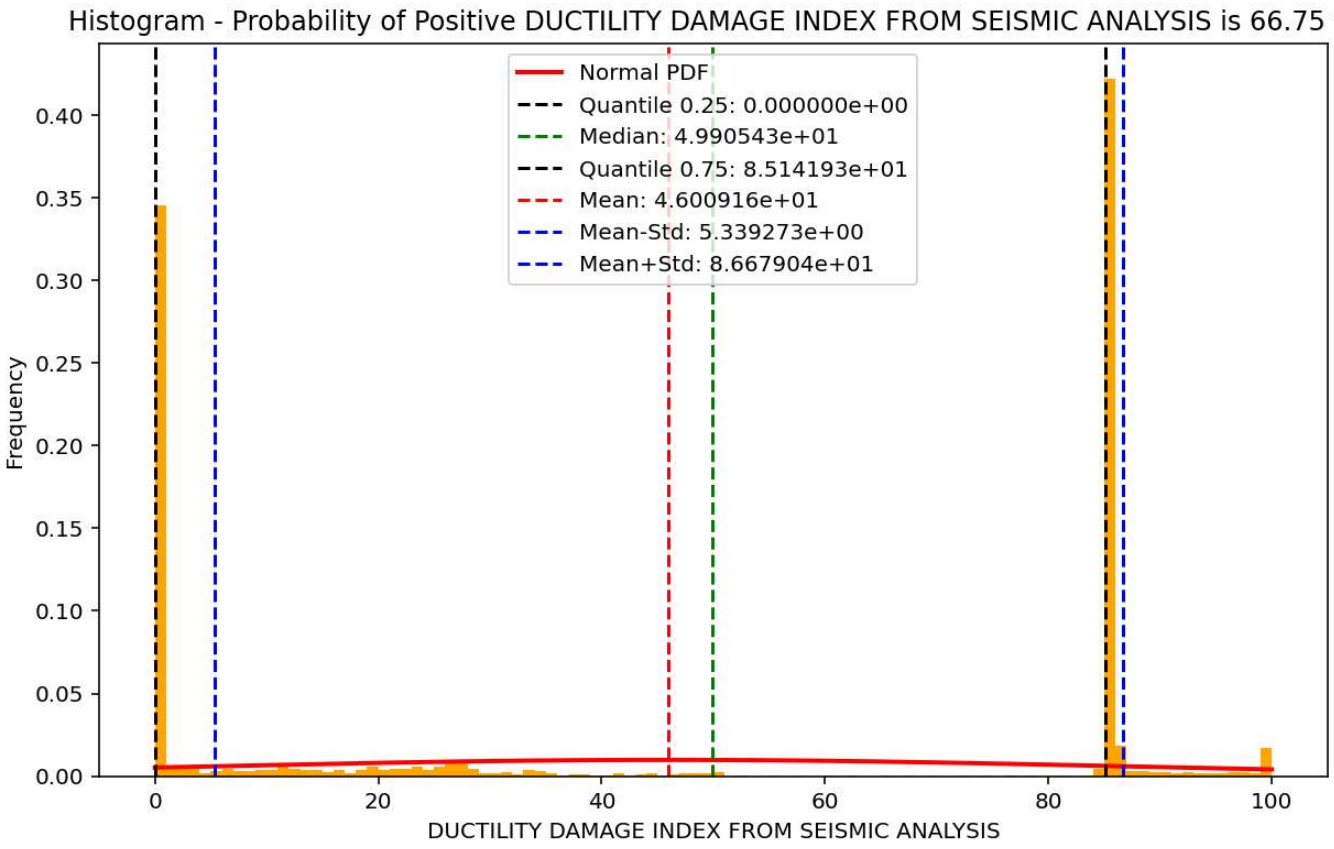

elements force vs Time During Seismic Analysis
Ele. 1 - MAX. ABS. : 1.0492e+05
Ele. 2 - MAX. ABS. : 2.2591e+04
Force [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

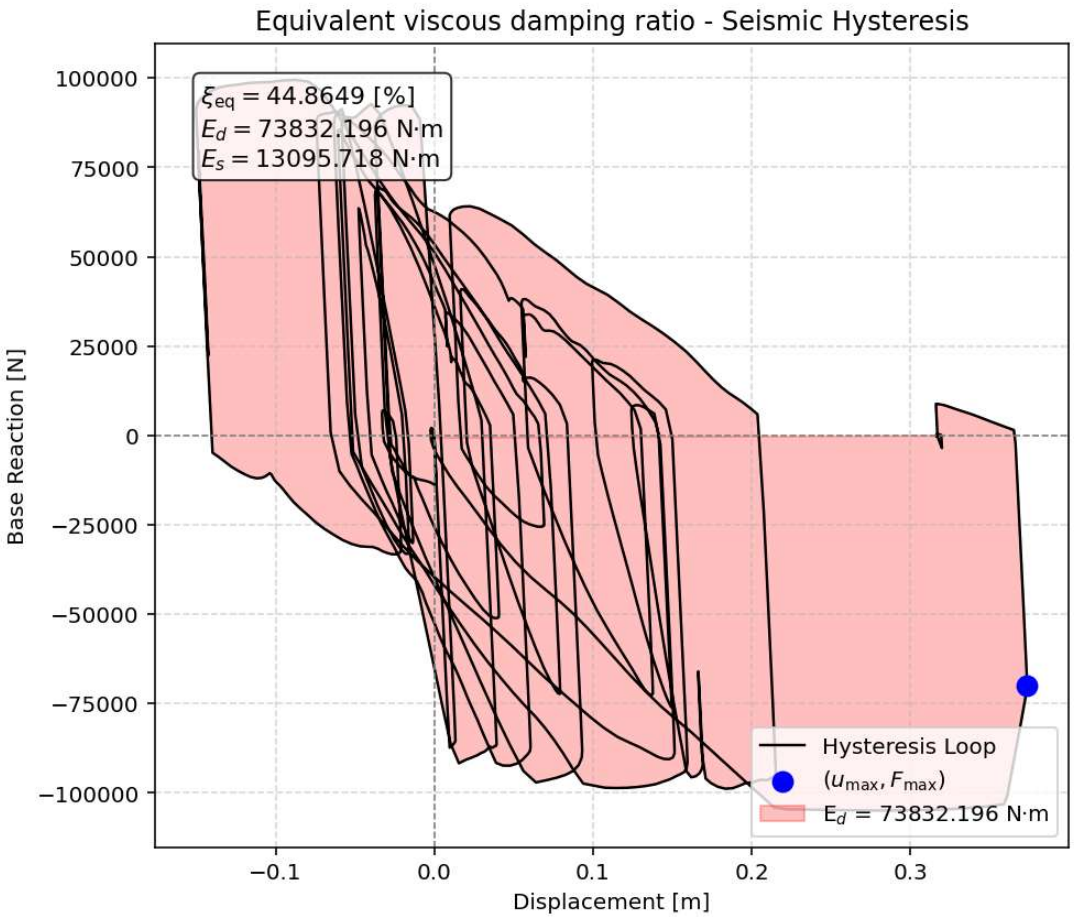

Node Displacements vs Time for During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.7355e-01
Node 3 - MAX. ABS. : 3.7759e-01
Displacement [m]
Time [s]
0.4
0.3
0.2
0.1
0.0
−0.1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

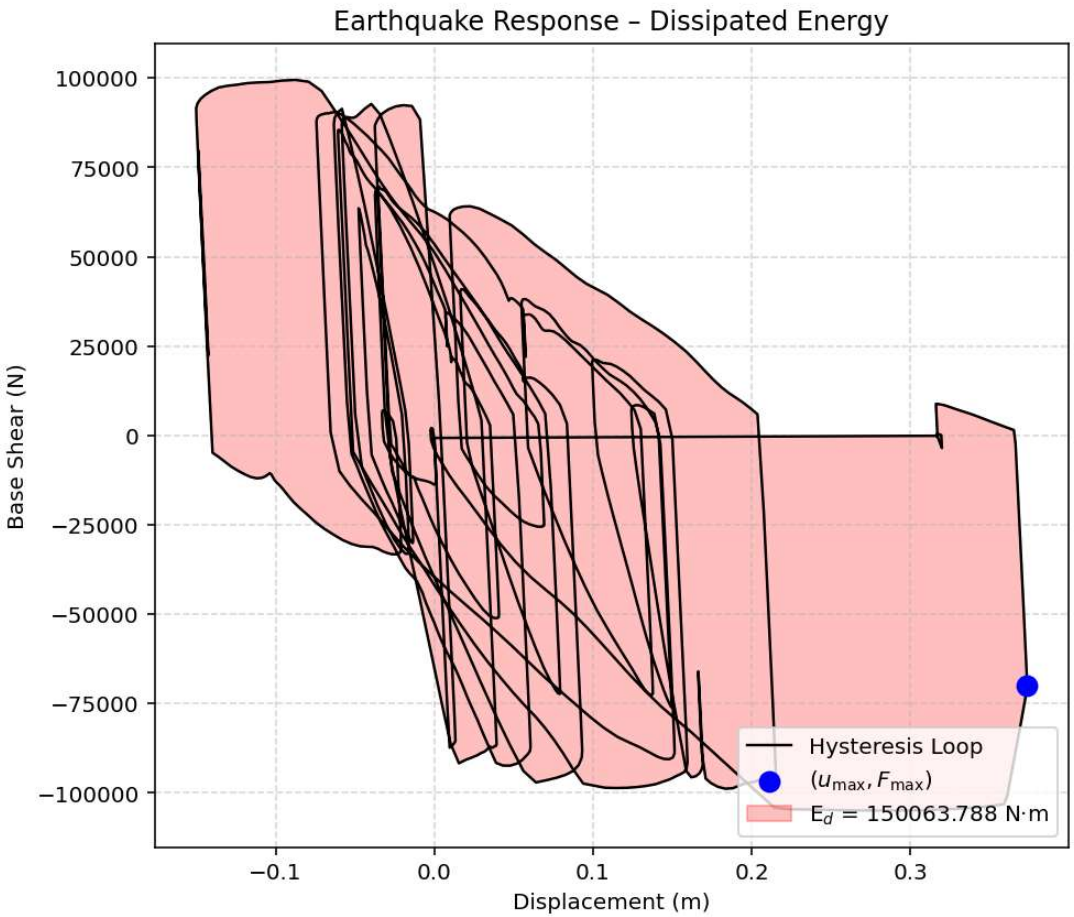

Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.1
0.0
0.1
0.2
0.3
Displacement [m]

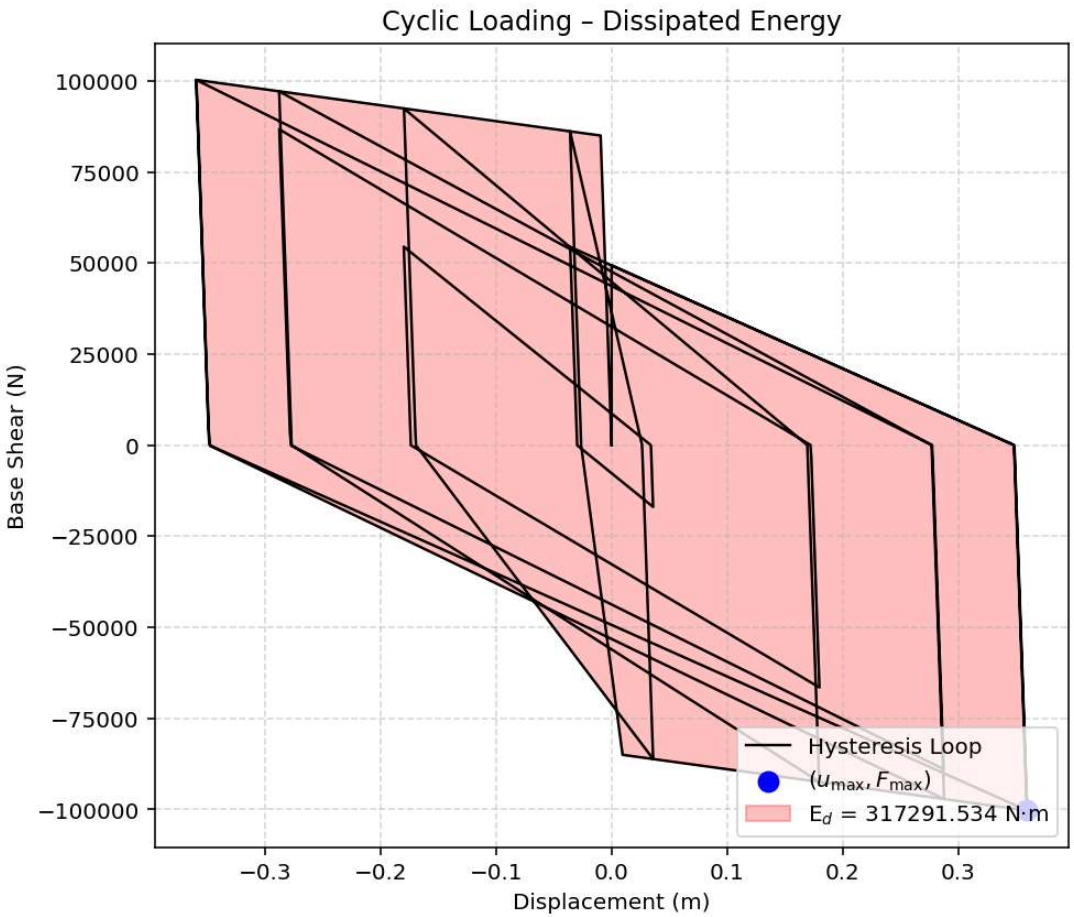

Histogram - Probability of Positive DISPLACEMENT FROM SEISMIC ANALYSIS is 81.00 %
Normal PDF
Quantile 0.25: 1.820247e-02
Median: 2.137239e-01
Quantile 0.75: 3.166144e-01
Mean: 1.729063e-01
Mean-Std: 1.749796e-02
Mean+Std: 3.283146e-01
Frequency
0
10
20
30
40
50
60
70
80
−0.1
0.0
0.1
0.2
0.3
DISPLACEMENT FROM SEISMIC ANALYSIS


Histogram - Probability of Positive REACTION FROM SEISMIC ANALYSIS is 48.00 %
Normal PDF
Quantile 0.25: -5.917093e+03
Median: -7.906991e+00
Quantile 0.75: 8.755359e+03
Mean: -1.983287e+03
Mean-Std: -4.293884e+04
Mean+Std: 3.897227e+04
Frequency
0.000200
0.000175
0.000150
0.000125
0.000100
0.000075
0.000050
0.000025
0.000000
−100000
−75000
−50000
−25000
0
25000
50000
75000
100000
REACTION FROM SEISMIC ANALYSIS


Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS is 66.75 %
Normal PDF
Quantile 0.25: 0.000000e+00
Median: 4.990543e+01
Quantile 0.75: 8.514193e+01
Mean: 4.600916e+01
Mean-Std: 5.339273e+00
Mean+Std: 8.667904e+01
Frequency
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0
20
40
60
80
100
DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS

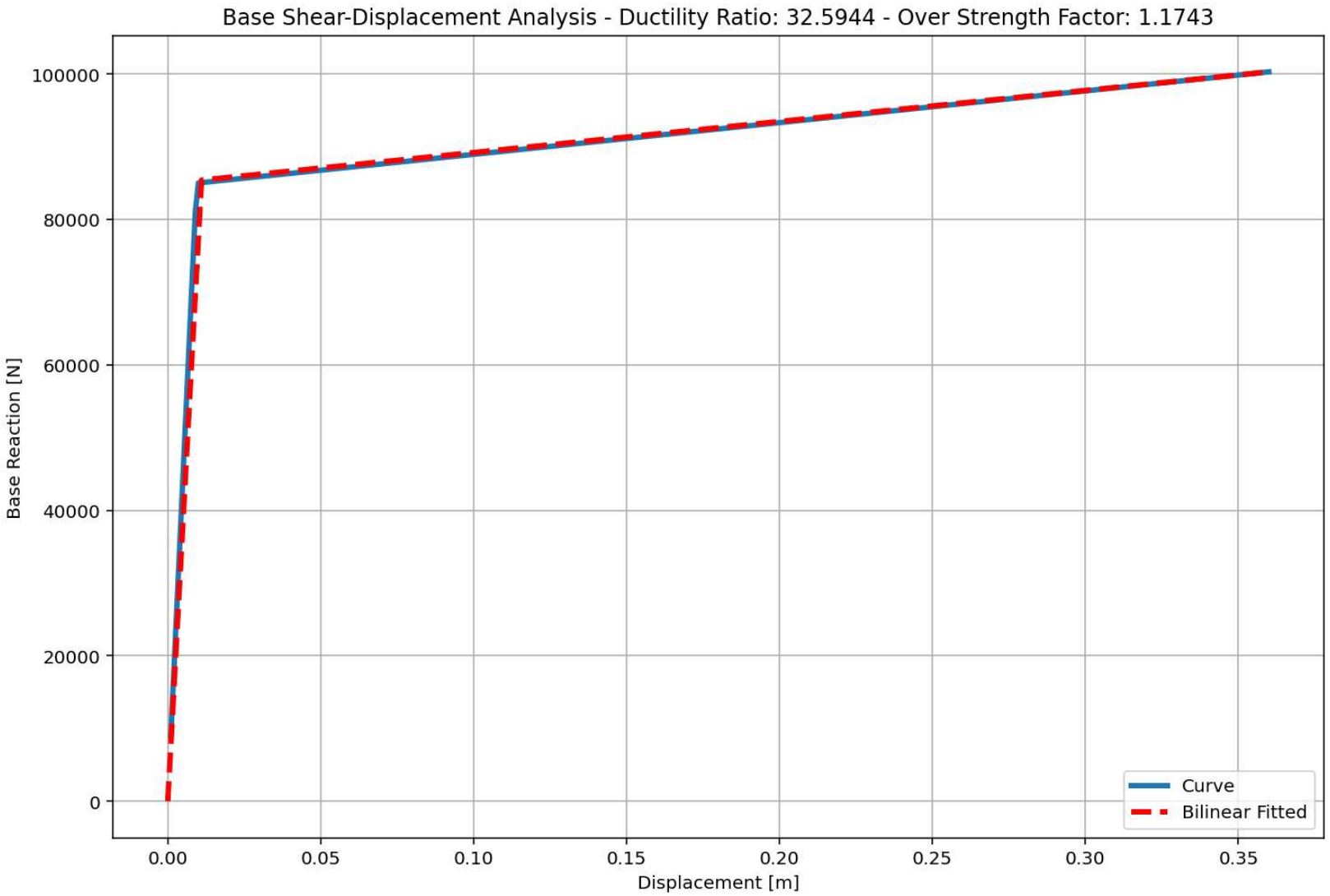

Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 44.8649$ [%]
$E_d = 73832.196$ N·m
$E_s = 13095.718$ N·m
Base Reaction [N]
Displacement [m]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 73832.196 N·m


Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 150063.788 N·m

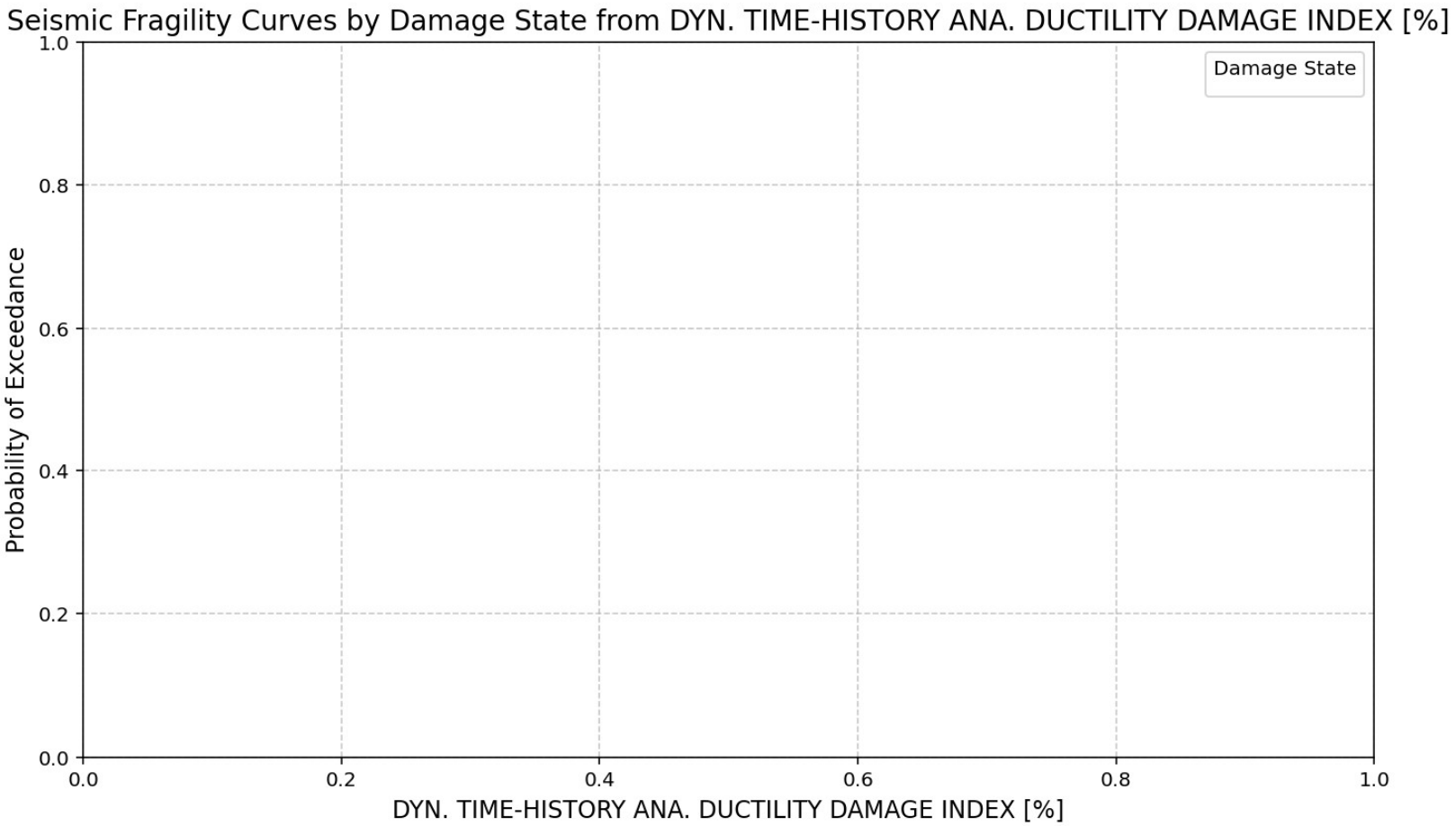

Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 317291.534 N·m

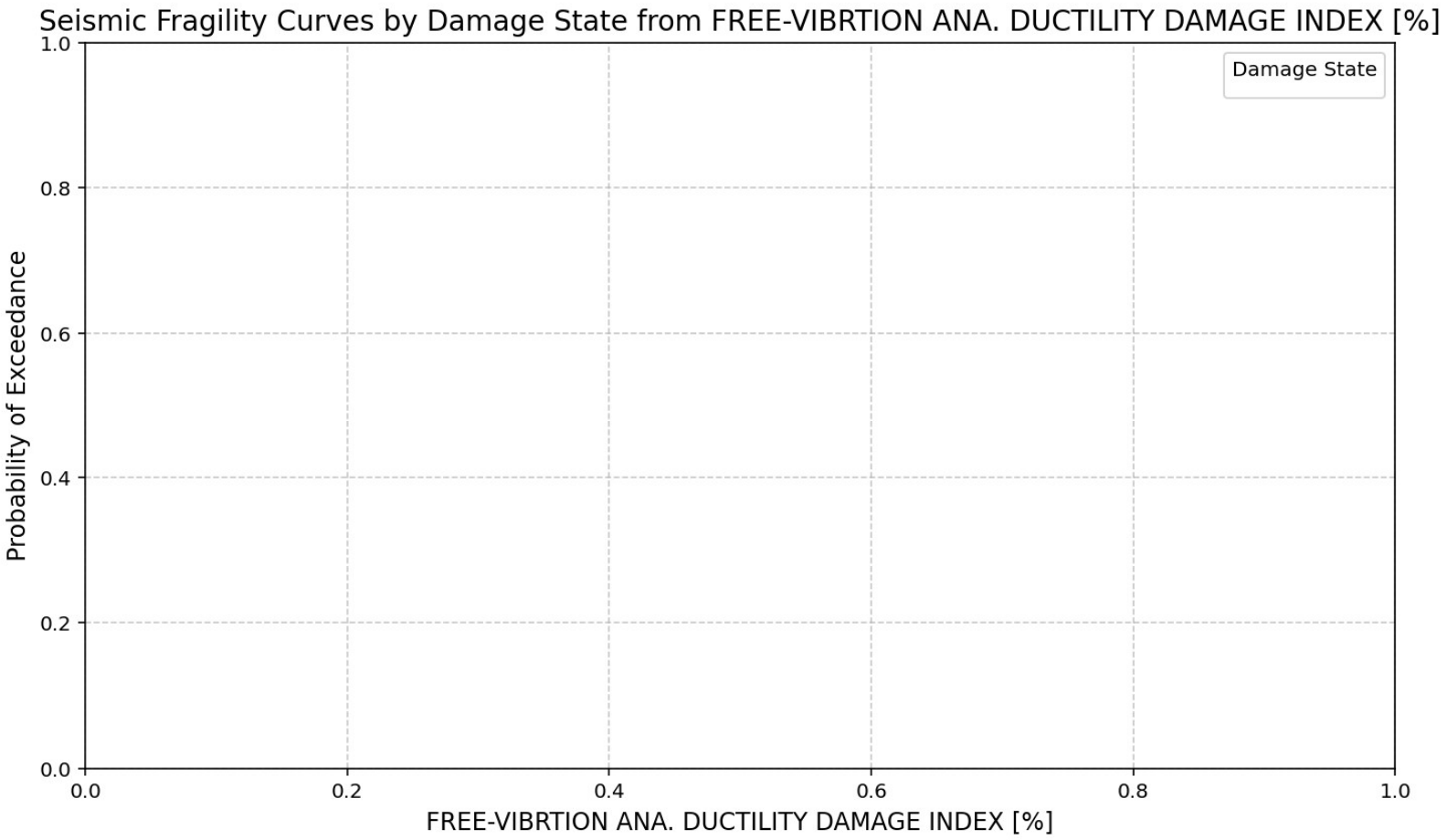
Dissipated Energy from Earthquake= 150063.79 N·m
Dissipated Energy from Cyclic Displacement= 317291.53 N·m

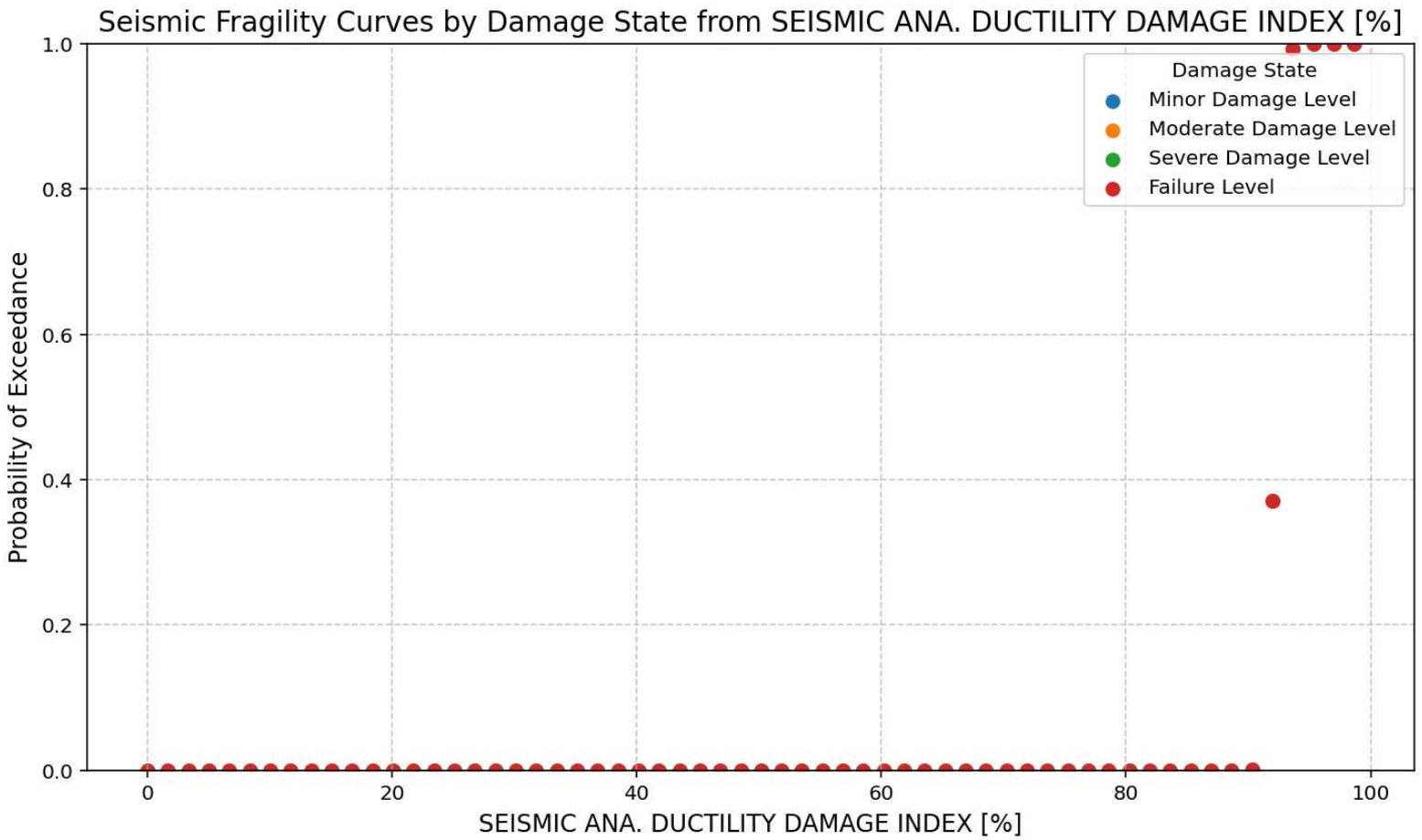
DISSIPATED ENERGY CAPACITY INDEX: 47.295 [%]
ZONE 5: MODERATE–HIGH DAMAGE


Base Shear-Displacement Analysis - Ductility Ratio: 32.5944 - Over Strength Factor: 1.1743
Base Reaction [N]
100000
80000
60000
40000
20000
0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Displacement [m]
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
--------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.10448514e-02 8.54147483e+04]
 [3.60000000e-01 1.00300000e+05]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :     7733444.75
 Structure Plastic Stiffness :     278611.11
 Structure Tangent Stiffness :     42656.63
 Structure Ductility Ratio :      32.59
 Structure Over Strength Factor:   1.17
+-------------------------------------------------+
Over Strength Coefficient (Ω0):     1.1743
Displacement Ductility Ratio (μ):    32.5944
Ductility Coefficient (Rμ):         32.5944
Structural Behavior Coefficient (R): 38.2746
Structural Ductility Damage Index in X Direction: 103.8821 (%)
```


Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]


Seismic Fragility Curves by Damage State from FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Probability of Exceedance
0.0
0.2
0.4
0.6
0.8
1.0
FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]


Seismic Fragility Curves by Damage State from SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Minor Damage Level
Moderate Damage Level
Severe Damage Level
Failure Level
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]